

Tron-Car RC Drivetrain and Chassis

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ABSTRACT

Tron-Car RC Drivetrain and Chassis

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The Project detailed in this engineering report is a remote-controlled Baja car that was used to compete in various events held on campus by the ASME chapter. This RC Baja car was split into two distinct problems: the Drivetrain and Chassis, and the Suspension and Steering. This report details the design, prototyping, and testing of a drivetrain that transfers torque from the RC car’s motor to the wheels for propulsion as well as a chassis that reliably holds all subsystems.

The Drivetrain and the Chassis were manufactured using a combination of traditional CNC manufacturing, laser manufacturing, and additive manufacturing methods. Materials selected for this project were 6061-T6 Aluminum and PLA plastic. Aluminum was used for its lightweight as well as its structural rigidity. PLA plastic is used for its strength and being cheap, making it best suitable for rapid prototyping.

Testing for this project was done on the CWU campus for ease of scheduling. The tests that were conducted consisted of an acceleration test, a head-on collision test, and a ramp climb test. The acceleration test resulted in accelerating from 0 to 10mph in under 5 seconds, deeming it successful. The head-on collision test resulted in a deflection of .5 inches, deeming it successful. The ramp test resulted in successfully climbing over the 45° ramp in under 3 seconds, deeming it successful.

Presentation Type:

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1. INTRODUCTION

a. Description

This project was chosen to create a vehicle capable of competing in the CWU ASME RC Baja competition. The project detailed in this proposal will use statics, dynamics, mechanics of materials, and physics calculations to develop an optimal solution for the baja competition. This competition consists of 2 events that include the Slalom-and-Sprint as well as the Baja Competition. This RC baja car seeks to create a low-cost, radio-controlled vehicle that can compete with success. The following project proposal will include documentation for the design and manufacturing for the drivetrain and chassis of the RC Baja competition car.

b. Motivation

This project was undertaken with the intention of creating a platform to learn how dynamics apply to vehicles, using this knowledge to develop skills related to RC inspection and survey tools.

c. Function Statement

For the RC Baja car, the chassis is the support for sub-assembly attachments, as for the Drivetrain, the drivetrain provides torque to the system for motion.

d. Requirements

The Baja RC Car will have the following specifications by the end of the project deadline:

1. Drivetrain assembly must be capable of sustained operation (for 2 minutes) at 20 mph without reaching point of failure.
2. Chassis must deflect less than .6 inches from a head-on impact at 15 mph.
3. Vehicle weight must be less than 10 pounds.
4. Drivetrain must be able to achieve a maximum speed of 30 mph
5. Drivetrain must be able to provide a minimum 0.2 Nm of torque to the wheels during initial acceleration.
6. Chassis must deflect less than .1 inches from normal due to weight of sub-assemblies.
7. Drivetrain sub-assemblies must be able to deflect from normal .2 inches due to impacts of 15 mph without yielding.
8. Drivetrain must be able to achieve a speed of 10 mph in under 10 seconds
9. Chassis must be capable of withstanding 20-40 Hz vibrations at a speed of 15-20 mph.
10. Drivetrain must be capable of providing torque to climb a 45° ramp.
11. The RC car must have a drag force lower than 120 lbf when at speeds between 10 mph and 30 mph.

e. Engineering Merit

This project will include calculations using formulas learned from the CWU Mechanical Engineering Technology Program. These calculations include Static calculations for forces acting on the chassis as well as the forces acting on the drivetrain, Dynamics calculations to determine forces acting on the system during impacts, and formulas from mechanics of materials for calculating stresses induced during operation, and physics calculations to determine the torque requirements for achieving minimum and maximum speeds for the success in the competition.

f. Scope of Effort

The scope of this proposal will only include drawing for the design and manufacturing of the Drivetrain and the chassis of the vehicle. All other parts and assemblies will be completed by either the designers' team or purchased through outside means.

g. Success Criteria

The success of the project was determined by how well the RC vehicle performs during testing and whether the vehicle completes the racing challenges set by the competition board, ie. Slalom-and-Sprint and the Baja competition. The vehicle was to be evaluated based on requirements set by both team members prior to manufacturing. This Project would be deemed successful if the vehicle achieved 2nd place during the Baja competition and achieved a Slalom-and-Sprint time of under 7 seconds.

h. Stakeholders

The Stakeholders of this project are as follows:

- Alex Amadio. Team member, designer, and builder.
- Federico Berrospe. Team member, designer, and builder.
- Charles Pringle, Mentor and Instructor.
- John Choi, Mentor and Instructor.
- CWU MET department, providing lab space for manufacturing and prototyping.
- Susan Donahue, donor of the Walter R. Kaminski Memorial scholarship, scholarship granted to Mechanical Engineering students for senior project funding.

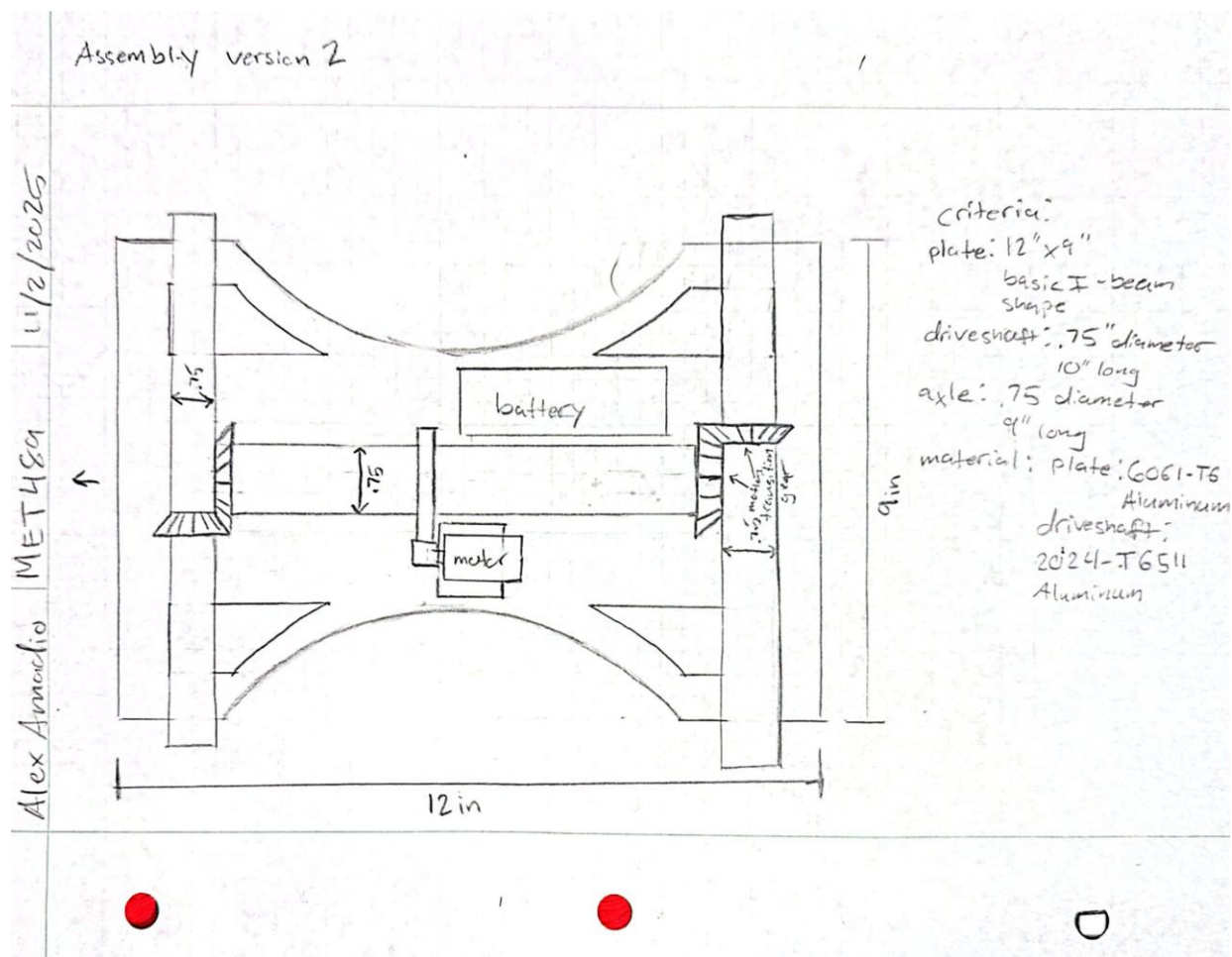
2. DESIGN & ANALYSIS

a. Approach: Proposed Solution

The problem this project seeks to solve is the need for an RC car capable of completing the CWU ASME RC Baja Competition. A decision matrix was utilized to gain a better understanding of what type of design is best suited to solve this problem. As detailed in Appendix F, the design matrix shows that an all-wheel drive drivetrain is best suited for this type of competition. During initial brainstorming sessions with the project partner and peers, it was suggested to use a belt system for the drivetrain as well as a 2-motor configuration, this was later scrapped as it violates the rules and regulations of the RC Baja competition.

b. Design Description

The current design uses a single central drive shaft that is connected to both the front and rear axle. These are connected using canted gear that fitted to the shaft of both. These gears will likely be covered by a housing unit, to prevent gears from wearing faster than designed from rocks, dirt, or other debris. The motor will use a similar canted gear system to rotate the central drive shaft; however, the motor will go through a gear reduction as to not to share the pinion gear in the motor, or the gears on the way to the drive shaft.



c. Benchmark

This project is benchmarked on an RC competition car that is commercially available and is sold by TENSSENX, an outdoorsman accessory online retailer. This commercially available vehicle advertises a top speed of 40 mph, a chassis made from aluminum, and a total weight of 1.2 pounds. All these advertisements are within the criteria for success and the requirements set for this project, with the top speed being lower than the benchmarked model.

d. Performance Predictions

Based on calculations done, the RC car's drivetrain is predicted to provide enough torque to achieve a maximum speed of 30 mph, while maintaining an operational speed of 20 mph. the RC car is predicted to be able to achieve 2nd place in the Baja competition, and a time of 7 seconds in the slalom-and-sprint.

e. Description of Analysis

The analysis done for this project was completed using formula from the CWU MET statics, to gain an understanding of what the weight on the wheel will be. This understanding allows for a better understanding of what types of wheels are needed to hold the RC car off the ground. Mechanicals of materials formula are used to better understand what the deflection of the chassis and the drivetrain due to components as well as impacts while at speeds of 15 – 20 mph. Physics and dynamics equations are used to find what the value of torque is acting on the drivetrain during operation, as well as during a cold start. Physics, dynamics, and Mechanical design calculations will also be performed to assess how much torque needs to be produced by the motor to climb a 45-degree ramp. Statics calculations will be used to create Free Body Diagrams to determine the forces on the wheels, through this calculation the proper wheel size can be determined.

f. Scope of Testing and Evaluation

The RC car will be tested and evaluated for whether it was successful by how it performs during testing in the spring as well as during the RC Baja Competition held by the CWU ASME chapter. The scope of the testing done for this is by how well the car can handle sharp turns while speeding up and slowing down the motor RPM during the Slalom. The car will also be tested on how well it can go from 0 – 10 mph during the Sprint section of the competition. Finally, the RC car's drivetrain and chassis will be tested on how well it can handle rough terrain and sudden drops and jumps during the Baja competition. Testing prior to this will be down to test its ability to climb a 45-degree ramp and how well it performs on subsequent impact.

g. Analysis

The requirements for this project will be met through the manufacturing and selection of components. The manufactured parts are designed using formulas learned through the CWU MET undergraduate program. The engineering merit of this project is to be exemplified using the CWU: RADD (Requirements, Analysis, Design Parameters, and Documentation) metric.

Some of the requirements found in section 1d were calculated using Plate deflection analysis, speed-torque relationship formulas, and Finite Element Analysis. Detailed below are analysis performed to understand what is to be manufactured to achieve requirements found in section 1d.

i. Analysis 1 – Structural Deflection and Weight on Wheels

This analysis is being conducted to satisfy design requirement 1.d.6, for the deflection on the chassis due to component weight. After calculations found in appendix A01, the chassis will deflect 0.00004 inches due to the weight of components. While the weight that the wheels experience is .3286 pounds on each of the front wheels and .3481 pounds on each of the rear wheels. The documentation for this analysis can be found in Appendix A01.

ii. Analysis 2 – Torsional Shear, angle of rotation due to torque, Driveshaft Diameter and Critical Speed Frequency

This analysis was conducted to determine the shear affecting the driveshaft, and the angle of rotation caused by the motor. The torque used for this calculation came from the minimum torque requirements defined in section 1.d.5 compared to the manufacturer's specifications. This analysis also looked at whether the chosen diameter for the drive shaft would break at maximum required speed as defined by Requirement 1.d.4. The shear value was calculated to be 576 lb*in with the angle of twist being 0.0004 degrees. This signifies that the drive shaft will not shear at due to torque from the motor. The drive shaft diameter is .25 inches, with the critical speed being 403513 RPM, this means that the drive shaft would need to spin at 403513 RPM before breaking or bending due to rotation. The critical frequency is 754225 Hz; this signifies that the drive shaft will break after experiencing vibration at the calculated frequency. The documentation for this analysis can be found in Appendix A02.

iii. Analysis 3 – Chassis Deflection due to 15 mph impact

This analysis was conducted to understand how the chassis design would deflect due to a head-on impact going 15 mph. Using deflection of a beam and the Euler's buckling equation. It was found that if the vehicle weighs 3 pounds or less, the chassis will experience a defection of .599 inches. This analysis satisfies the requirements found in section 1.d.2, however the value is extremely close to being out of spec. The vehicle would need to weigh less than 3 pounds to have this value see a reduction. The documentation for this analysis can be found in Appendix A03.

iV. Analysis 4 – Actual RPM and Gear Reduction

This analysis was conducted to gain an understanding of what RPM is coming from the chosen motor and how that RPM gets transferred to the Drivetrain. This was calculated using torque equations and an assumption of the motor's efficiency. The gear ratio was found using the gear ratio equation. Based on calculations found in Appendix A04, the motor transfers an RPM of 14592 RPM with a motor efficiency of 80%. This is an appropriate estimation of the motor efficiency as this encompasses if the motor is made poorly. For the gear reduction needed to get the RPM required for a maximum speed of 30 MPH, with 4-inch diameter tires, would be a

minimum of a 5-to-1 gear reduction. This means that for every 5 rotations from the motor the drivetrain would rotate once. This analysis was conducted to satisfy requirements found in 1.D.4. The documentation for this analysis can be found in Appendix A04.

V. Analysis 5 – Motor Power and Friction on wheels

This analysis was conducted to find the power that is generated from the motor during an incline this also looks at how much friction is needed from the wheels to propel up a 45-degree incline ramp. Based on calculations found in Appendix A05, with the use kinematics the friction force that was found is 68.31 pounds, and using power equations the power from the motor was found to be 1556.2 horsepower to go up a 45-degree incline. This satisfies the requirement 10 found in section 1.d. This motor power value is used during later analysis. The documentation for this analysis can be found in Appendix A05.

Vi. Analysis 6 – Damage Factor and Number of Cycles over time

This analysis was conducted to determine if the driveshaft can run continuously over a 2-minute span. This is found as a requirement in section 1.d.1. Using the power from the motor found during Analysis_05, as well as specs from chosen components, the damage factor was found. The damage factor is a ratio between the number of cycles of operation over the number of cycles the part can handle before yielding. Based on calculations found in Appendix A06 as well as standard tables for 2024 Aluminum for number of cycles when related to stress, the number of cycles the drive shaft can handle before yielding is 14000 cycles with the number of cycles over 2 minutes is 5826. This puts the damage factor at .441, this is less than 1 meaning the shaft will not yield during the operational 2 minutes. However, this details 2 minutes at 35 mph, whereas the requirement states that the driveshaft is operation over 2 minutes at 20 mph. This is lower than the calculation for 35 mph, this means that there is less stress affecting the drive shaft at 20 mph, thus significantly increasing the number of cycles before yielding. The documentation for this analysis can be found in Appendix A06.

Vii. Analysis 7 – Deflection of Driveshaft during a 15-mph impact

This analysis was conducted to gain a better understanding of the forcing acting on the driveshaft during a 15-mph impact. This satisfies the requirement 1d_7, that states that the drivetrain must not deflection more than .2 inches during impact. Based on calculations found in Appendix A07, it was found that the drive shaft deflects exactly .2 inches during impact. While this does satisfy the 1d-7 requirement, this doesn't fully account for a safety factor. The documentation for this analysis can be found in Appendix A07.

Viii. Analysis 8 – Chassis vibration at 15 mph

This analysis was conducted to understand what kind of vibration is affecting the chassis while at operational speeds. Found in section 1d-9, the requirement states that the chassis needs to withstand a frequency of 20-40 Hz. After completing analysis 8, calculations found in Appendix A08, it was found that the chassis experiences a frequency of 36 Hz on gravel. This is at the higher end of the frequency range which indicates that the car will vibrate more than needed. This vibration value, however, don't consider suspension from the shocks these values will likely

be 5-7 Hz lower with the consideration of shocks. This analysis satisfies requirement 1d-9. The documentation for this analysis can be found in Appendix A08.

Viii. Analysis 9 – Motor torque needed for acceleration.

This analysis was conducted to understand what the minimum torque that's needed to get to 10 mph within 10 seconds, as stated as a requirement in section 1.d.8. Based on calculations found in Appendix A09, the minimum torque that is needed is 3.538 Nm or about 2.61 lbf. This is significantly under the torque that the motor can produce based on manufacturer's specifications. Based on these calculations and the specifications from the vendor, the motor is judged to be able to produce the minimum torque to be able to accelerate the RC car to 10 mph within 10 seconds. This analysis satisfies requirement 1.d.8 for being able to reach 10 mph in 10 seconds. The documentation for this analysis can be found in Appendix A09.

X. Analysis 10 - Drag Force at operational speeds

This analysis was conducted to gain a better understanding of the drag force that is affecting the RC car at different operational speeds. Based on calculations found in Appendix A10, the drag force was found to be 12.33 lbf at 10 mph, 49.34 lbf at 20 mph, and 111.03 lbf at 30 mph. These values are under the acceptable maximum value detailed as a part of requirement 1.d.11. This indicates that with the assumption that the height of the car is 5 inches with the height added by the components, then requirement 1.d.11, for drag values being lower than 120 lbf, is satisfied. The documentation for this analysis can be found in Appendix A10.

h. Device: Parts, Shapes, and Conformation

The design for the car was chosen after completing different analysis of various aspects of the requirements. These aspects use a curved profile to prevent stress concentration from accumulating around joints. The chassis uses a 1-foot-long narrow body to prevent deflection due to components. The chassis uses a safety factor of 1.5, while the drivetrain has a safety factor of 2.

i. Device Assembly

The car uses a selected motor and battery to drive a student-built drivetrain to allow for locomotion, while an entirely student-built chassis provides the support for the entire project to come together. The chassis uses an aluminum frame with points to connect straps for vital components, like the motor housing and battery. The drivetrain uses a shaft with angled gears to convert rotational motion 90 degrees to the axles, to provide torque to the wheels at the same time.

j. Technical Risk Analysis

Some technical risk associated with this project would be around whether the gears for the drivetrain are properly tempered, this would cause a problem during operation if the gear's teeth sheared off rendering the car undrivable. Another technical risk would be around whether the motor has a bad magnet or the motor pinion was bad, this would make the pinion more brittle and susceptible to failure or the magnet would generate the torque needed for the RC car. The last technical problem would be with the battery, such as if the battery wasn't able to hold a charge or didn't have the advertised voltage output for the motor.

k. Failure Mode Analysis

During operation, the chassis experiences load caused by the weight of component sitting atop the chassis, this was found to not be an issue in analysis⁰¹. The drivetrain is more likely to fail during operation. The drivetrain experiences dynamic loads from rotational forces, maximum shear stress generated due to torque from the motor

l. Operation Limits and Safety

For the RC car, there is a level of safety concern around the drivetrain as this involves objects rotating at high RPMs. If not covered adequately then the drivetrain has a chance for clothing or hair to get caught. This is mitigated by covers that are situated around major gear hubs and around the driveshaft. The covers around gears are to reduce the chance of pinching to occur.

3. METHODS & CONSTRUCTION

a. Methods

This project was designed at CWU to complete the Baja RC competition at the end of the year. While it is designed at CWU, the manufacturing utilizes a mix of machines the student has access to. These machines include a mix of CNC and laser cutting methods for metal components as well as 3D Printing for plastic components. These methods were chosen based on careful consideration using a decision matrix as shown in Appendix F02 and Appendix F03.

Throughout winter quarter, the methods that were utilized to construct this project did not change much. The biggest change that occurred was the location of machines that needed to be utilized. During fall quarter, it was initially thought that much of the manufacturing would be done at the students' workshop, however, some of the machines that were needed were not operational when it was needed for some components. To solve this problem, the machine shop at CWU was utilized for work on the driveshaft.

i. Process Decisions

The decision matrix used to determine the best possible construction methods are detailed in Appendix F. This decision matrix in Appendix F02, detail mixing laser cutting and CNC methods will produce the best results for all metal components; these components include the chassis and the drive train. As for the Plastic components, Appendix F03 shows that 3D printing is the best choice for this application as it has the best adherence to tolerance as well as availability of the materials. Material selection for the chassis as detailed in Appendix F04, was chosen based on analysis seen in 2g as well as throughout Appendix A. The best possible material was considered from different 3D printed materials as well as aluminum, and Aluminum was chosen for its ideal strength cost and weight.

The laser cut parts will be using an xtools metal fabricator, these parts include the chassis frame and the bearing mounts. A CNC lathe will be utilized to manufacture the driveshaft and the axle components. Finally, 3D printing using a Bambu H2D/X1 carbon will be utilized to fabricate the non-structural components such as the motor mounts, the battery holder, and the covers. One of the main issues that was found during the construction of this project was that this is the first time that some of these tools were being turned on. Due to this a lot of calibration needed to be done to be able to effectively use the tool to cut to tolerance. This calibration was the only thing that really impeded the construction timeline. Other complications that were noticed during manufacturing was that the frame itself was not perfectly symmetrical. This was noticed after manufacturing for the first time and was needed to be recut.

Another thing that was noticed was after completing the 3D Printing process for AA-20-010, It was noted that the way that the part was initially produced could be redesigned to better optimize how the cover is attached to the chassis frame to better increase stability and make it easier to attach and remove for maintenance. Aside from AA-20-001 not being symmetrical, another complication that needed to be addressed is that the plate needed to be longer to

better accommodate the motor being larger than initially estimated, an estimation was used as the product manufacturer did not put the dimensions on the website when purchasing the required part, due to this the frame needed to be extended and as such the analysis associated with AA-20-001 needed to be updated to make sure the frame will continue to perform to requirements in 1d.

After some redesigning during winter quarter, the axle components were no longer needed for the final design of the RC car. Due to this redesign, the driveshaft was the only component that needed to be turned on the lathe.

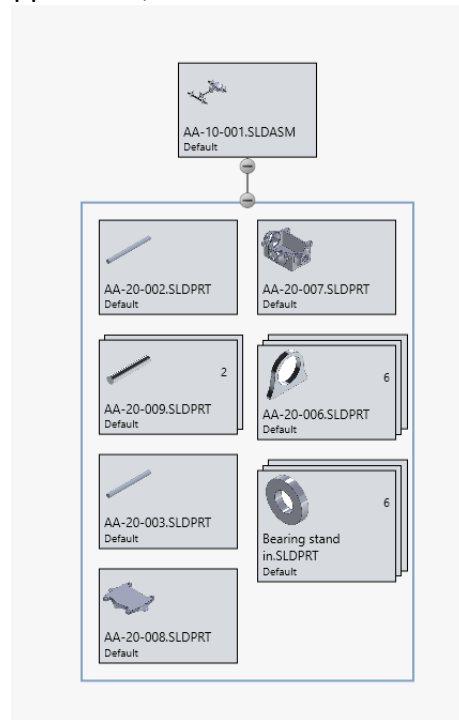
b. Construction

i. Description

The construction of the project will be done in multiple parts that correspond to designed parts found in appendix B. Many of these parts will be assembled into subassemblies, such as the front axle assembly, the rear axle differential assembly, and the motor-battery chassis assembly. Gears for each of these assemblies are a purchased part that has been detailed in appendix C. Many of these parts are made from 6061-T6511 Aluminum metal with fixture mounts being made from ABS/PLA plastic. When time is not an issue, parts will be made at the student's personal workshop. However, if time is an issue, then the parts will be made here at CWU. This being the case, however, 3D printed parts will always be made by the student personal 3D printer due to the student having a better understanding of the slicing program needed. As mentioned previously the construction took place in multiple parts, the first part is the purchasing of components like the motor and battery, the second part is the construction of the chassis subassembly to provide a stable platform for the rest of the car to mount to, the third part is the construction of the drivetrain subassembly this will connect to the chassis subassembly to provide locomotion, the final part is the final assembly with the project partners' subassemblies so that the final assembly is a complete and working prototype.

ii. Drawing Tree, Drawing ID's

The following is a subassembly used for the drivetrain of the RC Baja car. A larger and more full drawing tree can be found in appendix B; this details the entire RC Baja Car.



iii. Parts

Fall: to manufacture parts found in appendix B, there were 3 main manufacturing processes that were utilized to build the subassemblies. The first main process is a laser cutting that will use a MetalFab 1200W laser cutter. This is used for the chassis body (AA-20-001) and the bearing mounts (AA-20-006). The next process is the use of a CNC lathe for use on the drive shaft (AA-20-002) and axles (AA-20-003). The CNC lathe is used to keep the shaft and axles as close to circular as possible. The final process group makes use of 3D printing to build the mount components, the battery mount (AA-20-005) and the motor mount (AA-20-004). This process utilizes a Bambu Labs H2D printer to print in ABS/ASA plastic.

Winter: The manufacturing processes that were thought of during fall quarter are largely the same going into the winter quarter. The main difference in manufacturing processes was to the driveshaft component (AA-20-002). The initial thought was that the driveshaft would only be turned on the lathe, this was changed to add more processes to allow for gear attachments. This is due to a Partner Mill milling out flats for the gears to attach to. Another manufacturing process that needed to be changed was to print orientation for the parts. In 3D-printing the orientation of the part determines the strength of the part. For parts like the motor mount (AA-20-004), the orientation needed to be swapped to have a better fit for the mounting screws that were provided with the motor. Other parts that needed the print orientation to be changed was the Battery riser (AA-20-013), this needed to be flipped so that the screw holes would not sag.

iv. Manufacturing Issues

Fall: there are a couple of potential risks that can be foreseeable prior to manufacturing. One of the potential risks is learning how to utilize the CNC lathe as well as the laser cutter. Both machines are devices that the student has access to but had never utilized until this project. Another potential risk is the possibility of a print failure while 3D printing. While Bambu labs has a reputation for having a much lower failure rate than competitors, there is always a possibility of a failure, such as if the nozzle temperature is too low for the material or if the print bed needed to be heated

Winter: During winter quarter, there was a few manufacturing issues that occurred. The main thing that proved problematic was that the aerodynamic cover (AA-20-010) kept failing, this was found to be due to poor geometry. Another reason for this part failing was that the filament was not great, this caused the part to warp during cooling, this warping pulled the part from the build plate. To solve these problems, the printer used as well as the filament were swapped out. These attempts seemed to work, after they were done the print succeeded.

v. Discussion of Assembly

During winter quarter, the main subassemblies were created prior to attaching to the chassis frame (AA-20-001). The project partner completed the front suspension and steering block as the first subassembly. The chassis subassembly (AA-10-002) was completed after, this involved attaching the mounts and cover to the chassis frame (AA-20-001). Lastly the drivetrain (AA-10-001) would be attached alongside the rear suspension. All of the components are attached to the chassis frame using different sizes of screws. This could have been changed to make the assembling process more streamlined. During the process of making the assembly, there were many parts that needed to be added and some that were removed. The axle component (AA-20-003) was originally going to be manufactured, however this was rejected in favor of a purchased component. Parts that needed to be added to make the assembly work are the battery riser (AA-20-013) and the left and right bumper (AA-20-011 and AA-20-012). The left and right bumper were added to satisfy the need for it by the MET Professors. The final assembly should be able to reach a maximum of 30 mph without breaking, but the normal operational range is within the 15-25 mph range. The AA-20-011 and AA-20-012 were designed with the expectation that the deformation of the car is .6 inches due to head on impact as stated in Appendix A03.

4. TESTING

a. Introduction

The main testing approach will be the RC Baja Competition held at the end of the year for the CWU ASME chapter. This, however, has multiple race requirements that RC cars must be able to complete. These include the slalom-and-sprint and the Baja competition. Students will test the RC cars capabilities on mock tracks as well as on gravel and dirt roads, these tests will simulate the track conditions for the competition. Based on calculations and speed limits set by gear ratios, the RC car is predicted to perform well during the sprint tests. The car is also predicted to perform well during the Baja portion as values like the natural frequency are calculated for a gravel road.

Outside of the Baja competition testing, the RC car will undergo various tests that are related to the requirement found in section 1d. Based on calculations found in Appendix A, the car is predicted to perform to specification during the testing methods.

b. Method/Approach

To test the predictions made for the performance of the car, there will be various testing done. A maximum speed test will be used to find requirement 1d-4, this will require a stopwatch and 100 feet of flat land this will determine if the car can reach the maximum specified speed. During the speed test, there will also be an acceleration test that uses the same set-up to determine how long it takes to reach a specified speed, a purchased accelerometer will be used to determine the speeds. A crash test will be used to determine the reliability of the car for requirement 1d-2 and 7, this test will be the most destructive test that the car will undergo and will involve getting the car to 15 mph, determine using the accelerometer, then running into a concrete wall. Data will be collected based on pictures taken with a cellphone camera before the test and after the test, to determine how much deflection the RC car experiences. A scale will be used to determine the final weight of the car found in 1d-3. Lastly a student-made wooden ramp with a 45-degree incline will be used to determine how well the car can handle inclined surfaces found in 1d-10. The data collected will be how far up the ramp the car goes before slipping, the test would be determined successful if the car makes up the ramp. This ramp will be made the week prior to the test being completed to allow for adhesive to dry.

During spring quarter, the original testing methods were implemented. The tests chosen for the requirements found in section 1.d are an impact test, acceleration test, and a ramp friction test. These tests had a mix of initial success and failures. The Impact test had an initial failure. This failure was caused by a poor bumper design. This was fixed later to allow for better impact resistance; however, the bumper would have done significantly better with a material change to a full metal bumper. The acceleration test had an initial success and was able to get through all 5 trials. However, the initial assumption for the distance calculation was based on a full 60-foot runway, but it only needed about 15 feet to get to the required 10 mph. This assumption was due to the inaccuracy of the speedometer at a lower distance such as this test. Finally, the ramp friction test was a successful test, however the test did not go as expected. The front

suspension wasn't strong enough to keep the bumper very far off the ground. This caused the car to scrape along the lower section of the ramp, however after the tires took hold of the ramp, since the design is a rear-wheel drive, the car was able to reach to the top of the 45-degree incline. One of the ways that the scraping issue could have been fixed is the front being redesigned to allow for more of the suspension to be pushed down. The other way is to get strong shocks to keep the suspension in a downward position. Overall, the testing for this RC car was successful, the issues that were experienced seemed to be a result of poor design choices and not a testing process issue.

c. Test Process

The testing process for spring quarter included impact testing, Acceleration testing, and climbing over a ramp. For these tests to be successful, a large flat area with at most 100 feet in length, this was used for Acceleration testing as well as testing the steering for the slalom-and-sprint. A wall was utilized to test head-on impacts. Finally, a student-built wooden ramp was utilized to test the drivetrains' ability to climb an incline. These tests were done throughout the quarter and new parts were constructed as replacement parts since some parts were designed to take an impact and break. Items that were needed to perform these tests were an accelerometer, a stopwatch, a wooden ramp, and a wall.

d. Deliverables

Testing data was collected on a paper tablet to collect data during tests. This data was inputted into a document that was set up in the 1st week of the spring quarter. This testing data can be seen in Appendix G. This data includes pictures of before-and-after pictures of the RC Car during the impact testing. A screenshot can be found of the data document in Appendix G1.4, G2.4. and G3.4. The speed test was the most important test of the testing done for this RC car, this is because the motor speed needed to be fine-tuned to prevent the motor causing more damage than intended.

The first test that was conducted for the RC car was head-on impact testing. While the speed test is important, this test was conducted to gain a better understanding of the crash capabilities that the car had in the event the car crashed during the Slalom-and-sprint. This was also done with enough time prior to the Baja Competition, to be able to fix anything that broke.

For the impact testing, the requirement that was being tested was 1d.2 and 1d.7, however requirement 1d.7 is less of a focus as the drivetrain no longer goes the entire length of the car. The predicted result for this test was 0.6 inches from normal, the equation used to get this predicted value is the Euler's Buckling analysis equation for a rectangular cross section; this can be seen in appendix A03 for chassis deflection due to head-on impact. The actual value that was gathered was .632 inches from normal. This is above the requirement, attributed to the poor design of the bumper. Due to this poor design, the RC car could only survive 2 impacts before catastrophically failing. After the test was completed, the bumper design was changed to allow for better absorption of impact forces. During testing, it was more difficult than expected to get the car to impact straight on as the wind speed was a factor that was not accounted for during testing. After testing, the biggest issue was that some manufacturing defects were discovered. One of these defects caused the steering servo gear to become

striped and no longer work. This was solved because of the foresight to have a spare steering servo on hand. Another manufacturing defect was that it became clear which screws didn't have nylock nuts to secure them. This was solved by replacing some lost screws, that came off during testing and doing a pass replacing any structural nuts with nylock nuts, these nylock nuts allow for securing loose screws while also keeping them loose enough to allow for movement.

This test was run a second time after the bumper was redesigned. This new bumper fixed the issues that were caused by the original bumper and was able to get the deflection within the required deflection range.

The second test that was conducted was the acceleration testing. This test was conducted to gain a better understanding of how the car would be able to handle the Sprint section of the Slalom-and-sprint competition.

For the acceleration testing the requirement that was being tested was 1d.5 and 1d.8. The testable requirement was 1d.8 and was the focus of the testing, however 1d.5 states that the drivetrain must be able to provide a minimum 0.2 N/m of torque to the wheel during acceleration. There is not a good way of making this requirement a testable requirement, however the calculation for this can be seen in Appendix A09. As for requirement 1d.8, the drivetrain must be able to achieve a speed of 10 mph in under 10 seconds. This was a testable requirement and can be seen in Appendix G2. This test involved calculating the exact distance that the car would reach 10 mph and using a stopwatch to time the car. The calculation for this distance can be seen in Appendix G2.4. The result of this testing was that the car was able to achieve the required speed significantly under the required time, deeming this test to be a successful test. During testing there were not any issues that needed to be fixed during or after the test to improve the test. The only thing that could be found that could have been improved was to be more accurate with the predicted value.

The third test that was conducted was the ramp test. This test was done to ensure that the car would be able to handle jumps during the Baja competition that would decrease the overall time of the run.

For the ramp test, the requirement that was being tested was requirement 1d.10, the drivetrain must be capable of providing torque to climb a 45-degree ramp. The car was calculated to be able to climb the ramp with little issue. The calculation for this can be seen in Appendix A05. The testing can be seen in Appendix G3. This test was conducted twice, once with the bumper and once without the bumper. The reason for this is that while the car was able to climb the ramp with the bumper, it was more like the car was pushing itself up the ramp. Therefore, the second test was done to make sure the car was able to climb it without needing to be pushed. This is an issue that couldn't be solved during the testing period and would have needed to have been looked at for a redesign to make either the bumper smaller or the front section smaller to allow for the suspension to be able to have a wider range of articulation.

The final test that was conducted was the top speed test. This test was done to ensure that the drivetrain was not running faster than designed.

For the top speed test, the requirement that was being tested was requirement 1d.4, the drivetrain must be able to achieve a maximum speed of 30 mph. The car was calculated with the gear reduction to be able to achieve a maximum of 30 mph, this calculation can be seen in Appendix A04 and A05. The testing data can be seen in Appendix G4. This test was conducted between 50% motor power to 100% motor power. This was deemed not a successful test as the drivetrain was only able to achieve an RPM value of 2235 RPM, this converts to 26.6 mph. This is likely to be due to friction losses in the system being higher than expected. There are a few ways to be able to fix this issue, one way to fix this issue is to add more grease to the gears and axles, the other way to fix this issues are to redesign the arms to allow for more tolerance for the axle to pass through. This second fix is not something that is feasible in the time frame.

The final requirements were not tested, due to these requirements being untestable requirements. Requirement 1d.3 states that the vehicle weight must be less than 10 pounds, the final weight was 5.3 pounds which is under the requirement, meaning the requirement was met. Requirement 1d.6, the chassis must deflect less than .1 inches from normal due to weight of sub-assemblies, this requirement was met by the car not bending a significant degree from normal with the weight of everything being attached to the chassis plate. This was successful because it remains flat, the calculation for this can be seen in Appendix A01. Requirement 1d.9 states that the chassis must be capable of withstanding 20-40 Hz vibrations at a speed of 15-20mph. The calculation for this requirement can be seen in Appendix A08. This requirement is to ensure that the vehicle was able to drive over gravel without falling apart during the Baja section of the competition. Since the car was able to complete the Baja without falling apart this requirement was met. Requirement 1d.11, the RC car must have a drag force lower than 120 lbf when at speeds between 10 mph and 30 mph, was not able to be tested due to the wind tunnel and CWU being inoperable during the testing period. Therefore, this requirement could not be met or tested. The final requirement 1d.1 was able to be met as the car was driven along side of the operator as they were walking between classes and was able to keep up with the operator moving a 16 mph. the calculation for this requirement can be seen in Appendix A06. There was no official test done for this however, as stated before the car was able to keep up with the operator moving at 16 mph, meaning this requirement was met.

5. BUDGET

a. Parts

Parts are to be procured by one of 3 methods, either made in house, purchased from online retailer/warehouse, and made using student's personal workshop. Parts with longer lead times such as the motor and battery are to be ordered during early winter quarter and stored with the rest of the manufactured parts. Parts 20-002 and 20-003 are to be made in house using the CWU machine shop and Part 20-001 and 20-004 are to be made using a laser cutting process at student's personal workshop.

b. Outsourcing

Outsourcing will not be a necessary cost as the student's workshop has the capabilities to perform all processes outside of the need to purchase raw materials and advanced electronic components.

c. Labor

Student's personal hourly rate is \$12/hour as this is the rate for personal or school projects. This estimate is based on the students' estimation of capabilities.

d. Estimated Total Project Cost

This project is estimated to cost about \$800; this estimation includes project partner estimates. Much of the cost is detailed in Appendix C as the motor and battery are a significant portion of the cost. Labor is the second most expensive portion making up about \$300 of the estimated cost.

e. Funding Source

This project is funded by the Kaminski Scholarship Foundation, as well as funding from students' personal funds.

f. Winter Updates

fa: During winter Quarter, there was a major redesign of both the drive train and the chassis. Due to this redesign, there was no longer a need to manufacture the rear axles as these now needed to be purchased. This was updated in Appendix C and D. Additionally; there was a larger cost that was due to mistakes and recuts for the chassis frame. Due to these recuts, the cost of making the chassis frame was higher than expected.

fb: There were no issues with outsourcing as outsourcing was not used to make any part of this project. All Parts are either made by 3D-Printing on in-house printers or cut by machines owned by the student.

fc: Labor costs are accurate with the Gantt Chart in Appendix E.

fd: As this update was done during winter quarter there are no costs that needed to be addressed due to testing. The electronics test done to ensure component compatibility resulted in a successful test. This was the only test that was done during winter quarter that could have resulted in a need to change the costs or budget for new electronics.

fe: There is no change in funding from scholarships, however there was a change that the student needed to purchase parts out of pocket

g. Spring Updates

ga: During spring quarter, there was a major redesign to the drivetrain going from winter quarter to spring quarter. This major redesign, however, was not at any additional cost to the principle engineer as the material and gearing that were used for this redesign were material that was on hand at the time of the redesign. For testing, there would have been some additional costs associated with the ramp testing as described in appendix G3, however the material used for this ramp was scrape material that would have been thrown out regardless. There were some issues that came with the impact test, as described in appendix G1, these issues were solved with a small redesign to the bumper, this bumper redesign did not add any additional cost to the part itself.

gb: During spring quarter, there were no issues with outsourcing costs or any additional costs there were incurred by outsourcing. The only outsourcing cost would have been to use the student's personal shop for the project, and the price of gas to drive to and from this shop. If this cost was included in the budget, the project would have gone over budget by a significant margin.

gc: labor costs for spring quarter and throughout the senior year, are much higher than initially quoted during fall quarter, this is due to the number of iterations that were needed to reach a state that was satisfactory to the principle engineer, as well as faculty.

gd: As stated previously in section ga, during testing the bumper needed a redesign after the first test of the impact test. This redesign, however, did not increase the cost of the part by any significant margin. Otherwise, there were not any costs that were incurred by mistakes during testing.

ge: There were no changes to the funding sources throughout the spring quarter or during the senior year. Funding came from the Kaminski Scholarship Foundation as well as from the principle engineers' own pockets.

6. SCHEDULE

a. Design

Fall:

This section sees major headway made on the proposal of the project. Analysis is performed to gain a better understanding of what needs to be manufactured and how the Baja Car performs throughout testing. This performance analysis extends to the performance of the Car during the ASME CWU Baja Competition. There were many scheduling conflicts as the component selection for the car took more time than originally expected, this caused some errors to appear during analysis as the weight of certain calculations was slightly off. This was corrected as the component selection was finalized. Time estimates can be found in the provided Gantt chart, found in Appendix E; these time estimates were based on current ability and comprehension. The estimates themselves are close to the indicated values with some analysis taking slightly more or less time, usually by half a day.

Winter:

During this time, the design was fully finalized and ready for construction to start. Few major design changes occur to reflect current tooling availability. Throughout this project, from fall quarter to spring quarter, the estimated time investment comes out to about 100 hours for the design, conception, and analysis of the RC car during all quarters. For the winter quarter, the estimated time is about 80 hours. This is for the construction, prototyping, and revision of any parts. Lastly for spring quarter, this is estimated to be about 20 hours. This estimation is significantly lower than the other quarters as much of this quarter revolves around testing the car, the competition, and the presentation of the final car to the CWU MET faculty as well as to the SOURCE judges and onlookers. It was found during winter quarter that the design was not completed to be manufactured. A week was dedicated to fixing the design so that it can be best used for offroad applications. This design change was a major flaw that required a large setback in the manufacturing timeline. The design was supposed to be completed and ready for construction by the end of fall quarter, but due to these redesigns, the final design was completed 2/6/2026.

b. Construction

Fall:

Construction is fully planned out as to not have delays. These plans include the time for CNC components and parts that are milled on an engine lathe. These plans also include shipment dates and arrival times for purchased parts, such as the motor and the battery.

Winter:

During winter quarter, the construction of the RC car will begin. This includes the use of engine lathes for the driveshaft to get within tolerance for roundness, the use of CNC machinery to get the dimensions for the chassis within tolerance. Construction is going smoothly when the student could get into the workshop to manufacture parts. There were parts that needed to be manufactured more than once due to poor planning on manufacturing method. For the laser cutting, it took longer than expected to set up the laser cutter as the ambient temperature was lower than normal, thus creating a sub-optimal air compressor environment. This environment made some of the parts come out with more slag than expected. This slag made the post-processing time longer. For the 3D-printed parts, there were a couple of print failures that occurred after several hours of print time. This was a major problem as these print failures directly led to loss of money as well as time. This was solved over time as the part that failed was edited to make for better design. The use of the lathe made for easy construction of the driveshaft. Originally Task 4c in the Gantt chart in Appendix E was supposed to only take about 1 hour, this task ended up taking 4 hours due to needing a redesign as well as the laser not cutting the first time resulting in multiple cutting passes. As for Task 4b, this part is the motor housing which was not measured correctly. This part needs 1 hour to fully print, but due to this mismeasurement, the print will need to be reprinted which will add to manufacturing time.

c. Testing

Fall:

The testing for the RC car was done to test a few of the requirements found in section 1d. The schedule for these tests can be found in Appendix E section 6 and 7 of the spring quarter subsection. These tests include a crash test for the chassis, an acceleration test for the drive train, and an incline ramp test for the drivetrain as well. These tests test for requirement 1d.2, 1d.8, and 1d.10.

In terms of scheduling there were not that many issues that came up during spring quarter. The main issue that came up was working around work schedules. This was not something that could have been fixed very easily. The fix that was done for these scheduling issue was that the tests done for the drivetrain and chassis were done in the evening. This did take a lot of the risks out of each test as described in Appendix G, as many of these risks were about cars and pedestrians walking or driving into the testing area.

During the test period there were 2 changes that needed to be done for testing. These were during the impact testing, and the ramp testing. For the impact test, a second test was done after the bumper was redesigned to have no sharp corners in the design. This design allows for energy to get moved into the chassis. The better and more optimal change would be to also change the print orientation so that the layer line was not in the same plane as the impact plane. The second change to the testing was during the ramp testing. 2 tests were done for this test, one with the bumper and one without. This is due to the problem that the front suspension does not allow for a lot of ground clearance for the car. While the first test was successful, the second test shows that the bumper sticks out too much for the car to reliably be

able to climb the built ramp. a fix for this would be to swap the angle required for a smaller angle. However, this angle was chosen to ensure that the car could handle jumps at any reasonable angle.

7. PROJECT MANAGEMENT

Throughout this project there were many risks that were undertaken in order to ensure the successful completion of this project. While much of the risk is undertaken during the project, from financial, human, physical, and software resources. There is a level of risk associated with completing the project. This risk comes in the form of losing money to purchased components as the student documenting this proposal chose not to keep the RC upon completion, it will go to the students' project partner.

a. Human Resources

For the successful completion of this project, there will be many people aside from the Student and the Student's project partner. This includes Professor Charles Pringle, Doctor John Choi, and other CWU MET students and Facility. For project assistance, the student's father, Lou Amadio, will be asked to assist as this individual is the primary owner of many of the machines there are planned to be utilized. The risks with receiving outside assistance are that individuals may not understand the scope of the project and assistance may not be helpful to the completion of the project. Another risk that may come up is scheduling conflicts may occur. The risks will be managed by asking a few days, about 3-4 days, in advance if the students' father is available to keep the workshop open. This advanced notice will be needed for the student as well as the student must commute to the workshop to make use of the resources.

b. Physical Resources

As mentioned prior, there are many machines that are being planned on being used. These include access to a 3D printer (Bambu Labs H2D), A CO2 laser cutter, an all-in-one CNC machine, an all-in-one Lathe, Drill Presses, Engine Lathes, and a metal bender. As many of these machines that are accessible to the student, there is a risk associated with many of these machines, and that is that they are owned by the students' father, therefore meaning that the student may not have access to these machines without prior permission and scheduling. Other physical resources are raw materials such as aluminum sheets and aluminum rods; the risks associated with these items are whether the vendor has it in stock or lead times. Should the risks with physical resources become a hinderance to the success of the project, then the fall back will be the resources and facilities provided by the CWU MET department.

c. Soft Resources

Unlike with physical resources, there are not as many software resources being used. The primary use of software is GitHubs' Jekyll (JECK-uhl) program and Virtual Studio Code (VSCode). These programs are employed to code and develop the required webpage deliverable. The biggest risk with this method of webpage development is that troubleshooting must be done by the student. This troubleshooting may incur missed deadlines leading to a delay in schedule. Another piece of software that is being employed is the RC controller board/program as this is used to be able to control the RC Baja Car and is vital to the success of the project. The risks with this piece of hard/software are that because it is vital it is also prone to breaking during use or shorting due to contamination. For designing components and drawing tree found in Appendix B, SolidWorks 2025 was utilized. The risks involved with this program are crashes

delaying the design of components and assemblies. Useless saved extensively, there isn't a fix for this type of risk as a crash could occur at any point during the design process.

d. Financial Resources

The project sponsor is committed to providing monetary support for components that cannot be manufactured by the student. This support extends to raw materials, electronics, and tires. The students will also be using their own funds to finance the project should the project go over budget at any point throughout senior year.

8. DISCUSSION

a. Design

Throughout fall quarter, analysis was carried out to get a better understanding of the design parameters that needed to be met or the car to perform to specifications outlined in section 1.

Throughout the fall quarter, the design underwent 2 major redesigns. The first redesign detailed an update to the drive shaft and axles. Based on initial analysis, the driveshaft and axles needed to be .25 inches in diameter to withstand the rotational forces that are acting on the parts. However, later analysis of the lifetime cycles decided that a drive shaft of .25 inches diameter would not be enough to prevent failure. Thus, an increase in diameter was needed. The increase went from .25 inches to .75 inches for the part to withstand a minimum operational time of 2 minutes.

The second major redesign during fall quarter was going back through every part and updating each model to reflect a current process method. This includes updating the battery mount as this part was originally being laser cut and bent from aluminum but after careful consideration, it was decided to 3D Print it from PLA instead. The other redesign is to the bear mount, the cover, and the frame as all the parts originally had no fixture holes to be able to connect everything to the frame.

The last major redesign was to the placement of the drivetrain and the cover; the cover needs to be able to cover all parts and subassemblies. This action is to reduce the drag experienced by the car, hopefully reducing the work needed by the motor to reach specified speeds. The other major redesign was to the placement of the drivetrain system. Initially, this system was to be placed on the underside of the frame to give it enough ground clearance, however this was changed to the top side of the frame because the differential housing was too big for the chosen wheels.

Some risks that needed to be overcome during the design process were that some of the programs being used were crashing, not allowing the proctors to view the webpage after turning it in. The fix that was found for this was to sit down for multiple hours figuring out where the codebase broke, turning out the codebase broke from a single dependency not being updated.

b. Construction

Throughout winter quarter, it was noted that manufacturing the car would be the easiest step of the process. This, however, was the wrong assumption.

During winter quarter the student didn't like how the chassis was originally designed causing a need to completely redesign the chassis frame/body. This setback then raised some concern about the dimensions of the motor; the inclusion of the motor also caused the project to

become out of budget. Due to both the chassis redesign and the budget, a new motor was sought out.

The reason for the chassis redesign is because the original design did not allow for enough clearance for the wheels and suspension to move during operation, which is a basic function for an off-road RC Car, this was overlooked during the initial design phase.

The redesign of the chassis frames also caused a need to redesign the drivetrain subassembly. Due to this the rear suspension also needed to be redesigned to accommodate, this is done by the project partner. This redesign was started because of overlooking the need for the car to bounce, the original design did not account for this.

During the construction phase of the project, the original parts were produced using a laser cutter, the convenience of having access to a laser allowed for more complex shapes to be produced in a relatively short amount of time. This allowed for a short manufacturing timeframe. In addition to laser cutting, the inclusion of 3D-printed parts in materials like PLA or PETG allow for complex shapes that have similar or better stability and strength to cast plastic or vacuum formed plastic.

Something that wasn't helpful with the construction of the project is that the project is being manufactured during winter quarter. During these months, it is significantly more difficult to keep the laser cutter warm enough that the compressed air, that is used for the cutting environment, won't freeze. This freezing leads to an increase in the amount of slag that gets produced on the part that's being cut. This slag also is a result of trying to cut metal that is too large for the machine to handle. Due to the slag build up it increases the production time when removing this slag.

While constructing many different components for the RC car during winter quarter, it was elected to use a different manufacturing method for the driveshaft (AA-20-002). While the driveshaft itself has changed in terms of the design. The base stock was changed from ¼ inch round stock to an M6 screw. This swap allows for the gears to be securely attached to the driveshaft as well as prevent slipping during operation.

While manufacturing, AA-20-015 and AA-20-16 were rendered obsolete as the ring and spur gears no longer need to have adapters due to the change to the drive shaft (AA-20-002).

The chassis frame (AA-20-001) needed to be recut multiple times throughout winter quarter. This was because of impatience for cutting 6061-T6 aluminum. This led to a higher material cost that didn't need to be wasted. Something that was considered after the fact was 3D-printing a temporary chassis frame to check fitting, then cut in metal.

c. Testing

While testing the RC Tron-car, there were a few issues that arose. These issues came about because of poor design.

During the Impact testing as seen in Appendix G1, the car survived 1 impact before breaking on the second impact. There were no changes to the procedure that was made during the testing period. The value that was gained from the 1st crash was .632 inches which is outside of requirement 1.d, The chassis must deflect less than .6 inches from a head on impact. After performing image analysis on the bumper, it was concluded that the problem was with the design of the bumper. The bumper was redesigned to have a larger fillet along the connection point of the perpendicular impact zone and connection plate. This larger fillet prevents stress concentration from building up. Another issue that was noticed is that the layer lines of the 3d printed bumper were along the impact plane, this is not a good thing because it allows the print to split during impacts. The way to fix this is to change the print orientation to make either the layer lines to be perpendicular to the impact plane, or at an angle to it. This was not done at the time of testing due to time constraints for the bumper; however, these changes were later implemented. These modifications were needed to allow the bumper to absorb more energy and prevent that energy from being transferred into the chassis plate.

The impact test was rerun with the new modified bumper, as seen in Appendix G1. This new bumper design was able to withstand more force allowing for a second measurable impact. With this modified bumper design both measurable impacts were within the required deflection seen in section 1d.2, The chassis must deflect less than 0.6 inches from head-on impact at 15 mph. The deflection from these measurable impacts were 0.582 inches and 0.59 inches. This can be seen in appendix G1.3.

During the acceleration test as seen in Appendix G2, the car was able to successfully pass 5 trials. This test used a calculated distance that was based on an exponential acceleration and velocity. There are a couple of issues that arose from this test that led to a possibly inaccuracy of the data. The predicted value for this test was 10 seconds from the requirement for the car that it must reach 10 mph within 10 seconds, as seen in section 1d.8. However, the recorded value for this test was an average of 2.48 seconds to hit 10 mph. This indicates a couple of things. This means that either the gear ratio of drivetrain was calculated incorrectly or manufactured with the incorrect gear ratio, when the drive train was swapped from a driveshaft design to a direct gear drive. This inaccuracy would cause the car to be much faster than intended. A possible reason why the wrong gears were used is out of necessity, the drivetrain swap was within a tight window of operation therefore gears that were on hand were used. The main fix would have been to rebuild the drivetrain with the correct gear ratio and rerun the test; however, this would have taken too much time.

During the ramp test as seen in appendix G3, the car was required to make it over a 45-degree angle. A total height and length of 20 inches was chosen this put the ramp's length long enough that the car would be entirely on the ramp as it drove up it. Unlike the other tests chosen, this

was done on the sidewalk to avoid the need to watch for passing pedestrians and cars. The requirement that this test was being done for can be seen in section 1d.10, the drivetrain must provide enough torque to climb a 45-degree ramp. This test consisted of 5 trials, the car was able to climb the ramp each time making this test a successful test, however there were a few issues that arose during testing. The first and main issue was that each trial barely passed each time, while this does not impede the car ability to climb the 45-degree, the fix would be to change the requirement itself to be a lower angle. This lower angle would ensure that the car could handle ramps and jumps without issue. Another is that there was so much vibration that the car lost a few screws during the testing period, this was fixed by replacing the screws with ones that were on hand at the time. A better fix for this issue would have been to tighten the nuts with a proper locking paste rather than relying on nylon lock nuts. This test was run twice for a total of 10 times run, the reason for this was that the bumper was a major impedance for the test, while the test was successful with the bumper attached, a second test run was run with the bumper removed. This second test was more successful than the first test, this is however, this is not something that can be fixed within the scope of this report as the issue this second test was solving is a suspension issue as the front of the car did not have enough ground clearance with the bumper attached to get the wheels securely on the ramp. This second test can be seen in appendix G3.3

9. CONCLUSION

a. Design

This project serves as the Capstone project for the CWU MET program. The project chosen was an RC car that is to compete in the ASME CWU chapter Baja Competition. This proposal details the design, construction, and testing of the Chassis and drivetrain components. These components serve a specific purpose for the success of the RC Baja car. The chassis is the support for sub-assembly attachments, as for the Drivetrain, the drivetrain provides torque to the system for motion. The analysis chosen for these components was chosen based on the design requirements in section 1-d, these analyses include deflection across the chassis, the required torque and gear reduction to reach a specific speed, and the torsional shear that is acting on the driveshaft. The analysis contributes to the success of the project, by providing the minimum requirements for the purchased parts and constructed components. These analyses were performed using formulas learned through the CWU MET program such as statics formulas for determining the deflection on the chassis, dynamics to understand the forces acting on the car during impacts, and physics to understand the minimum acceleration for launching. Resources needed for this project to be successful are easily accessible by way of the student personal workshop as well as through online vendors who provide access to commonly used metals and 3D-printing materials. To determine whether is project was a successful senior project or not the car will undergo a series of testing that was defined by the student, such as a crash test, acceleration test, max speed test, and an endurance test. The car will also compete in the ASME CWU Baja competition that is held during the spring quarter, and it will determine to be successful if the car finishes the race in 2nd place during the Baja competition and 7 seconds or less in the Slalom-and-Sprint race.

b. Construction

The construction of the RC car saw many changes between fall and winter quarters. The original design used a 4-wheel-drive system, this got changed to a 2-wheel-drive system. Towards the end of winter quarter, the design of the drivetrain needed to be modified due to purchased gears not interacting with each other. The construction method for components did not change between fall and winter quarter. The Chassis frame (AA-20-001) needed to be recut multiple times to reflect the number of times the chassis frame was altered. This could have been done differently to keep the material costs down and allow for more material to be on hand in the event of a mistake or failure. After the winter quarter was concluded, the drivetrain needed a full redesign and re construction. This included swapping the system from a driveshaft design to a direct gear design, this change allowed for the car to function as intended as well as be able to be tested during spring quarter.

c. Testing

After redesigning the drivetrain during the break between winter and spring quarters. The RC Baja Tron-Car was a mostly successful design that was able to be tested to ensure that the car would be able to reach the benchmarked product. Upon testing, however, the Tron-Car was not

able to reach close to the benchmark. The car, however, was mostly success compared to the stated requirements. The car was tested in four requirements that were of vital importance to the success of the vehicle, 1d.2: Chassis must deflect less than .6 inches from a head-on impact at 15 mph, 1d.4: Drivetrain must be able to achieve a maximum speed of 30 mph, 1d.8: drivetrain must be able to achieve a speed of 10 mph in under 10 seconds, and 1d.10 drivetrain must be capable of providing torque to climb a 45° ramp. Tests for each of these requirements can be found throughout Appendix G. After some redesign of the front bumper during test G1 for requirement 1d.2, the car was successful. The tests for each requirement were successful; however, the car was not able to achieve a maximum speed of 30 mph or be within 10% of this metric. This was not able to be redesigned and retested as time went short and the conclusion for how to go about redesigning and fixing the problem was not achievable at the time of testing. The predicted performance of this project was to be able to achieve each requirement, however some requirements were not able to be tested due to not having the facilities available, or being a non-procedural metric, such as weight. For the success criteria the car was predicted to be capable of achieving 2nd place during the Baja competition and achieve a time of under 7 seconds during the Slalom-and-Sprint. After competing in the ASME Baja competition, the car was able to achieve a podium 1st place during the Slalom race and the Sprint race, with a final completion time of 6.4 seconds during the Sprint, however the Slalom was not as successful with a completion time of 55 seconds. During the Baja competition the final podium finished 2nd place with a time of 1 minute and 4 seconds. This puts the final finish at 1st place for the completion, making the vehicle more successful than predicted.

10. ACKNOWLEDGEMENTS

- The Kaminski Scholarship Foundation provided initial funds to the principal engineer to be able to purchase components and material for this project.
- Ranch Hand Robotics provided equipment and advice to the principal engineer during the winter quarter for construction and as well as pointing out flaws in the design during the redesign process.
- Charles Pringle mentored the principal engineer throughout the project on the report as well as testing methods.
- John Choi mentored the principal engineer on questions and concerns with the ASME Baja rulings.
- Federico Berrospe worked with the principal engineer on the project throughout the year.
- Matthew Thien proofread the testing report for the principal engineer, checking for spelling mistakes.
- Paige Henderson used their experience to look over the drivetrain redesign during winter quarter to check to see if the principal engineer was missing anything during the design process.

References

- <https://traxxas.com/3384-bl-2s-brushless-3300-kv-motor>

Traxxas parts were used to make up a majority of the drivetrain hardware such as the motor, electronic speed controller, and the battery. Axles were also purchased from Traxxas for ease of compatibility. Traxxas was used during this project as a local RC store carries the components making lead times very short and the principal engineer did not need to wait for online shipments.

- <https://www.youtube.com/watch?v=F40ZBDAG8-o>

This video was used to gain a better understanding of what a differential is in a car and why it is important to include into a 4-wheel vehicle. This video shows that without the differential, the vehicle would not have been able to operate during the Slalom race portion of the Baja competition.

- <https://www.youtube.com/watch?v=OpNEocKyeRs>

This video shows how to construct a differential that is compatible with metric gears. The contents of the show how to build a differential using purchased components.

- Machine Elements in Mechanical Design by: Rober L. Mott, Edward M. Vavrek, and Jyhwen Wang

This is the mechanical design textbook used for sourcing equations used for the analysis of this project.

APPENDIX A - Analysis

Appendix A01 - Structural Deflection & Weight on Wheels

Alex Amadio

ME1 484

10/6/2025

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Analysis 01: Structural
given: $D_{alum} = 2.7 \text{ g/cm}^3$ (based on initial assumptions)
battery weight: 196g
subject to change

weight of 2 motor: 0.4816 - 70.218 Kg

find: force on wheels (end points)

deflection due to

assume: equilibrium

perfect square plate

equal Distribution of components

method: static/moment equations

Navier's Double-Sine Expansion

Rayleigh-Ritz single term worked estimate

Solution:

$$w(x,y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} W_{mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)$$

$$W_{mn} = \frac{Q_{mn}}{D \left(\left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \right)^2}$$

$$W_{max} = \frac{Q_{eq}}{D \pi^4 \left(\frac{1}{a^2} + \frac{1}{b^2} \right)^2}$$

bending strain energy

$$U = \frac{D}{2} \int_0^a \int_0^b (\nabla^2 w)^2 dx dy$$

$$U = D \frac{\pi^4}{8} \left(\frac{1}{a^2} + \frac{1}{b^2} \right) ab w^2$$

work done by load

$$V = \int_0^a \int_0^b q w dx dy = q w \frac{ab}{4}$$

$$W_{max} = \frac{Q_{eq}}{D \pi^4 \left(\frac{1}{a^2} + \frac{1}{b^2} \right)^2}$$

$$Q_{eq} = \frac{P_c + 2P}{ab}$$

P_c = weight from battery

P = weight from motors

$a = 12$

$$D = \frac{E t^3}{12(1-\nu^2)}$$

E = young's modulus

t = thickness

ν = poisson's ratio

Appendix A01 – Structural Deflection & Weight on Wheels Cont. (2)

Alex Amadio

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Solution cont.:

$$\omega_{max} = \frac{\epsilon_{ee}}{D\pi^4 \left(\frac{1}{a^2} + \frac{1}{b^2} \right)^2}$$

$$\epsilon_{ee} = \frac{P_c + 2p}{ab}$$

$$D = \frac{Et^3}{12(1-\nu)^2}$$

$$\nu = 0.33 \text{ (Aluminum 6061-T6)}$$

$$P_c = 196 \text{ kg} \cdot 9.81 = 1,922.76 \text{ N}$$

$$E = 69 \text{ GPa} \rightarrow 69.0 \times 10^9$$

$$p = .218 \cdot 9.81 = 2.13858 \text{ N}$$

$$a = 12'' \times 0.0254 \rightarrow 0.3048 \text{ m}$$

$$b = 9'' \times 0.0254 \rightarrow 0.2286$$

$$t = .0125'' \times 0.0254 \rightarrow 0.0003175 \text{ m}$$

$$D = \frac{(69.0 \times 10^9 \text{ Pa})(.0003175 \text{ m})^3}{12(1-0.33)^2} = 1291.24$$

$$\epsilon_{ee} = \frac{P_c \times (2)(p)}{.3048 \cdot .2286} = 118.03$$

$$\omega_{max} = \frac{118.03}{(1291.24)\pi^4 \left(\frac{1}{.3048^2} + \frac{1}{.2286^2} \right)^2} = 0.00000105 \text{ m} \rightarrow 0.0000413 \text{ in}$$

the chassis will deflect 0.0000413 in, due to weight of proposed components at 1/8 thickness

Appendix A01 – Structural Deflection & Weight on Wheels Cont. (3)

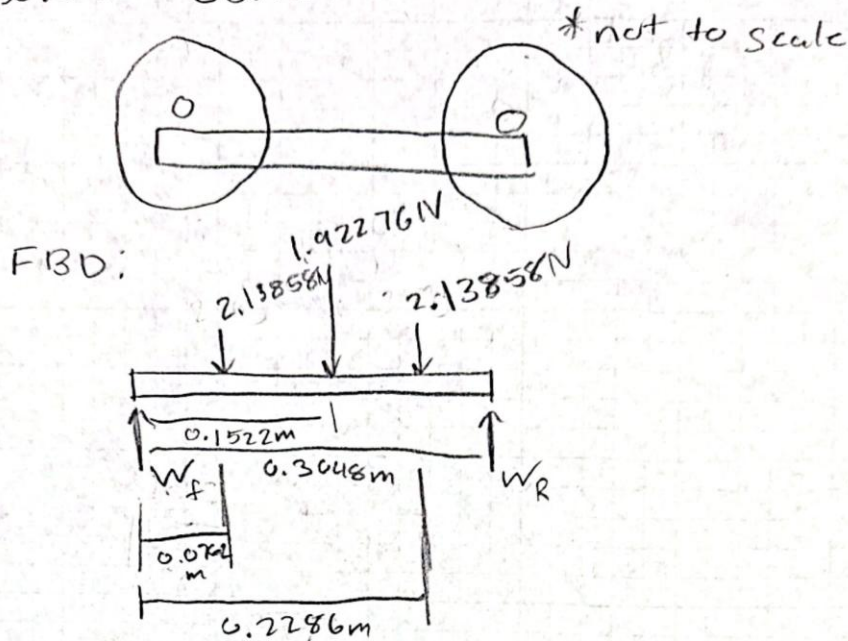
Alex Amadio

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solution cont.:



$$\sum F_x = 0 = W/A$$

$$\sum F_y = 0 = W_f + W_R - 2.13858 - 1.92276 - 2.13858$$

$$\sum M_{W_f} = 0 = (2.13858 \text{ N})(0.0762 \text{ m}) + (1.92276 \text{ N})(0.1522 \text{ m}) + (2.13858 \text{ N})(0.2286 \text{ m}) - (W_R)(0.3048 \text{ m})$$

$$W_R = \frac{0.9445 \text{ Nm}}{0.3048 \text{ m}} = 3.0987 \text{ N}$$

$$W_R = 3.0987 \text{ N} \times 0.22481 \text{ lbs} \quad \boxed{W_R = 0.6966 \text{ lbs/2}}$$

$$0 = W_f + 3.0987 - (2)(2.13858) - 1.92276$$

$$W_f = 3.10122 \text{ N} \rightarrow 0.22481 \text{ lbs} \quad \boxed{W_f = 0.6971 \text{ lbs/2}}$$

the wheels currently experience 0.3486 lbs on both of the front wheels and 0.3481 lbs on both back wheels

Appendix A02 – Torsional Shear, angle of rotation due to torque, Driveshaft Diameter and Critical Speed Frequency

Alex Amadio

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Analysis 02:

given: $\rho = 0.0975 \text{ lb/in}^3$ $G = 11.5 \times 10^6 \text{ psi}$
 $L = 10 \text{ in}$
 $E = 10 \times 10^6 \text{ psi}$
 $T = .1475 \text{ lb/ft (from motor specs)} \rightarrow 1.77 \text{ lb/in}$

find: - shear stress on drivetrain before gear reduction

- Drivetrain diameter to withstand 30MPH

assume: 3 in ϕ wheels

gear reduction down to a torque of

1.77 lb/in

method: shear stress analysis

critical speed analysis

Solution:

$$\tau = \frac{T}{J}$$

$$\phi = \frac{TL}{JG}$$

$$\therefore N_{cr} = (4.76 \times 10^6) \frac{r}{L^2} \sqrt{\frac{E}{\rho}}$$

$$(r = .125 \text{ in})$$

$$N_{cr} = (4.76 \times 10^6) \frac{.125}{10^2} \sqrt{\frac{10 \times 10^6}{0.0975}}$$

$$N_{cr} = 6025.7492 \text{ RPM}$$

$$f_{cr} = \frac{N_{op}}{60} = 753225 \text{ Hz}$$

$$N_{op} = .75(N_{cr}) = 4519349.4 \text{ RPM}$$

$$\frac{N_{op} \cdot 3}{336} = 403513 \text{ MPH}$$

this means the drive shaft with a diameter of .25 inch will not fail at 30mph with a critical frequency of 753225 Hz

Appendix A02 – Torsional Shear, angle of rotation due to torque, Driveshaft Diameter and Critical Speed Frequency Cont. (2)

Alex Amadio	MET489	10/12/2025	7/2
solution cont. $\tau = \frac{Tr}{J} = \frac{1.77 \cdot (1.125)}{J} = 576 \text{ lbin}$ $\phi = \frac{TL}{JG} = \frac{1.77(10 \text{ in})}{JG} = 0.0004024^\circ$ $J = \frac{\pi r^4}{2} = \frac{\pi (.125)^4}{2} = 0.000382495 \text{ in}^4$			

Appendix A03 – Chassis deflection due to 15 mph impact

	Alex Amadio MET429 10/14/2025 find: deflection due to head-on impact given: $V = 15 \text{ mph}$ $m \approx 3 \text{ pounds}$ (based on estimates) $b = 5.2 \text{ in}$ (based on drawing in appendix B06) $t = .125 \text{ in}$ $L = 12 \text{ in}$ $E = 10 \times 10^6 \text{ psi}$ $\phi = .25 \text{ in}$ $\nu = .33$ (for Aluminum) assume: perfectly elastic collision method: Kinetic energy Deflection solution: $\delta_{\max} = \sqrt{\frac{mv^2}{K_{\text{bend}}}} = \sqrt{\frac{(3)(15^2)}{1880.79}} = 0.599 \text{ in}$ $K_{\text{bend}} = \frac{48EI}{L^3} = \frac{48(10 \times 10^6)(0.006770833)}{12^3} = 1880.79$ $I = \frac{bt^3}{12} = \frac{5.2(.125)^3}{12} = .006770833 \text{ in}^4$ $F_{\max} = K\delta = (1880.79)(.599) = 1126.74 \text{ lb}$ at a speed of 15 mph the car will deflect .599 inches with a peak impact force of 1126.74 pounds	2/1
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Appendix A04 – Drivetrain RPM and Motor-Drivetrain Gear Ratio

Alex Amadio	MEET489	10/14/2025	7/1
given: voltage: 7.6V (from Battery specs) KV = 2400 (from motor specs) wheel $\varnothing = 4$ in find: RPM and required gear ratio assume: no slipping between gears 80% efficiency method: RPM calculations Solution: $\text{RPM} = (KV)(V)(\eta) = (2400)(7.6)(.80) = \boxed{14592 \text{ RPM}}$ $\text{MPH} = \frac{\text{RPM} \times \left(\times 60 \left[\frac{\text{min}}{\text{hour}} \right] \right)}{12 \left[\frac{\text{in}}{\text{foot}} \right] \times 5280 \left[\frac{\text{feet}}{\text{mile}} \right]} = \frac{(14592)(12.57)(60)}{(12)(5280)} = 173.7 \text{ mph}$ $C = \pi \varnothing = \pi(4) = 12.57 \text{ in}$ 173.7 mph with no gear reduction gear reduction for target speed of 30 mph $G = \frac{N_{\text{motor}}(\pi \varnothing)}{1056 V_{\text{target}}} = \frac{14592(\pi(4))}{1056(30)} = 5.7$ gear reduction for target speed of 35 mph $G = \frac{14592(4\pi)}{1056(35)} = 5$ $G = 5$ with safety in mind			

Appendix A05 – Motor Power and Friction on Wheels at an Incline

Alex Amadio	MET 489	10/26/2025	7/1
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Given: $N = 2200 \text{ RPM}$ $i = 5$
 $\emptyset_{\text{diameter}} = 3 \text{ in}$ $g = 32.2 \text{ ft/s}^2$
 45° incline $m = 3 \text{ pounds}$

find: motor power and friction on wheels

assume: constant torque
 $\eta = 80\%$

method: $P = T\omega$
 $T = F_{\text{drive}} \times r$

Solution:

$$\omega = \frac{2\pi N}{60} = \frac{2\pi (2200)}{60} = 231 \text{ rad/sec}$$
$$T = \frac{F_{\text{drive}} \times r}{(i)(\eta)} = \frac{(68.3)(3.75)}{(5)(.8)} = 6.2$$
$$F_{\text{drive}} = mg \sin \emptyset = 3(32.2)(\sin 45) = 68.3 \text{ lb}$$

the wheels need to generate 68.3 lb of friction

$$P = 6.2 \times 231.06 = 1556.2 \text{ hp}$$

$P_{\text{motor}} = 1556.2 \text{ hp to go up a } 45^\circ \text{ incline}$

Appendix A06 – Damage Factor and Number of Cycles Over Time

Alex Amadio	MET 489	10/26/2025	7/2
given: $RPM = 14592/5 = 2918.4$ $\rho_{alum} = 2.7 \text{ g/cm}^3 \Rightarrow .00475 \text{ lb/in}^3$ $A = 0.0491 \text{ in}^2$ $L = 10 \text{ in}$ $P_{motor} = 490.71$ $r = .375 \text{ in}$ $\phi = .75$ find: The damage factor assume: perfect horizontal shaft Doubling Driveshaft Diameter methods: Miner's Rule Torsional shear (recalculation) bending moment Fig. 5-10 Solution: $\omega = \frac{2\pi(2918.4)}{60} = 305.61 \text{ rad/s}$ $\omega_c = \rho A r \omega^2 = .284 (.442)(.375)(305.61)^2 = 4396.5$ $M_{max} = \frac{\omega_c L^2}{8} = \frac{4396.5 (10)^2}{8} = 54956.24$ $\sigma_b = \frac{54956.24 (.375)}{I} = 468164.24$ $I = \frac{\pi (.375)^4}{4} = .0155$ $T = \frac{5252 \cdot 15.56 \cdot 2}{14592} = .52419$			

Appendix A06 – Damage Factor and Number of Cycles Over Time Cont. (2)

Alex Amadio	MET 484	10/27/2025	2/2
$\tau = \frac{T_c}{J} = \frac{356.4(2375)}{J} = 6320.62$ (off from previous calculation but within parameters)			
$\sigma_{eq} = \sqrt{468164.24^2 + 3(14521.2)^2}$			
$\sigma_{eq} = 468292.227 \text{ psi} \rightarrow 47 \text{ ksi}$			
$J = \frac{\pi (.375)^4}{2} = 0.0311$			
$D = \frac{N}{N_i} = \frac{5826 \text{ (revolution over 2 minutes)}}{14000 \text{ (Fig 5-10)}}$			
$N = 2913 \times 2 = 5826$			
$D = .416 < 1$			

Appendix A07 – Deflection of Driveshaft at 15 mph Impact

	Alex Amadio	MET 489	11/2/2025	1/2
	given:	$d = .75 \text{ inches}$	$V = 15 \text{ mph}$	
		$E = 10 \times 10^6 \text{ psi (2024-T6511 Aluminum)}$		
		$L = 10''$		
		$K = 1.0 \text{ (Pinned-pinned)}$		
	find:	deflection on drive shaft	during 15 mph	head
	assume:	perfectly elastic collision		
		pinned-pinned		
		simply supported beam		
	method:	Euler critical buckling load		
		slenderness ratio		
		kinetic energy		
		simply supported		
	solution:	$\lambda = \frac{L}{\sqrt{\frac{I}{A}}} = \frac{10}{\sqrt{\frac{0.015532}{0.44179}}} = 53.3 < 100$		
		$I = \frac{\pi (.75)^4}{64} = 0.015532 \text{ in}^4$		
		$A = \pi (.375)^2 = 0.44179 \text{ in}^2$		
		$P_{cr} = \frac{\pi^2 EI}{(KL)^2} = 15329.47 \text{ psi} \rightarrow 15.33 \text{ Ksi}$		
		$F = \frac{\frac{1}{2} m V^2}{8} = \frac{\frac{1}{2} (3) (15)^2}{8} = 16.87$		

Appendix A07 – Deflection of Driveshaft at 15 mph Impact Cont. (2)

Alex Amadio

MET 489

11/2/2025

2/2

solution cont.

$$\delta_{\max} = \frac{1687(10)^3}{48(10 \times 10^6)(I)} = 0.2 \text{ inches}$$

$$\boxed{\delta = 0.2 \text{ inches}}$$

$$1687 < 15329$$

the driveshaft will not buckle more than
0.2 inches at 15 mph

Appendix A08 – Chassis Vibration at 15 mph

Alex Amadio

MET 489

11/2/2025

1/2

Given: $V = 15 \text{ mph} \rightarrow 6.7056 \text{ m/s}$

$d_{\text{wheel}} = 4 \text{ in} \rightarrow 0.1016 \text{ m}$ $\zeta \approx 1$

$b = 9 \text{ in} \rightarrow 0.2286 \text{ m}$ $A_{\text{rod}} = 0.001$

$t = 0.125 \text{ in} \rightarrow 0.003175 \text{ m}$ $A_{\text{ground}} = 0.003$

$L = 12 \text{ in} \rightarrow 0.3048 \text{ m}$

$E = 71 \text{ GPa} \rightarrow 71 \times 10^9 \text{ Pa}$

$m = 3 \text{ pounds} \rightarrow 1.360777 \text{ Kg}$

find: can the chassis withstand 10-30 Hz at 15 mph

assume: Basic oil-filled shock

method: excitation frequency

chassis stiffness

Natural frequency

transmissibility
response Amplitude

Solution:

$$f_{\text{wheel}} = \frac{V}{\pi d} = \frac{6.7056}{\pi (0.1016)} = 21.01 \text{ Hz}$$

$$I = \frac{bt^3}{12} = \frac{0.2286 (0.003175)^3}{12} = 6.097 \times 10^{-10}$$

$$K = \frac{48(71 \times 10^9)(6.097 \times 10^{-10})}{(0.3048)^3} = 73378.85$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{73378.85}{1.360777}} = \boxed{36.96 \text{ Hz}}$$

Appendix A08 – Chassis Vibration at 15 mph Cont. (2)

Alex Amadio	MET U89	11/2/2025	2/2
Solution cont. $r = \frac{\omega_{wheel}}{\omega_n} = \frac{21.01 \text{ Hz}}{36.96 \text{ Hz}} = 0.57$ $T(r) = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}} = \sqrt{\frac{1 + (2(0.1)(0.57))^2}{(1 - (0.57)^2)^2 + (2(0.1)(0.57))^2}}$ $T(r) = 1.47$ displacement amplitude road $x_{response} = A_{road} \cdot T(r) = (0.001) \cdot 1.47 = 0.0015 \text{ m}$ gravel $x_{response} = A_{gravel} \cdot T(r) = (0.003) \cdot 1.47 = 0.0044 \text{ m}$			

Appendix A09 – Motor Torque Needed for Acceleration

Alex Amadio	MET 489	11/9/2025	2/1
Given: $D = 4 \text{ in}$ $t = 10 \text{ seconds}$ $V = 10 \text{ mph} \rightarrow \frac{10560}{\pi(4)} = 840 \text{ RPM}$ $P_{\text{motor}} = 1556.2 \text{ hp}$ $a = 12 \text{ in} \rightarrow .3048 \text{ m}$ $b = 9 \text{ in} \rightarrow .2286 \text{ m}$ $m = 3 \text{ lbs} \rightarrow 1.36 \text{ kg}$ find: how much torque from motor is needed to reach 10 mph in 10 seconds assume: no slip method: $\omega_f = \text{RPM} \times \frac{2\pi}{60}$ $T = I\alpha$ Solution: $\omega_f = 840 (\text{RPM}) \times \frac{2\pi}{60} = 87.96 \text{ rad/s}$ $\alpha = \frac{87.96 \text{ rad/s}}{10 \text{ sec}} = 8.796 \text{ rad/s}^2$ $I = \frac{m(a^2 + b^2)}{12} = \frac{1.36(.3048^2 + .2286^2)}{12} = 0.0165$ $T = I\alpha = 0.0165 \times 8.796 = 0.145134 \text{ Nm}$ $\omega_{\text{motor}} = 87.96 \times 5 = 439.80 \text{ rad/s}$ $T_{\text{motor}} = \frac{1556.2}{439.80} = \boxed{3.538 \text{ Nm}}$ motor needs 3.538 Nm to reach 10 mph in 10 seconds			

Appendix A10 – Drag Forces on the Car

Alex Amadio | MET 489 | 11/9/2025

21

Given: $C_d = 0.4$

$$P = 0.07647 \text{ lb/ft}^3$$

$$A = 9 \text{ in} \times 5 \text{ in} = 45 \text{ in}^2 \rightarrow 3.75 \text{ feet}^2$$

$$V_1 = 10 \text{ mph} \rightarrow 14.66 \text{ ft/s}$$

$$V_2 = 20 \text{ mph} \rightarrow 29.33 \text{ ft/s}$$

$$V_3 = 30 \text{ mph} \rightarrow 44.00 \text{ ft/s}$$

find: drag forces

assume: 5 in height with height of components
method: drag force equation

solution:

10 mph:

$$F_d = \frac{1}{2} \rho C_d A V^2$$

$$F_d = \frac{1}{2} (0.07647) (.4) (3.75) (14.66)^2$$

$$F_d = 12.33 \text{ lbf}$$

20 mph

$$F_d = \frac{1}{2} (0.07647) (.4) (3.75) (29.33)^2$$

$$F_d = 49.34 \text{ lbf}$$

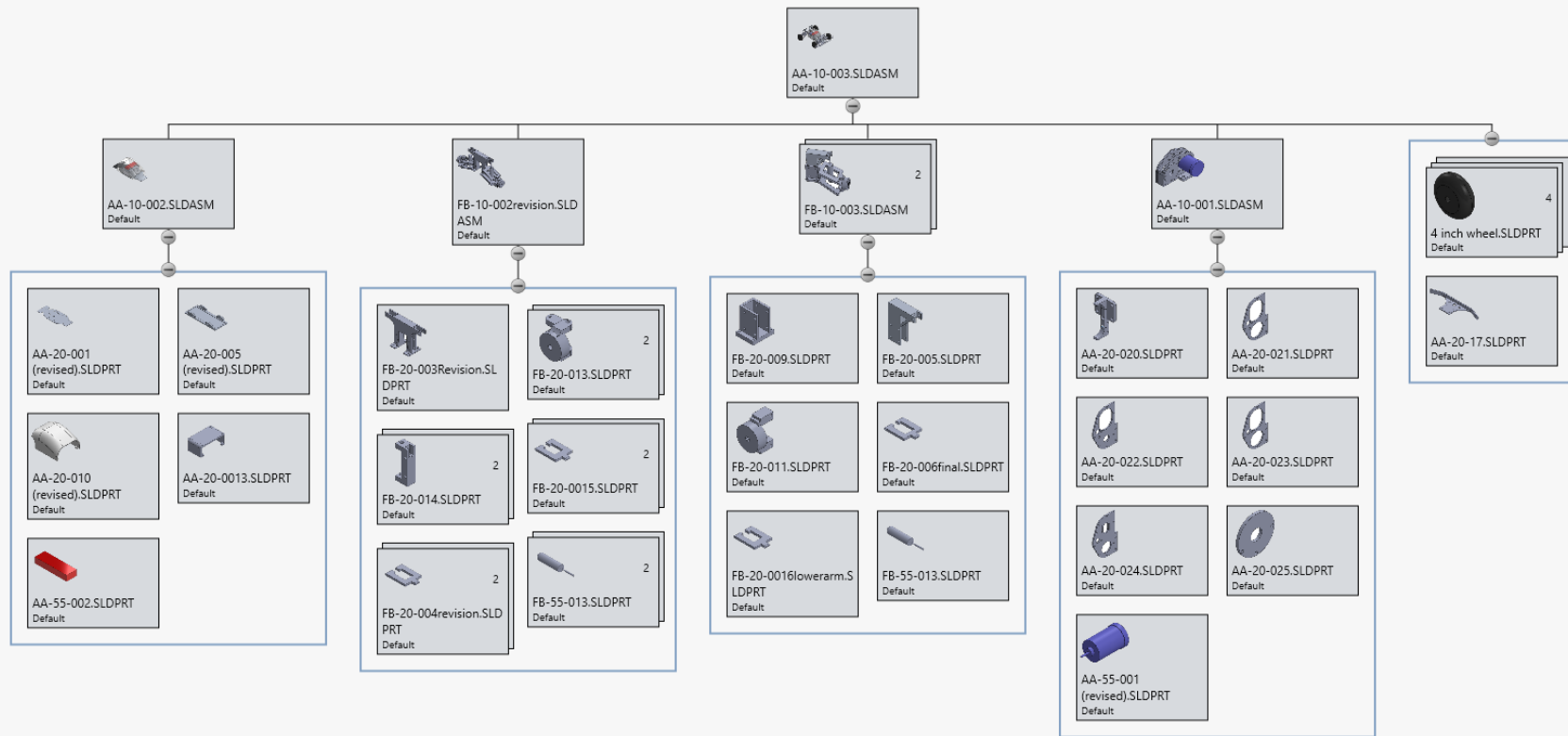
30 mph =

$$F_d = \frac{1}{2} (0.07647) (.4) (3.75) (44)^2$$

$$F_d = 111.03 \text{ lbf}$$

APPENDIX B - Drawings

Appendix B01 – Drawing Tree



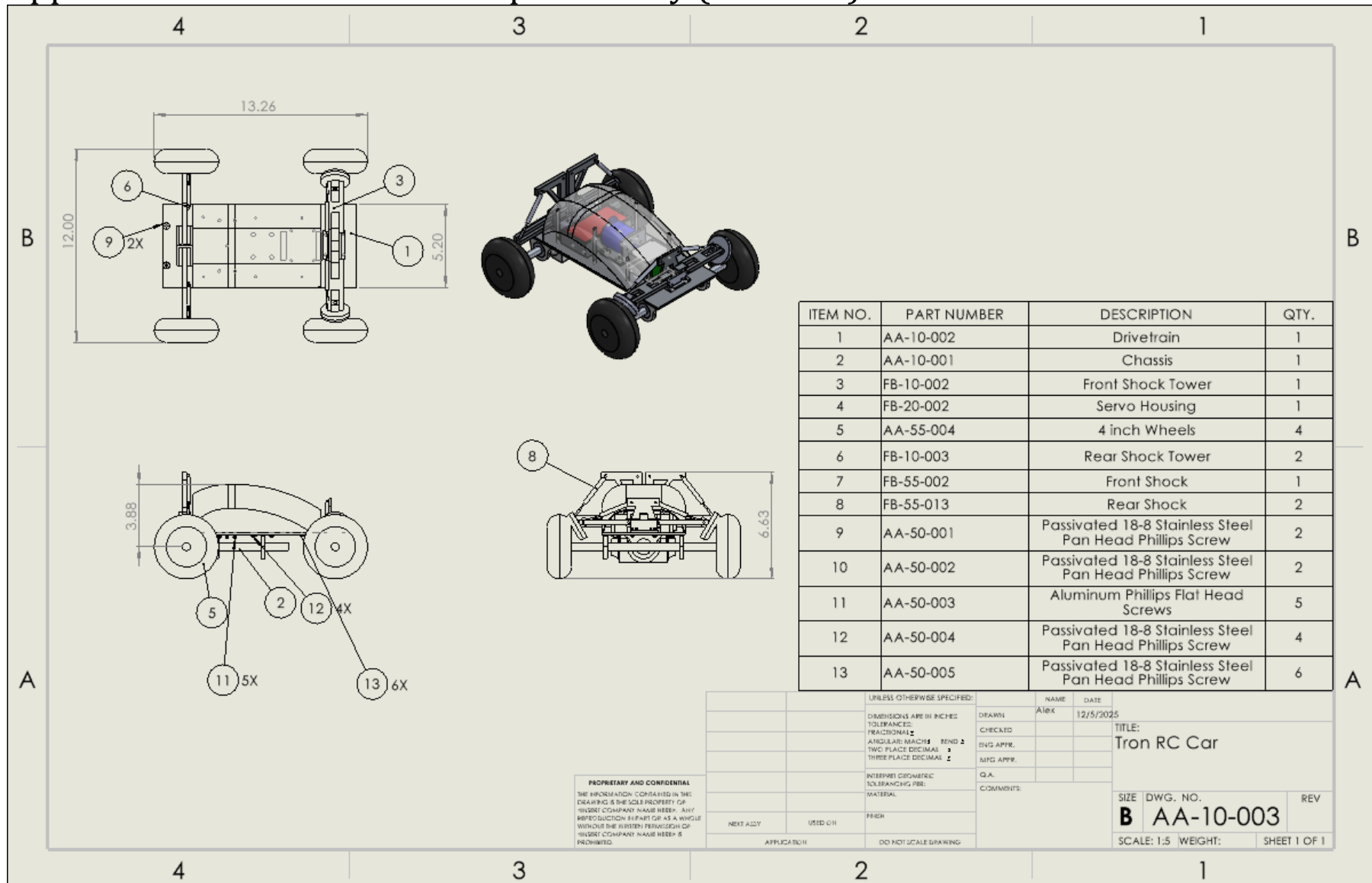
Appendix B02 – Drawing Index

Table B02. Drawing Index

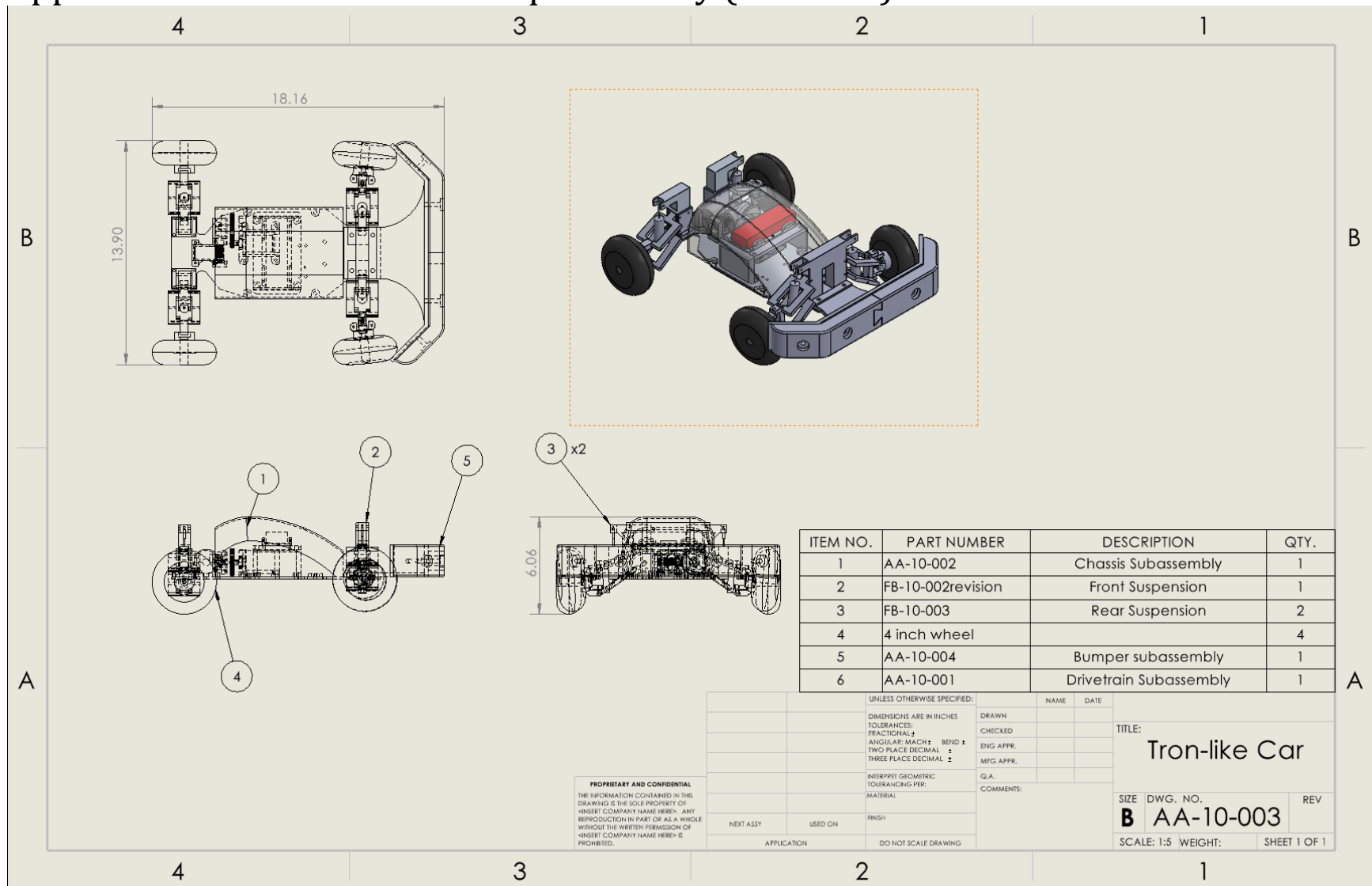
Drawing Assignment Num.	Drawing #(s)	Date submitted
Upload: DWG 1	AA-20-001	10/15/2025
Upload: DWG 2	AA-20-002	10/21/2025
Upload: DWG 3	AA-20-003	10/29/2025
Upload: DWG 4	AA-20-004	10/29/2025
Upload: DWG 5	AA-20-005	11/5/2025
Upload: DWG 6	AA-20-006	11/5/2025
Upload: DWG 7	AA-20-007	11/12/2025
Upload: DWG 8	AA-20-008	11/12/2025
Upload: DWG 9	AA-20-009	11/19/2025
Upload: DWG 10	AA-20-010	11/19/2025
Upload: DWG 11	AA-20-011	2/6/2026
Upload: DWG 12	AA-20-012	2/6/2026
Upload: DWG 13	AA-20-013	2/6/2026
Upload: DWG 14	AA-20-014	2/6/2026
Upload: DWG 15	AA-20-015	3/4/2026
Upload: DWG 16	AA-20-016	3/4/2026
Upload: DWG 17	AA-20-017	3/4/2026
Upload: DWG 18	AA-20-018	3/4/2026
Upload: DWG 19	AA-20-019	3/4/2026
Upload: DWG 20	AA-20-020	4/7/2026
Upload: DWG 21	AA-20-021	4/7/2026
Upload: DWG 22	AA-20-022	4/7/2026
Upload: DWG 23	AA-20-023	4/7/2026
Upload: DWG 24	AA-20-024	4/7/2026
Upload: DWG 25	AA-20-025	4/7/2026
Upload: Subassembly 1	AA-10-001	11/19/2025

Reupload: Subassembly 1	AA-10-001	3/13/2026
Final Upload: Subassembly 1	AA-10-001	4/7/2026
Upload: Subassembly 2	AA-10-002	12/5/2025
Reupload: Subassembly 2	AA-10-002	3/13/2026
Upload: Top assembly	AA-10-003	12/5/2025
Reupload: Top assembly	AA-10-003	3/16/2026
Final Upload: Top assembly	AA-10-003	4/7/2026
Upload: Subassembly 4	AA-10-004	3/16/2026

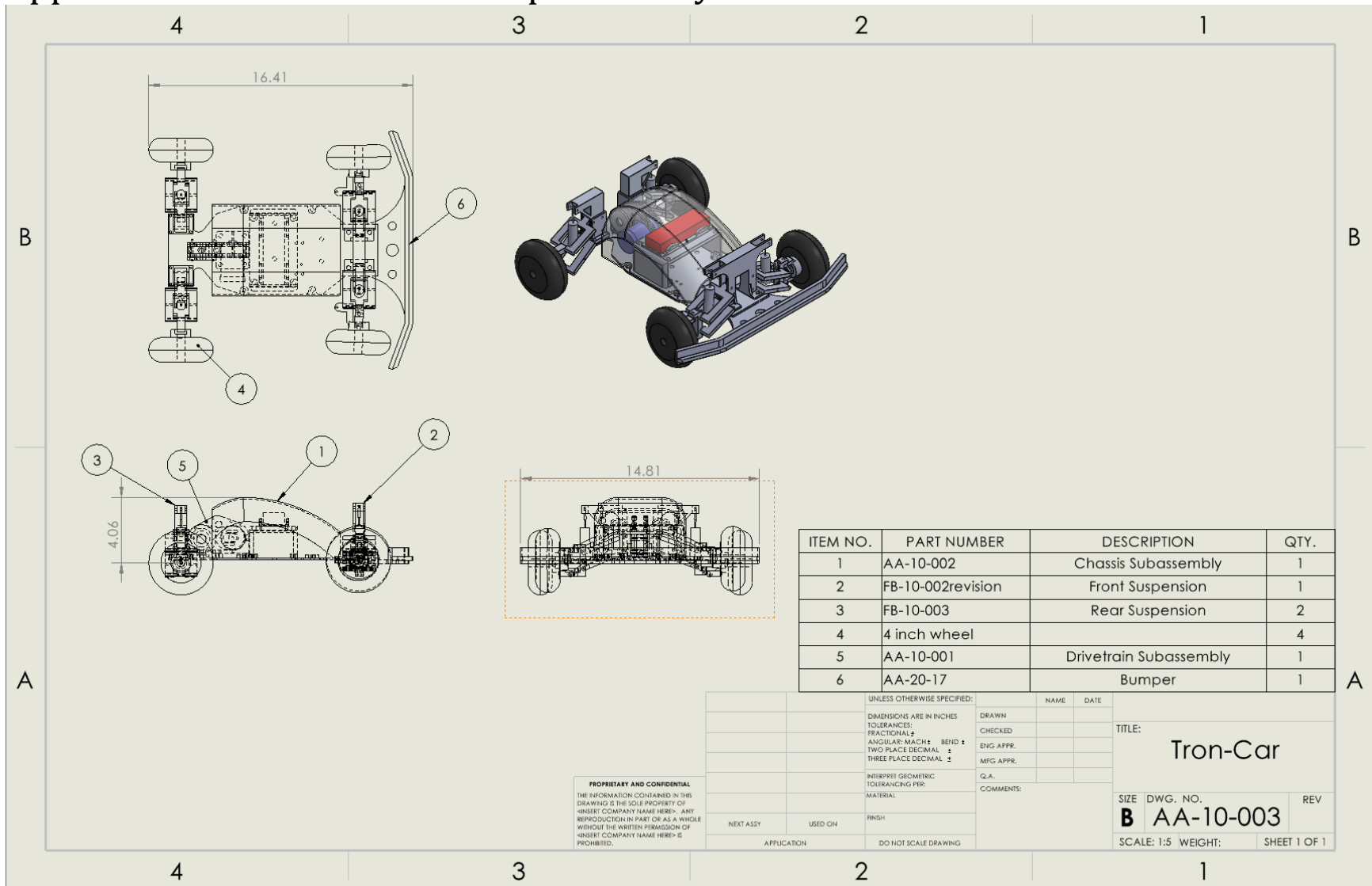
Appendix B03 – AA-10-003 – Top Assembly (Obsolete)



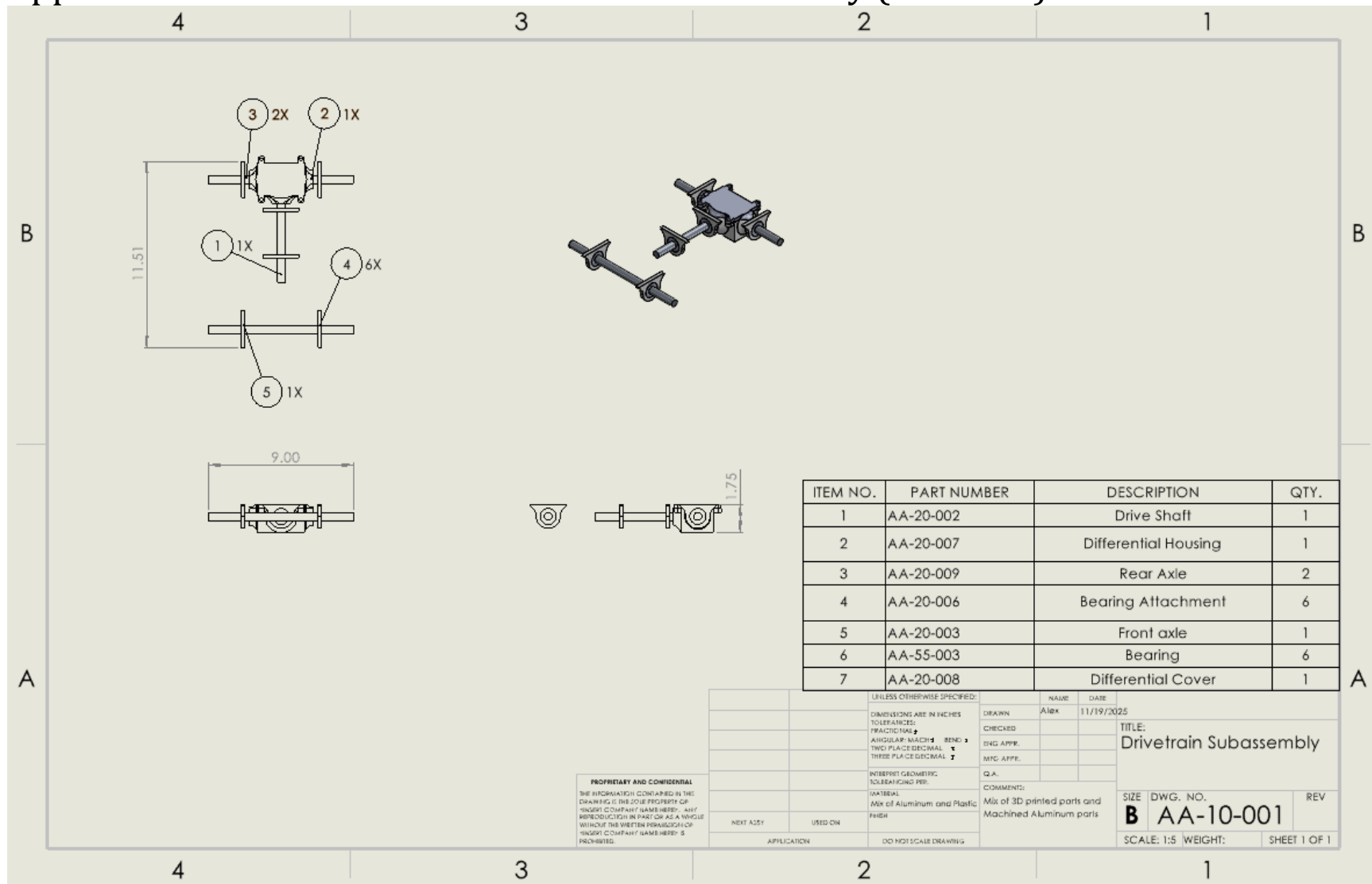
Appendix B04 – AA-10-003 – Top Assembly (Obsolete)



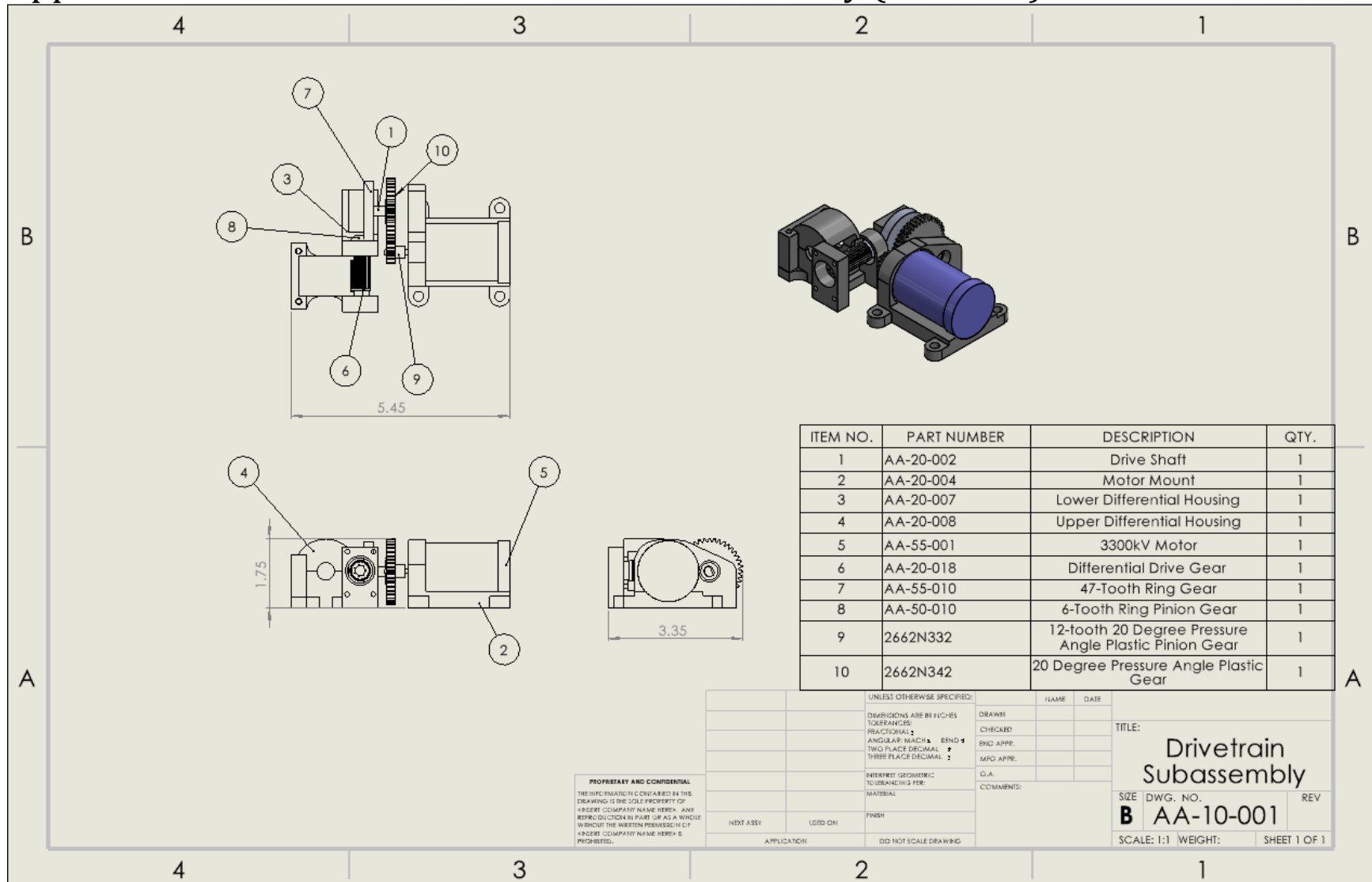
Appendix B05 – AA-10-003 – Top Assembly



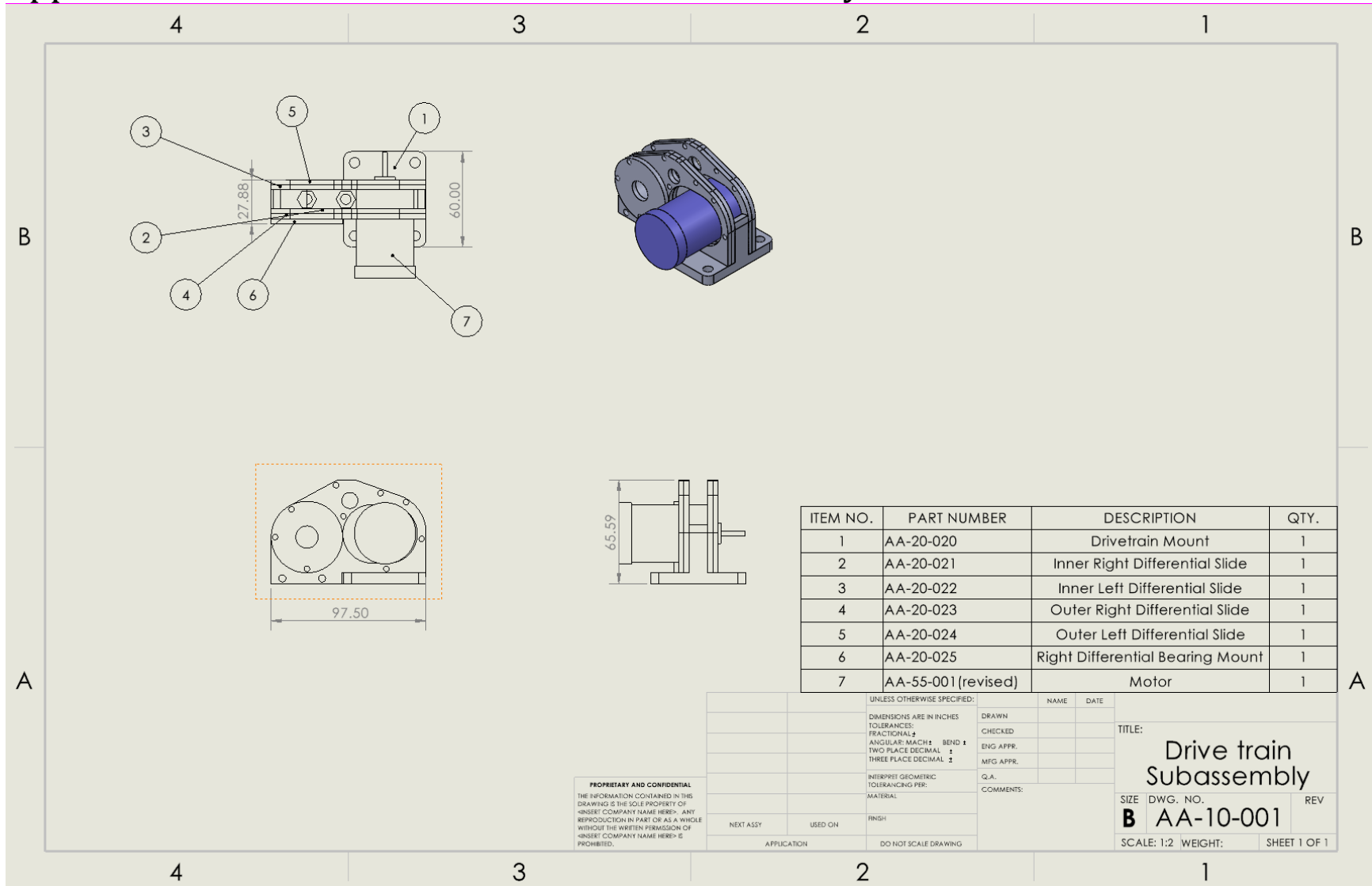
Appendix B06 -AA-10-001 – Drivetrain subassembly (Obsolete)



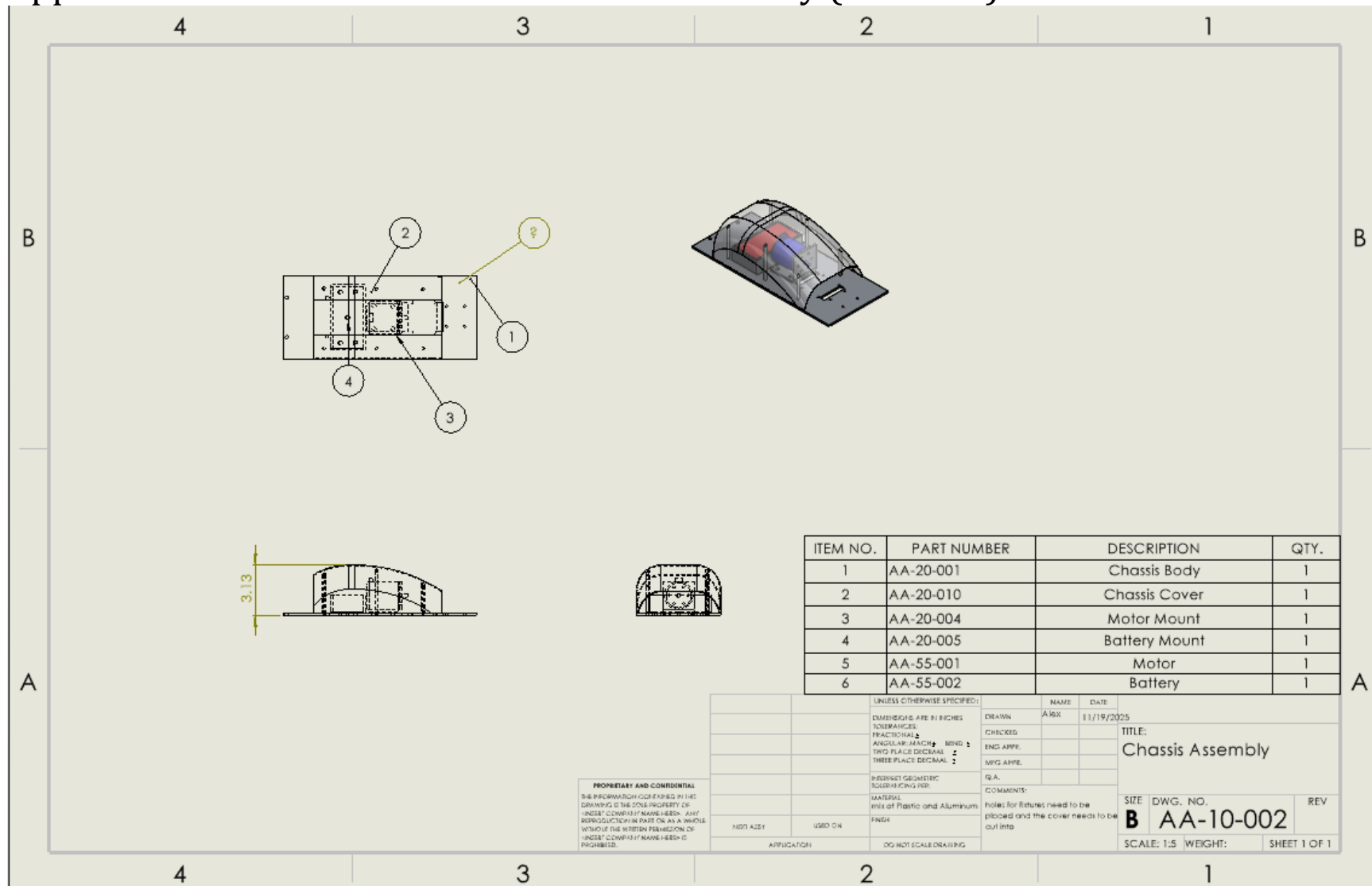
Appendix B07 -AA-10-001 – Drivetrain subassembly (Obsolete)



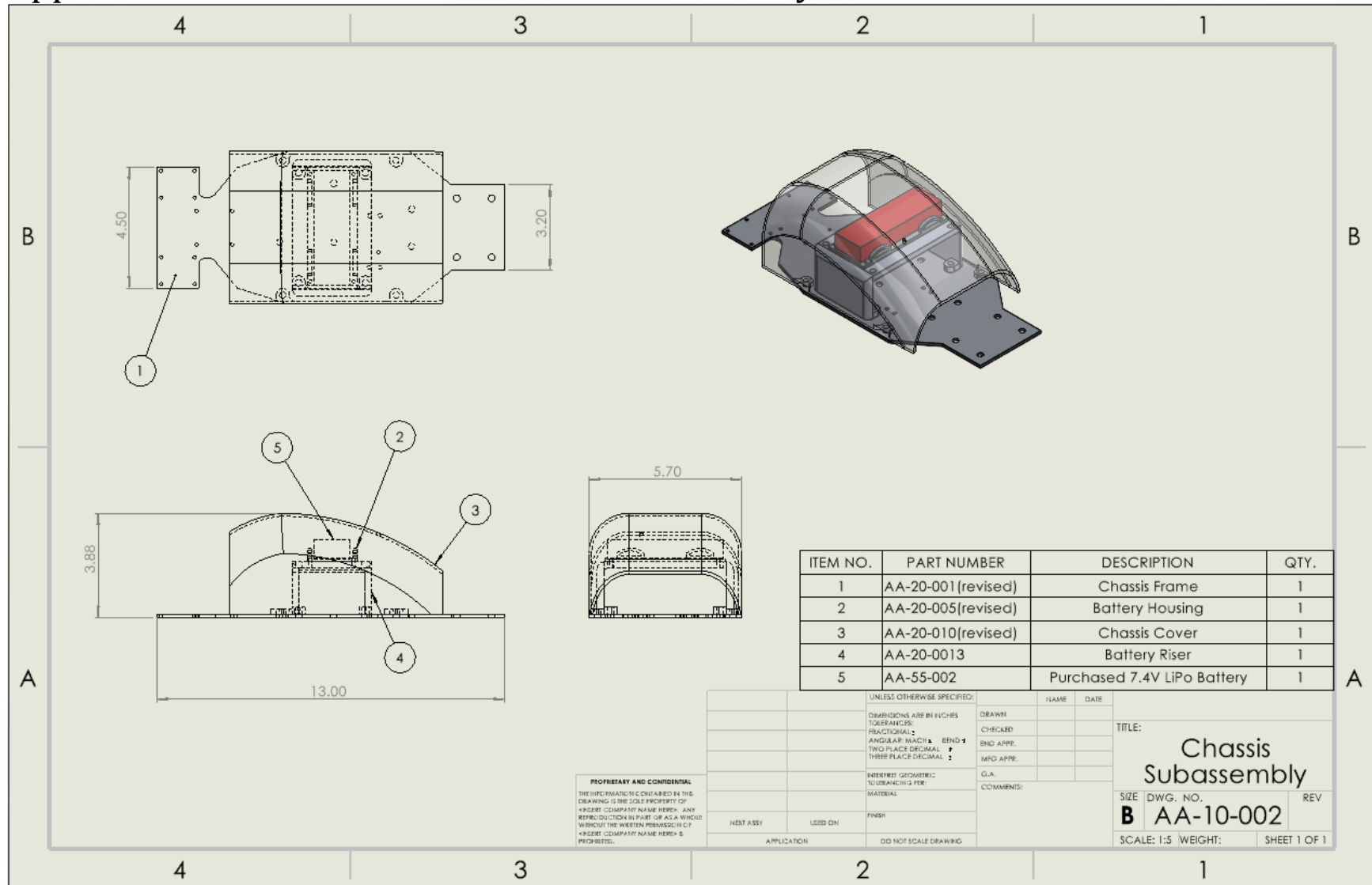
Appendix B08 –AA-10-001 – Drivetrain subassembly



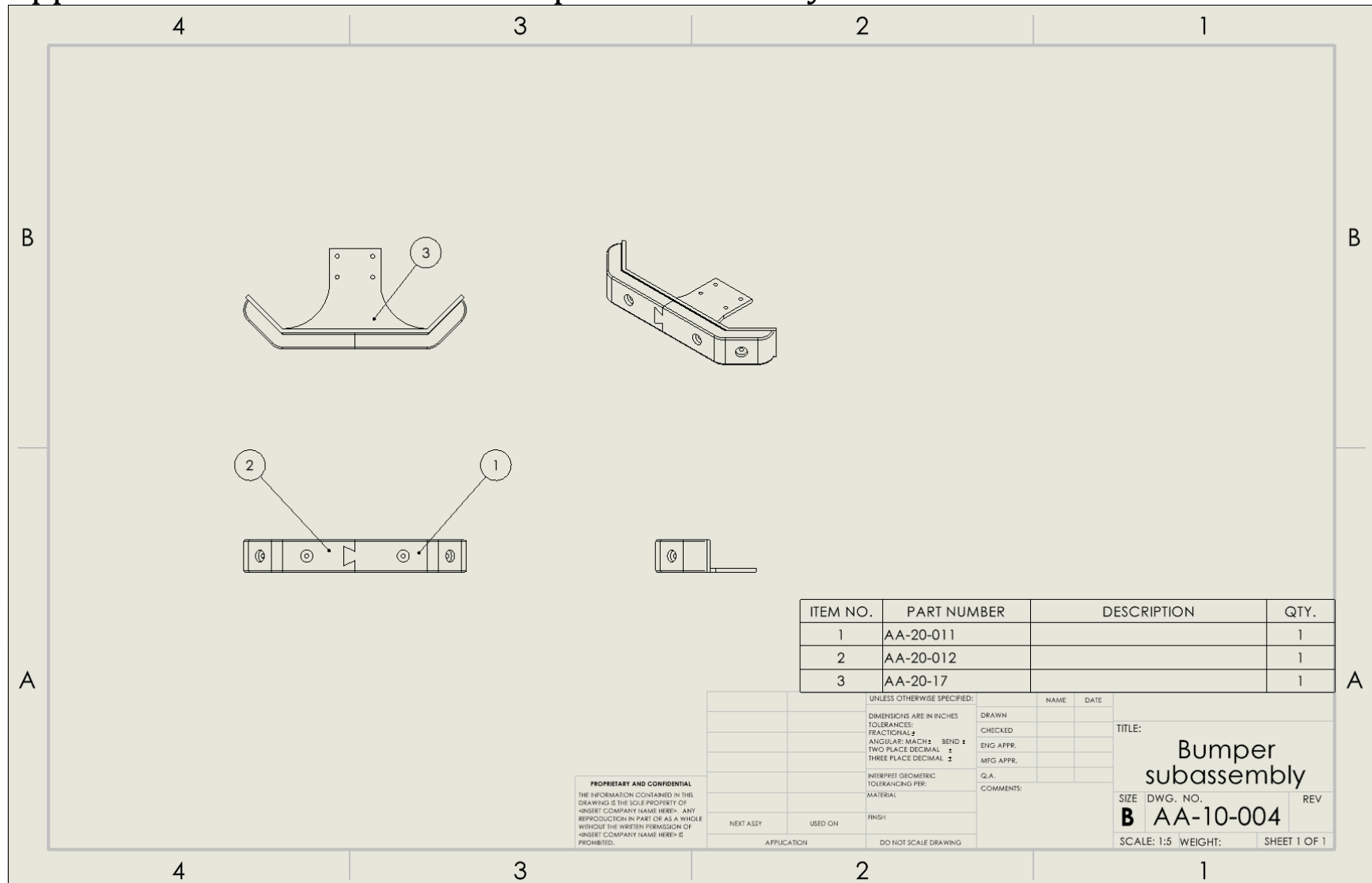
Appendix B09 –AA-10-002 – Chassis subassembly (Obsolete)



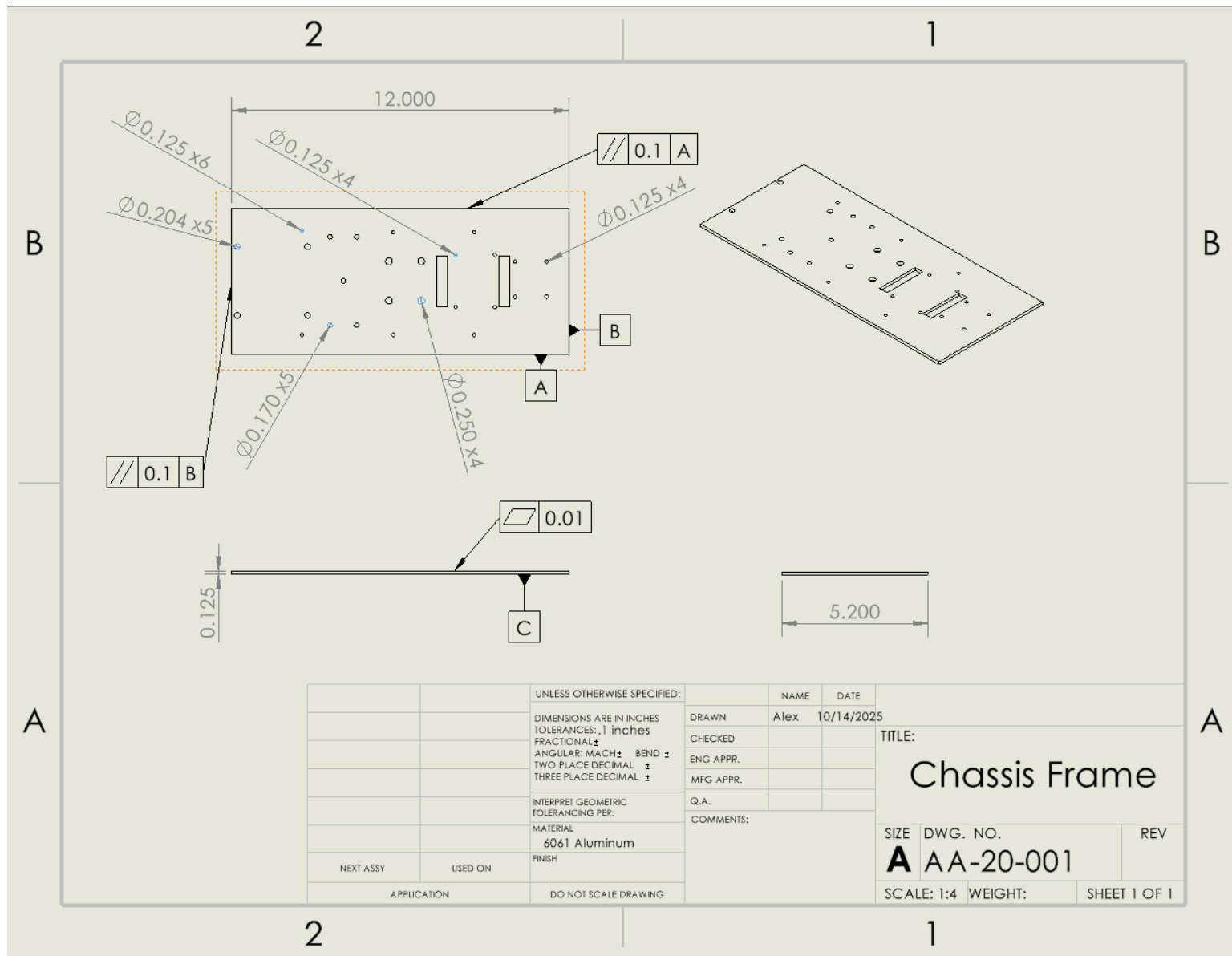
Appendix B10 -AA-10-002 – Chassis subassembly



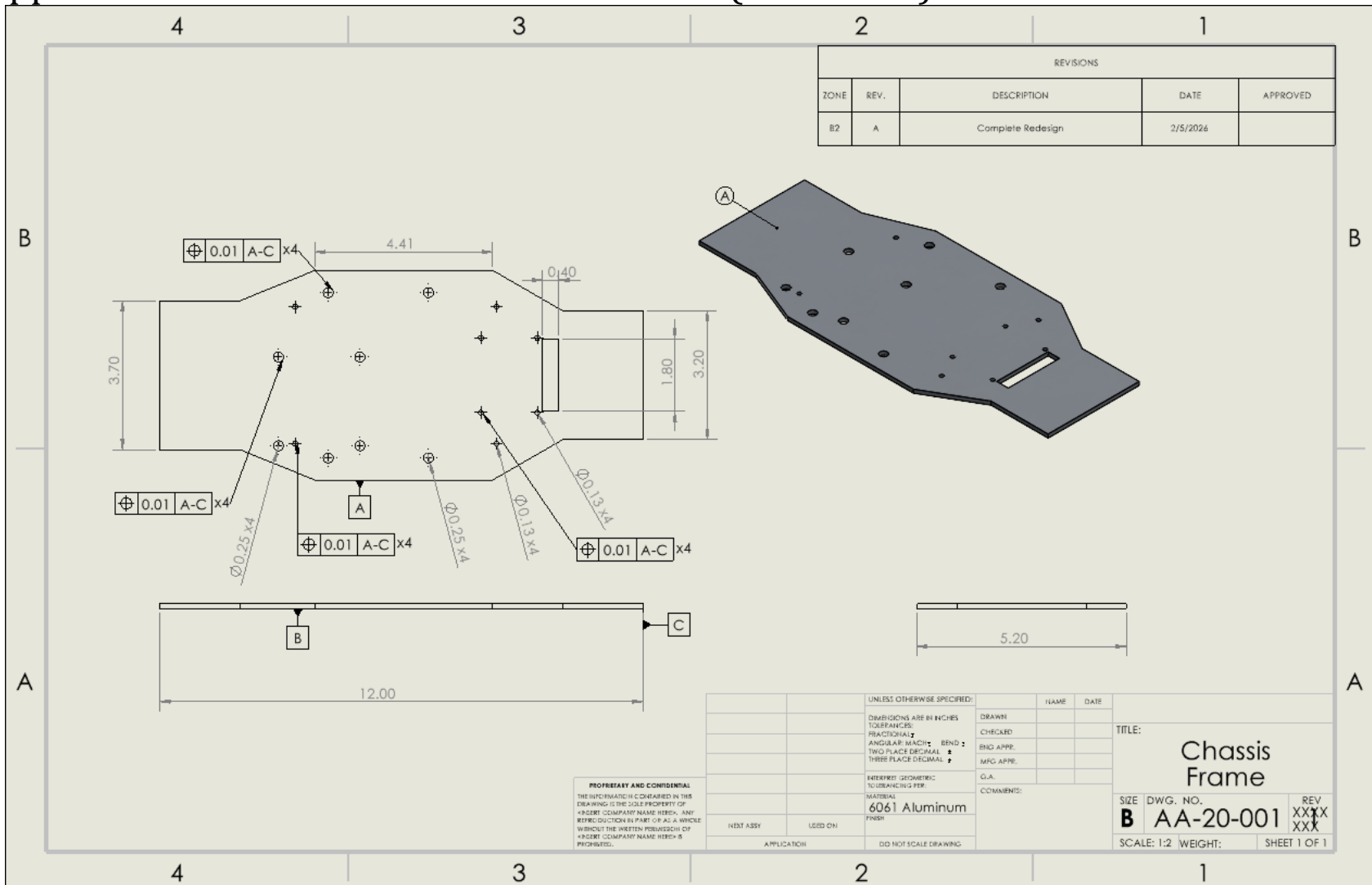
Appendix B11 -AA-10-004 – Bumper subassembly



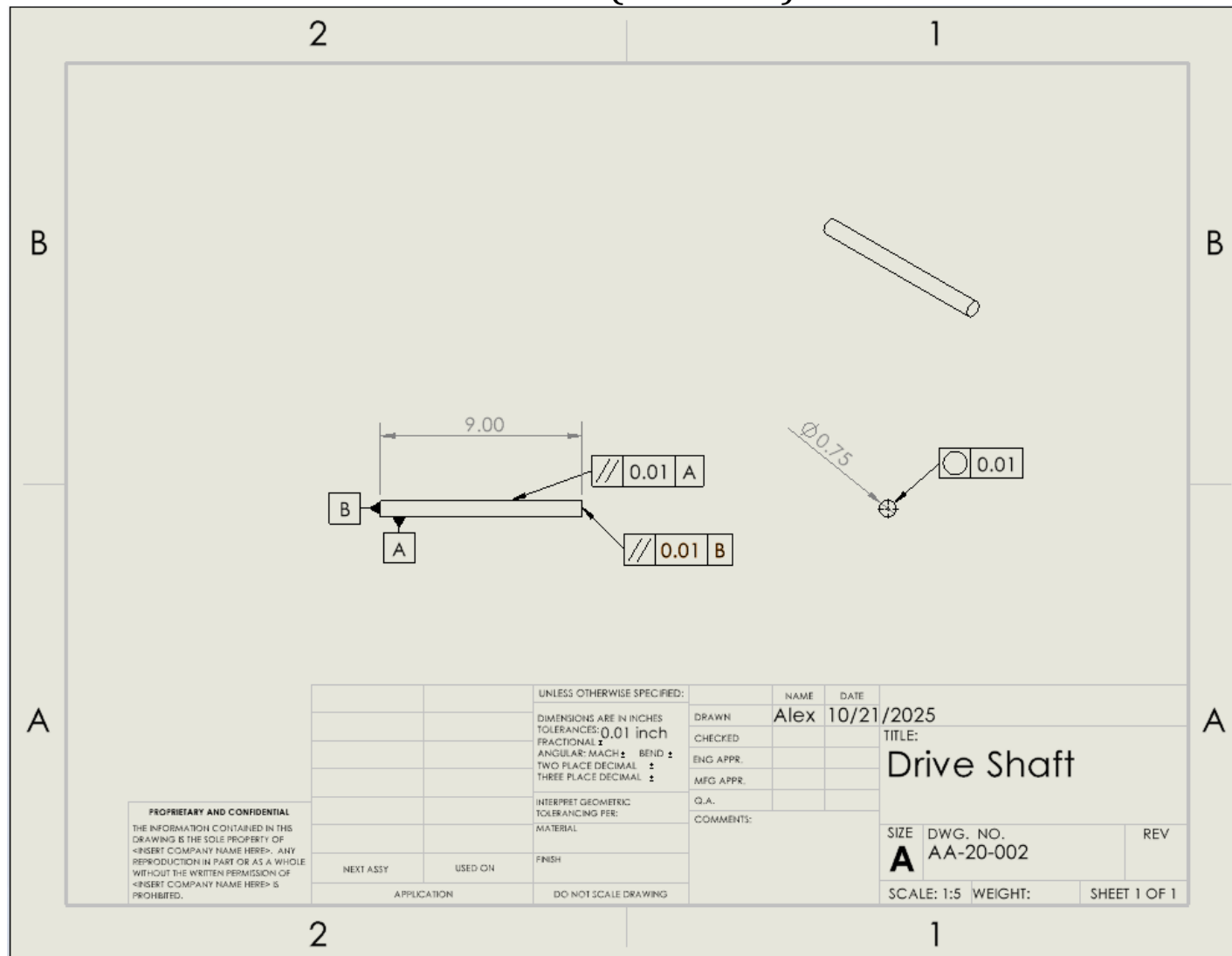
Appendix B12 – AA-20-001 – Chassis Frame



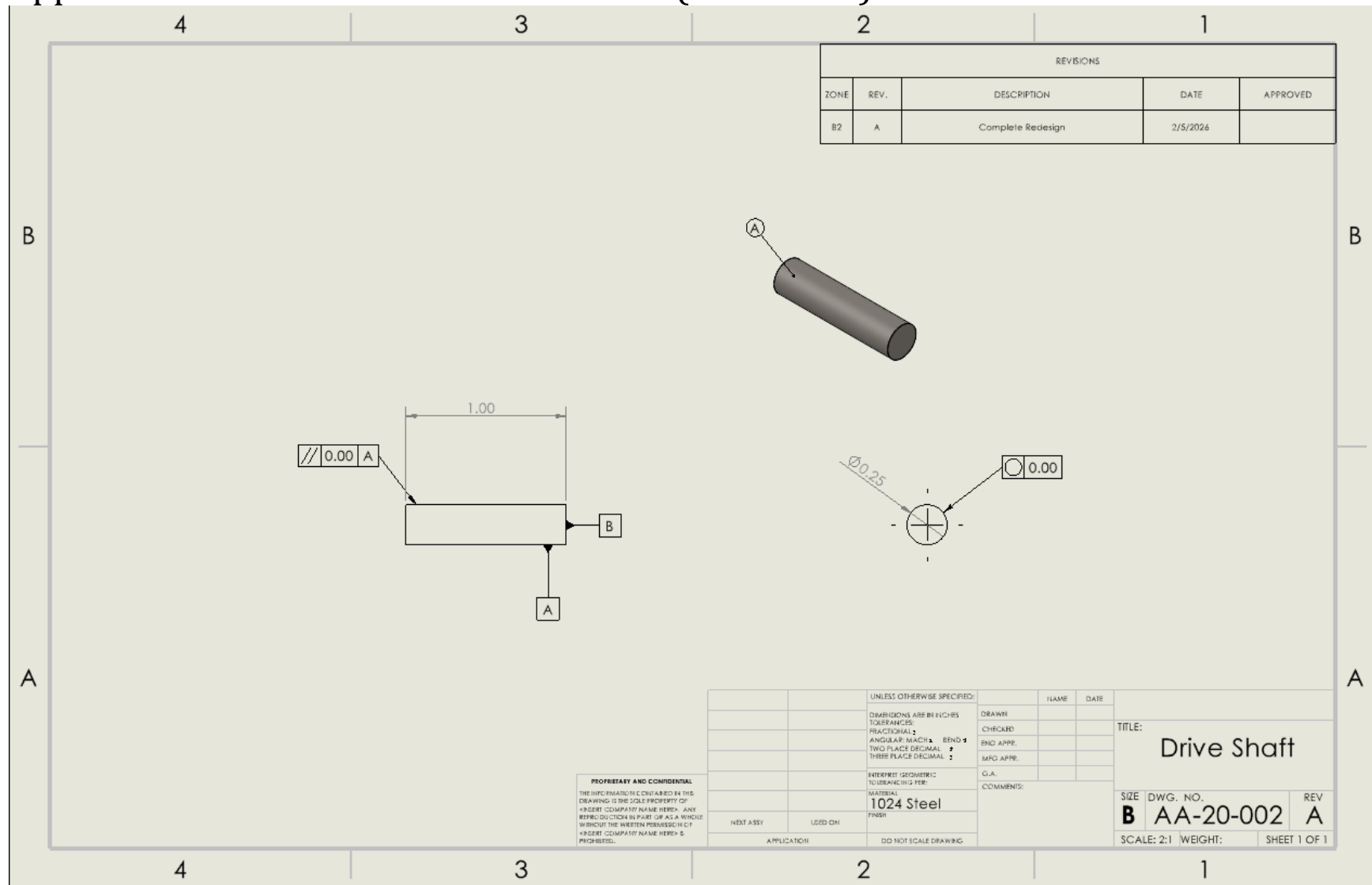
Appendix B13 – AA-20-001 – Chassis Frame (Revision 1)



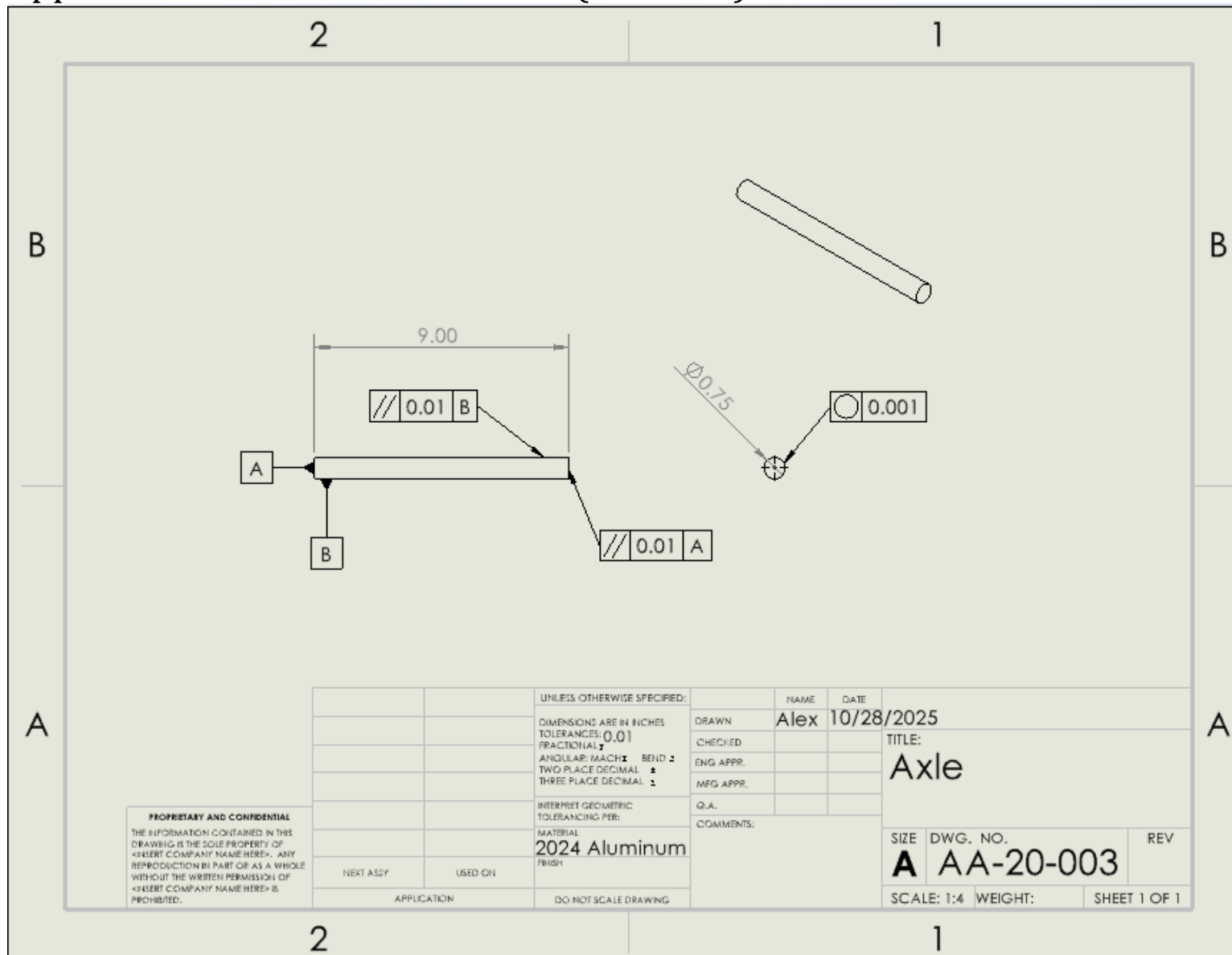
Appendix B14 – AA-20-002 – Drive Shaft (Obsolete)



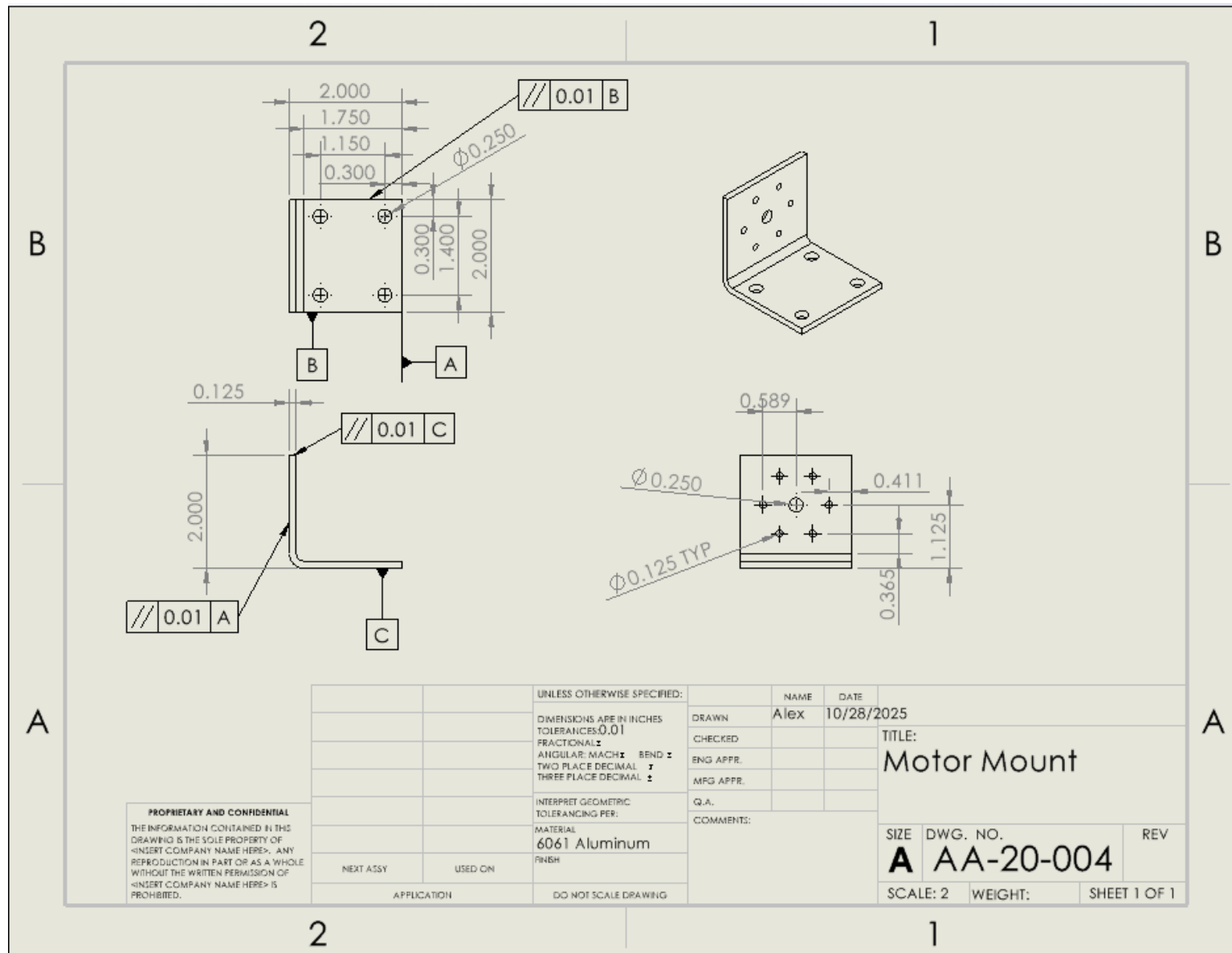
Appendix B15 – AA-20-002 – Drive Shaft (Revision 1)



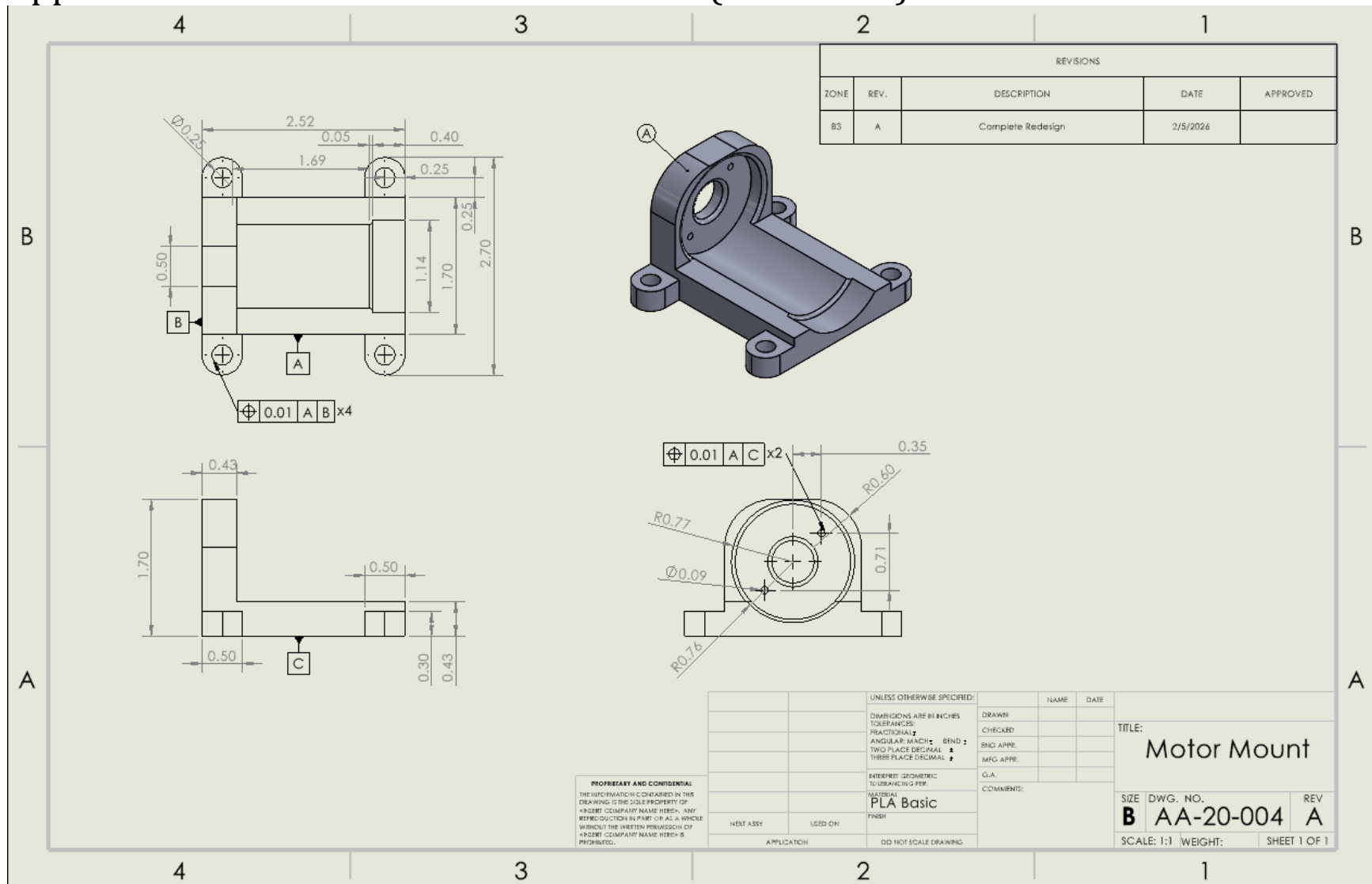
Appendix B16 – AA-20-003 – Axle (Obsolete)



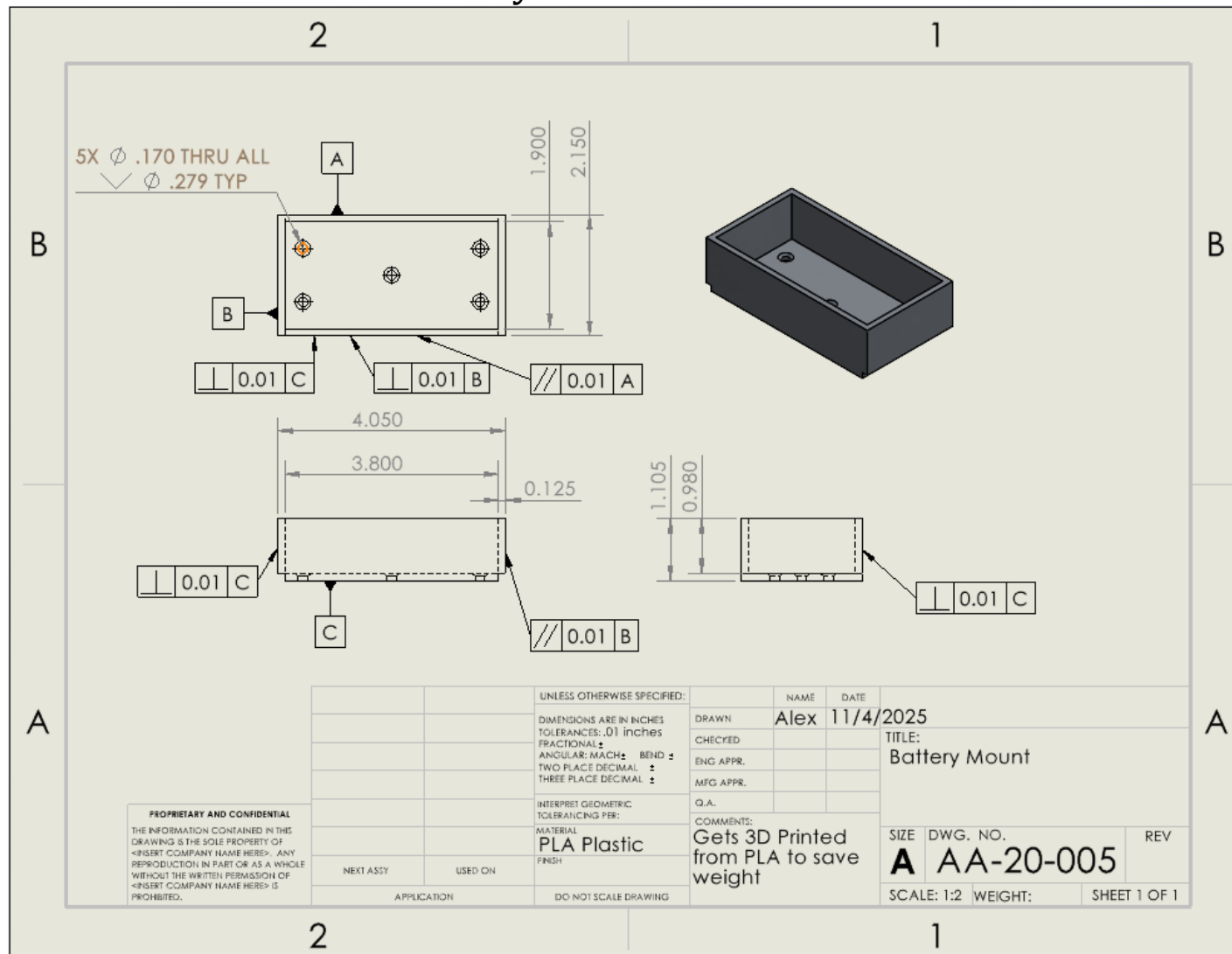
Appendix B17 – AA-20-004 – Motor Mount



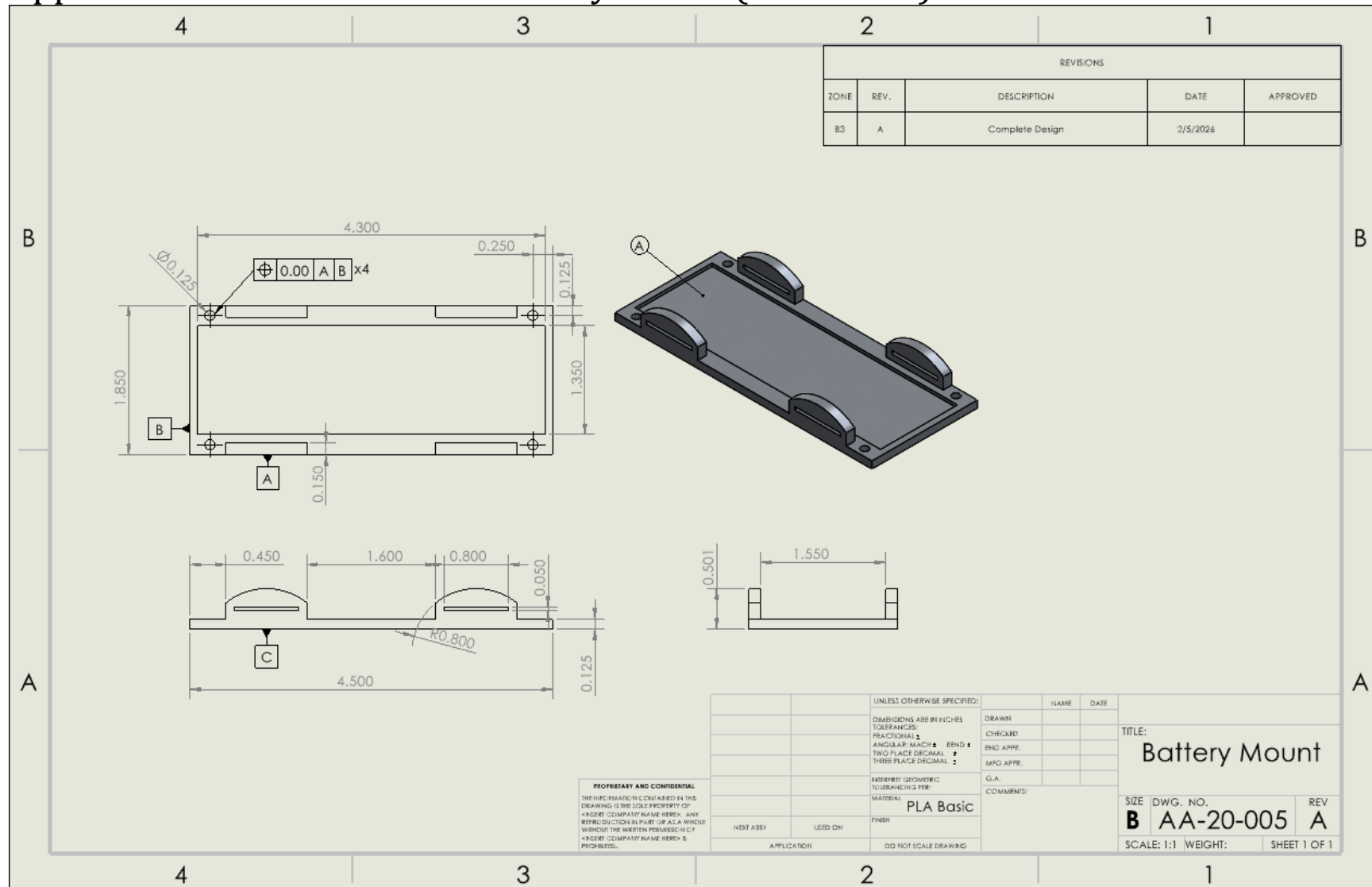
Appendix B18 – AA-20-004 – Motor Mount (Revision A)



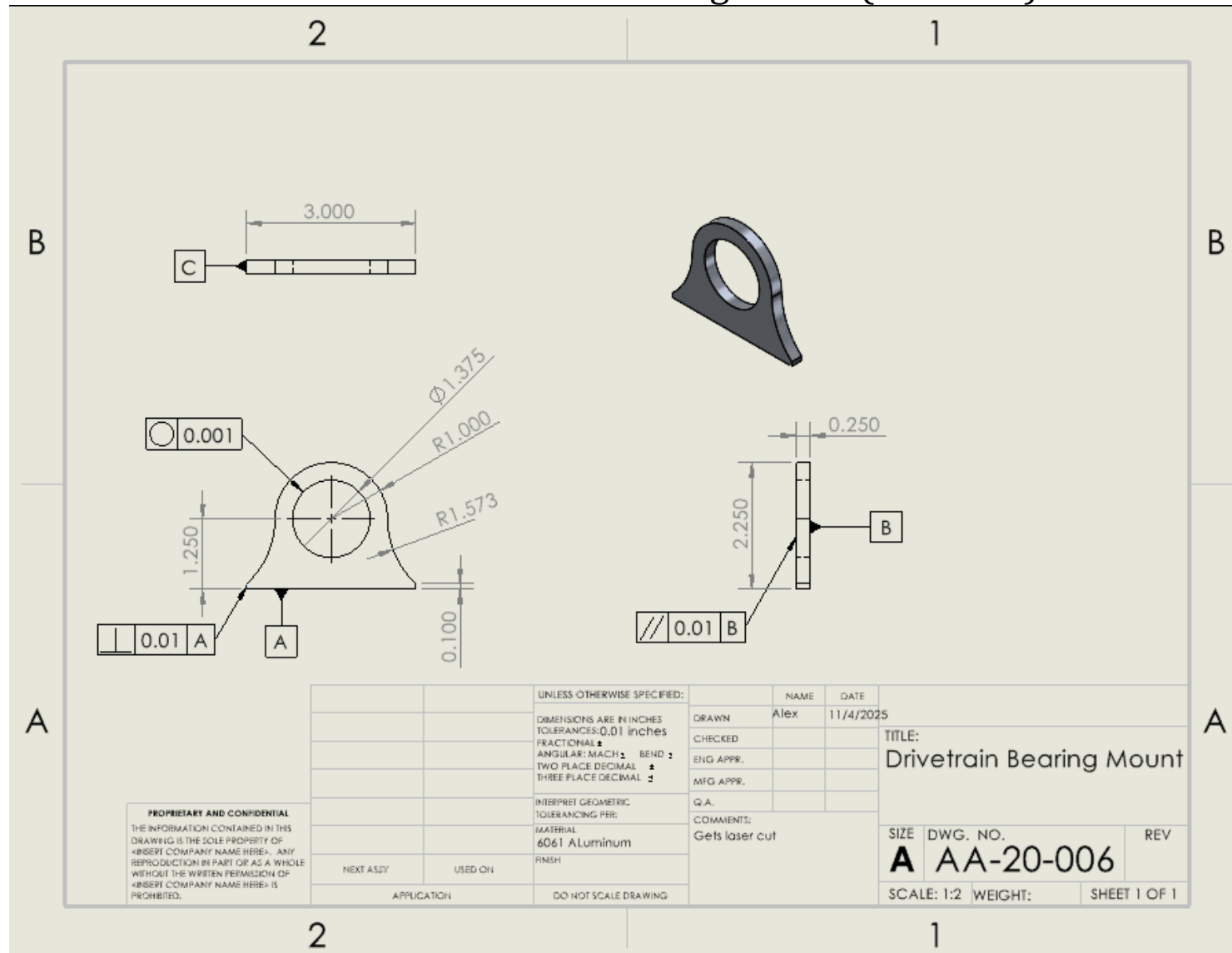
Appendix B19 – AA-20-005 – Battery Mount



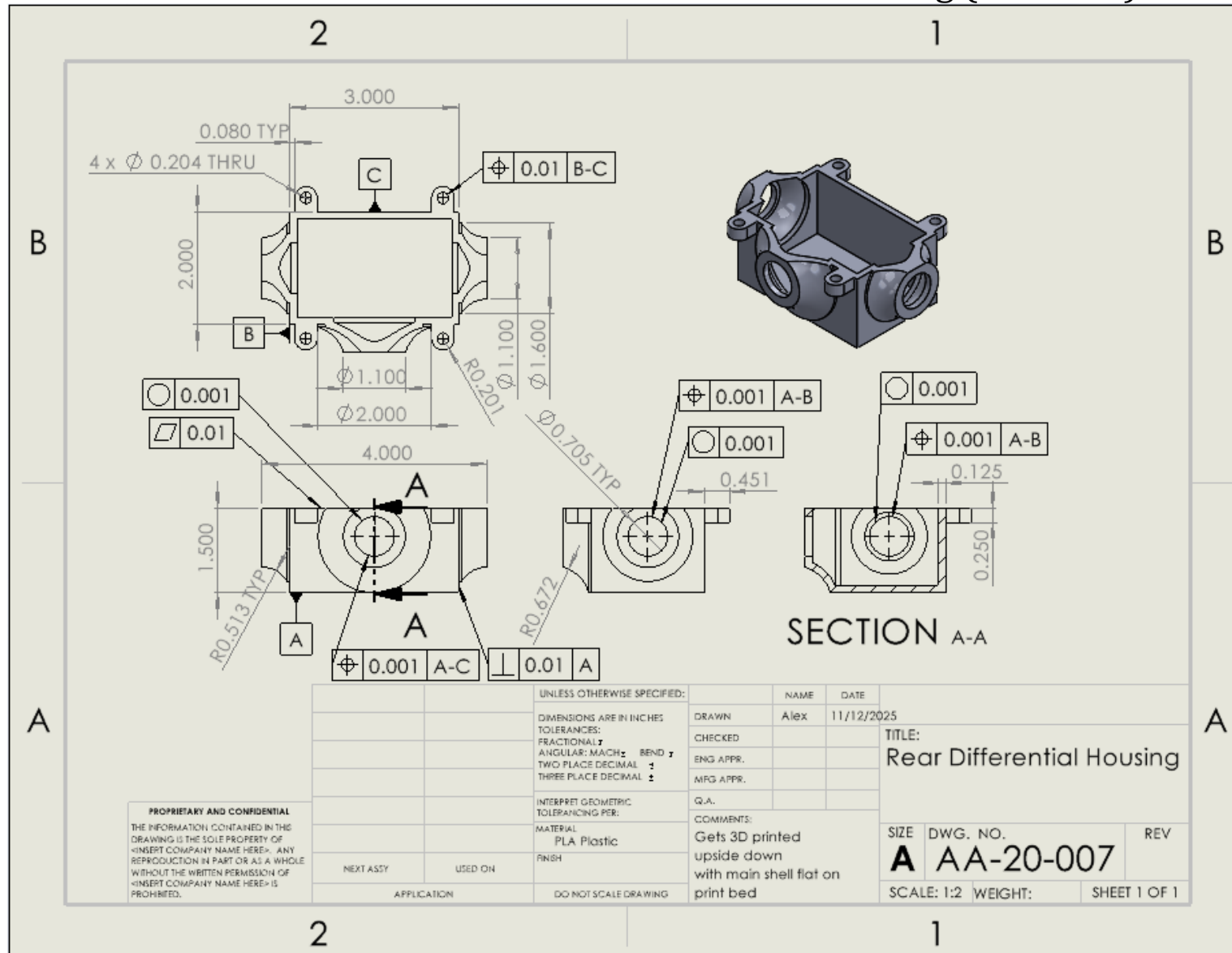
Appendix B20 – AA-20-005 – Battery Mount (Revision 1)



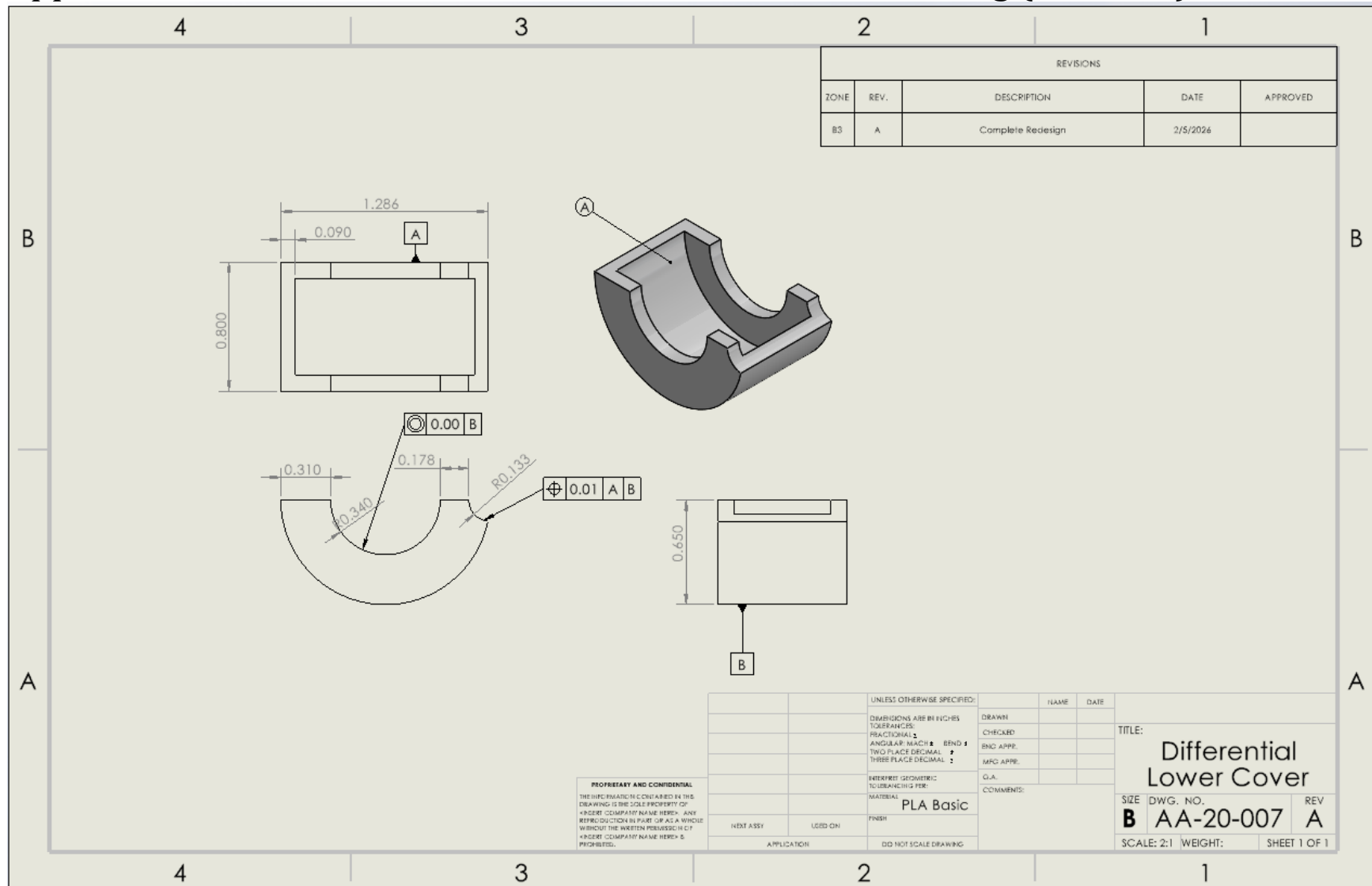
Appendix B21 – AA-20-006 – Drivetrain Bearing Mount (Obsolete)



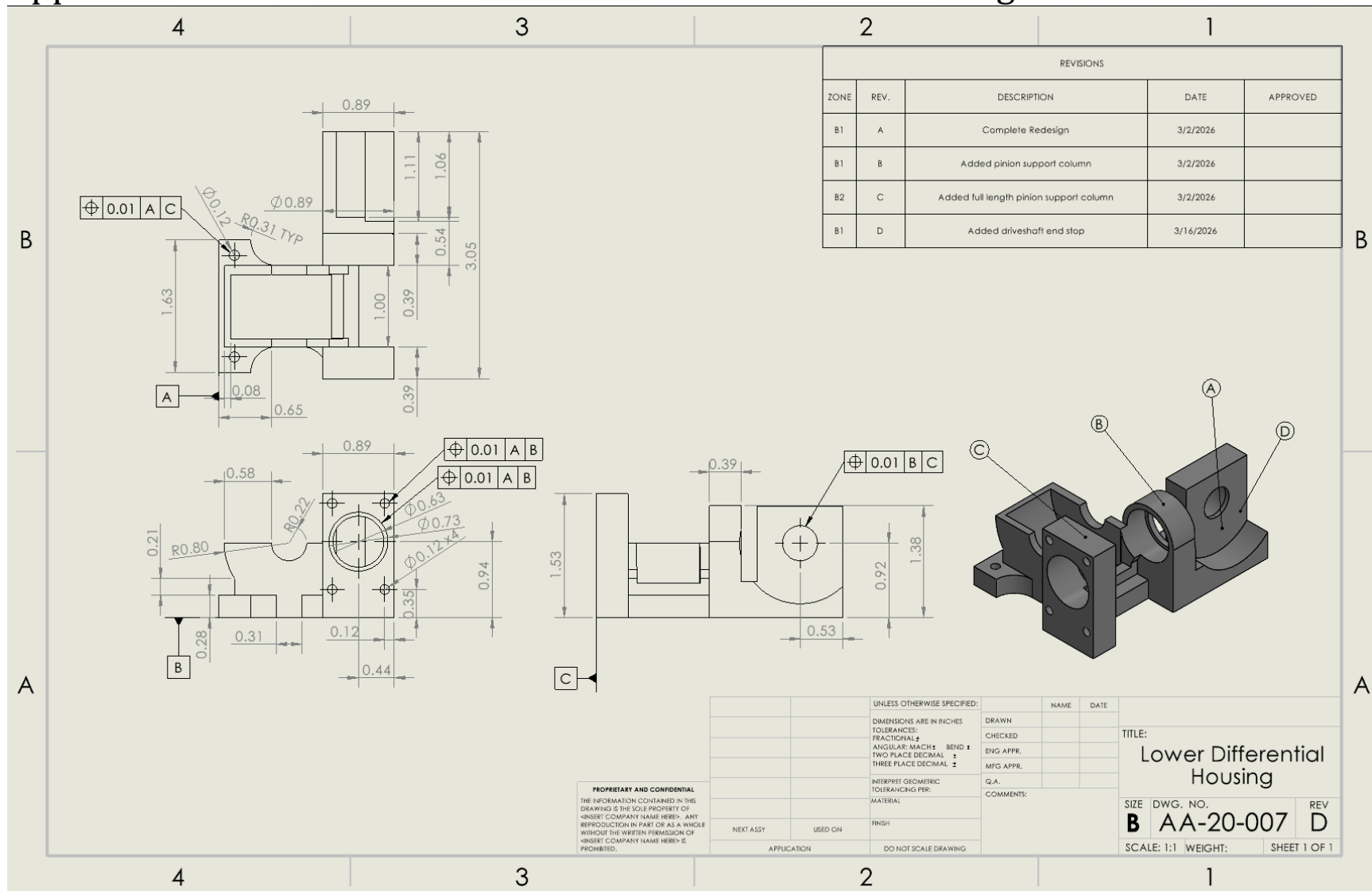
Appendix B22 – AA-20-007 – Rear Differential Lower Housing (Obsolete)



Appendix B23 – AA-20-007 – Rear Differential Lower Housing (Obsolete)



Appendix B24 – AA-20-007 – Rear Differential Lower Housing

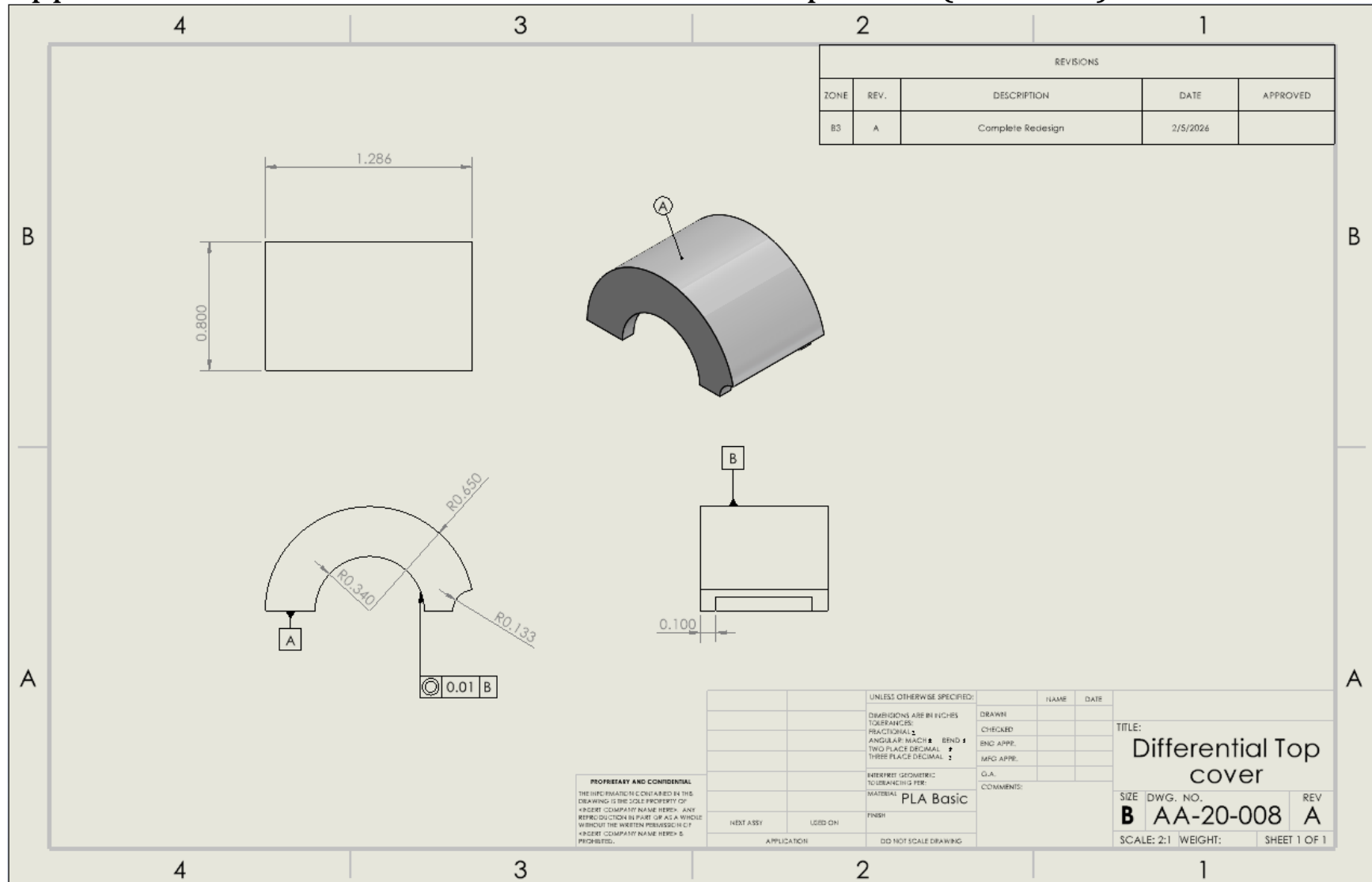


PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF [INSERT COMPANY NAME HERE]. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF [INSERT COMPANY NAME HERE] IS PROHIBITED.

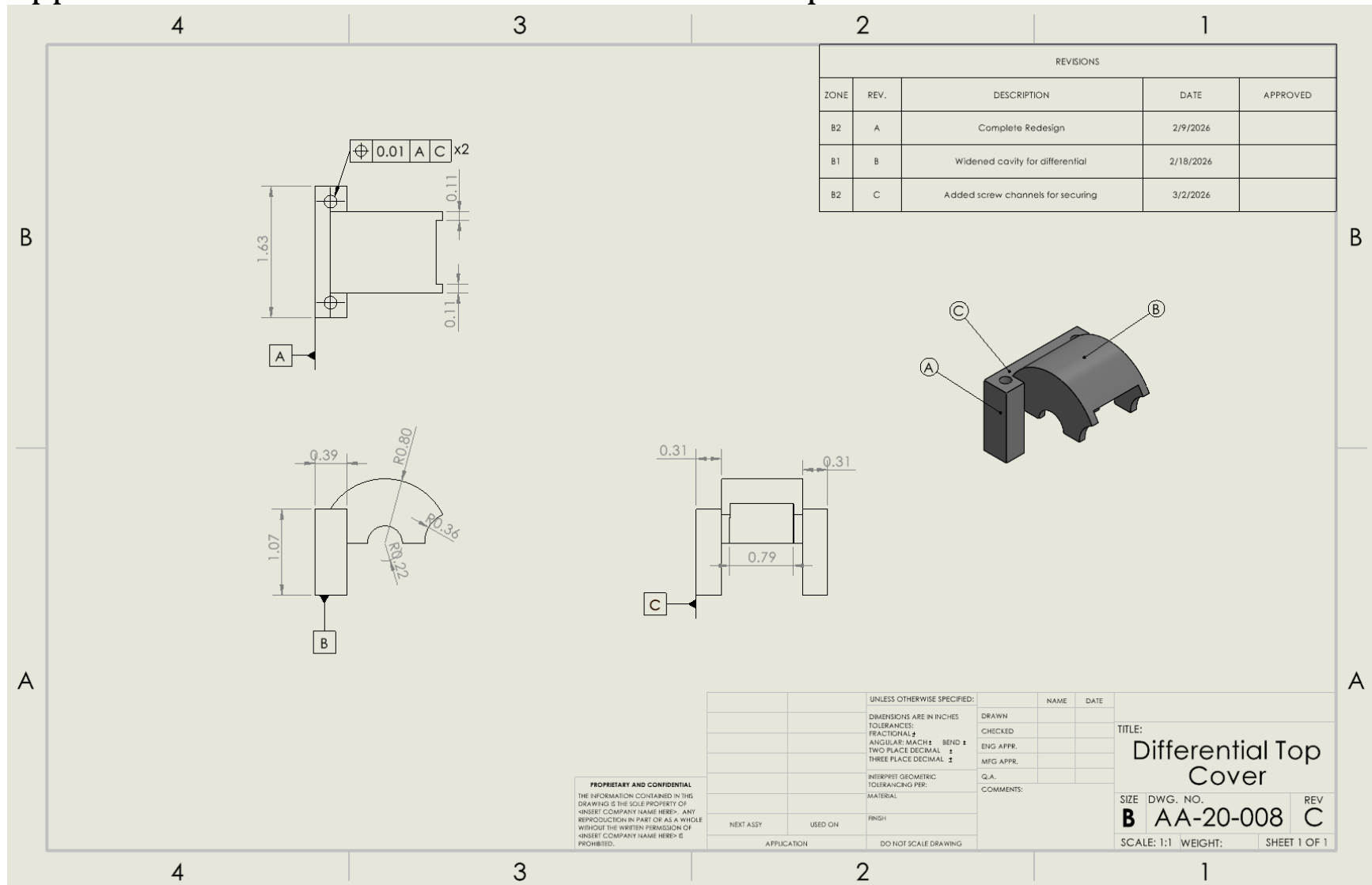
UNLESS OTHERWISE SPECIFIED:		NAME	DATE
DIMENSIONS ARE IN INCHES		Alex	11/12/2025
TOLERANCES:			
FRACTIONAL $\pm$			
ANGULAR MATCH $\pm$ BEND $\pm$			
TWO PLACE DECIMAL $\pm$			
THREE PLACE DECIMAL $\pm$			
INTERPRET GEOMETRIC TOLERANCING PER:			
MATERIAL			
PLA Plastic			
FINISH			
DO NOT SCALE DRAWING			

TITLE:		SIZE	DWG. NO.	REV
Rear Differential Cover		A	AA-20-008	
SCALE: 1:2		WEIGHT:		SHEET 1 OF 1

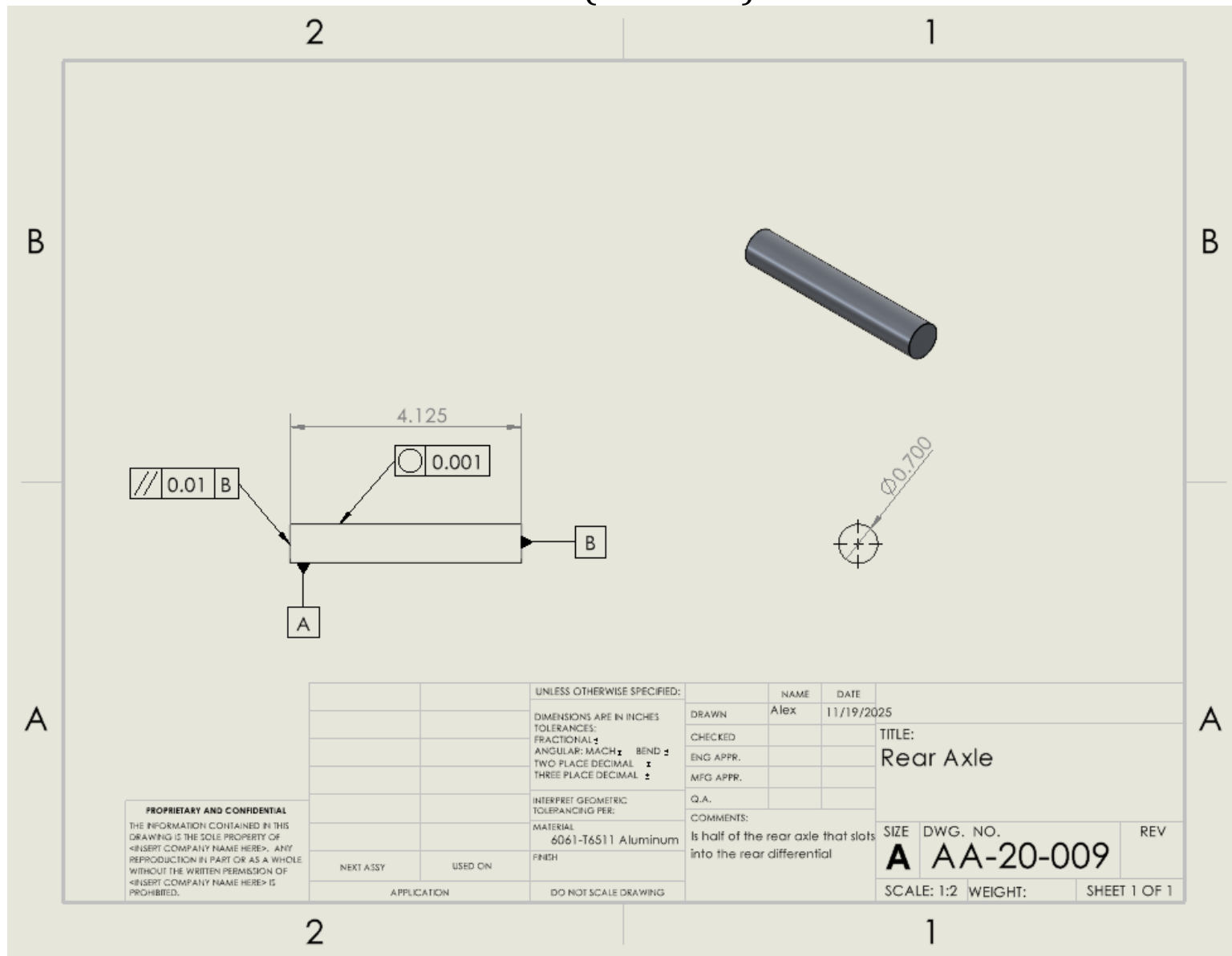
Appendix B26 – AA-20-008 – Rear Differential Top Cover (Obsolete)



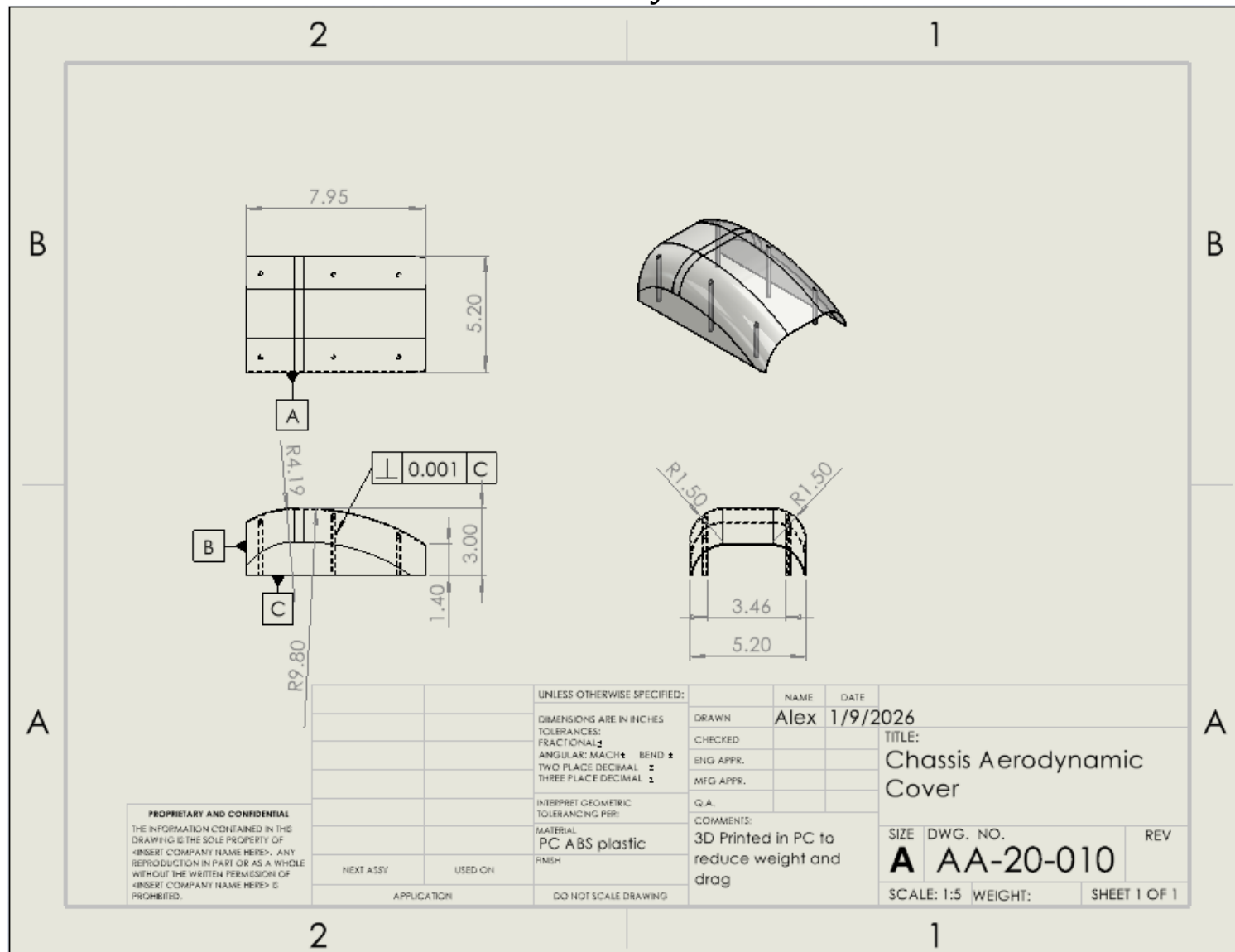
Appendix B27 – AA-20-008 – Rear Differential Top Cover



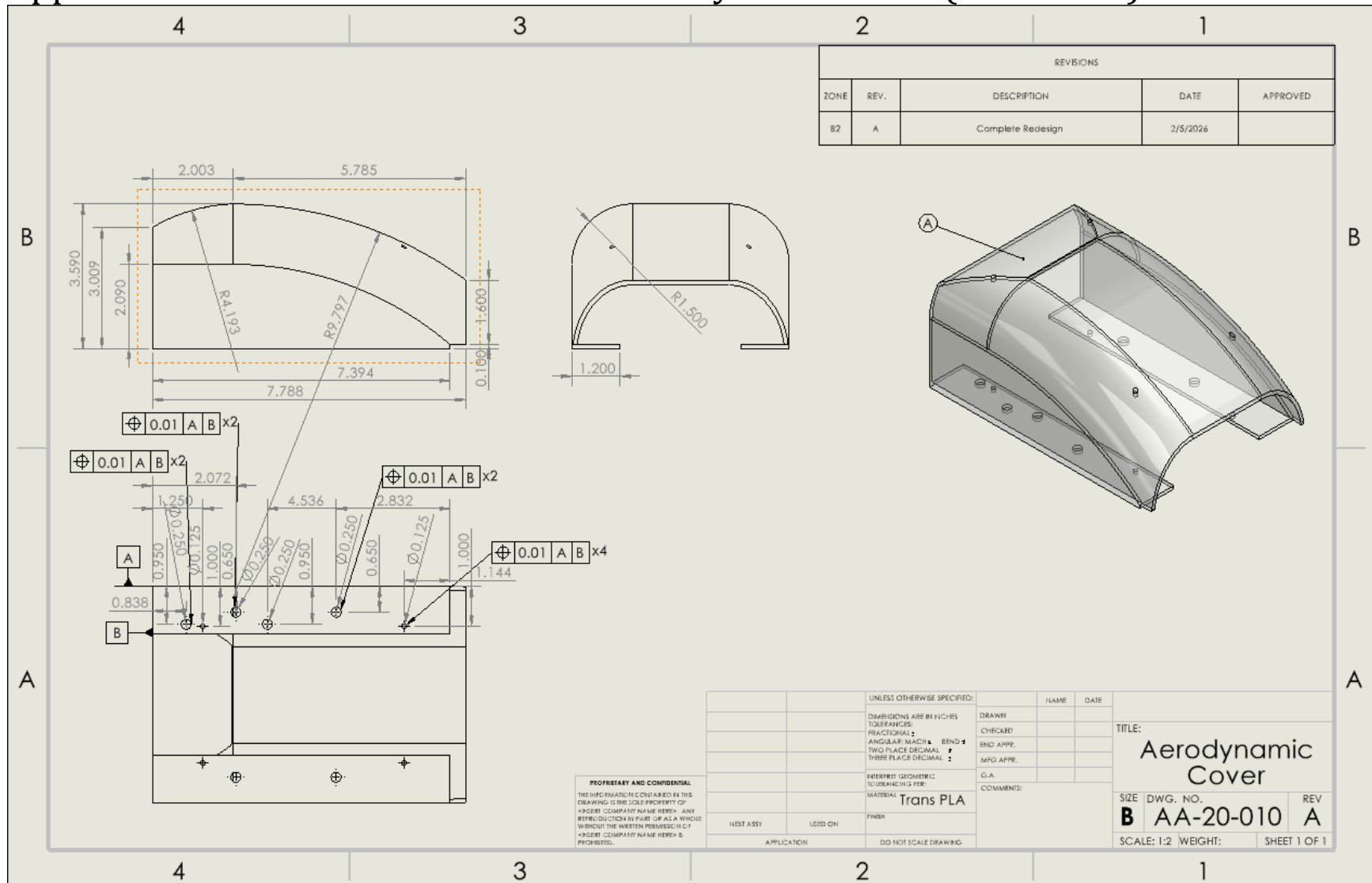
Appendix B28 – AA-20-009 – Rear Axle (Obsolete)



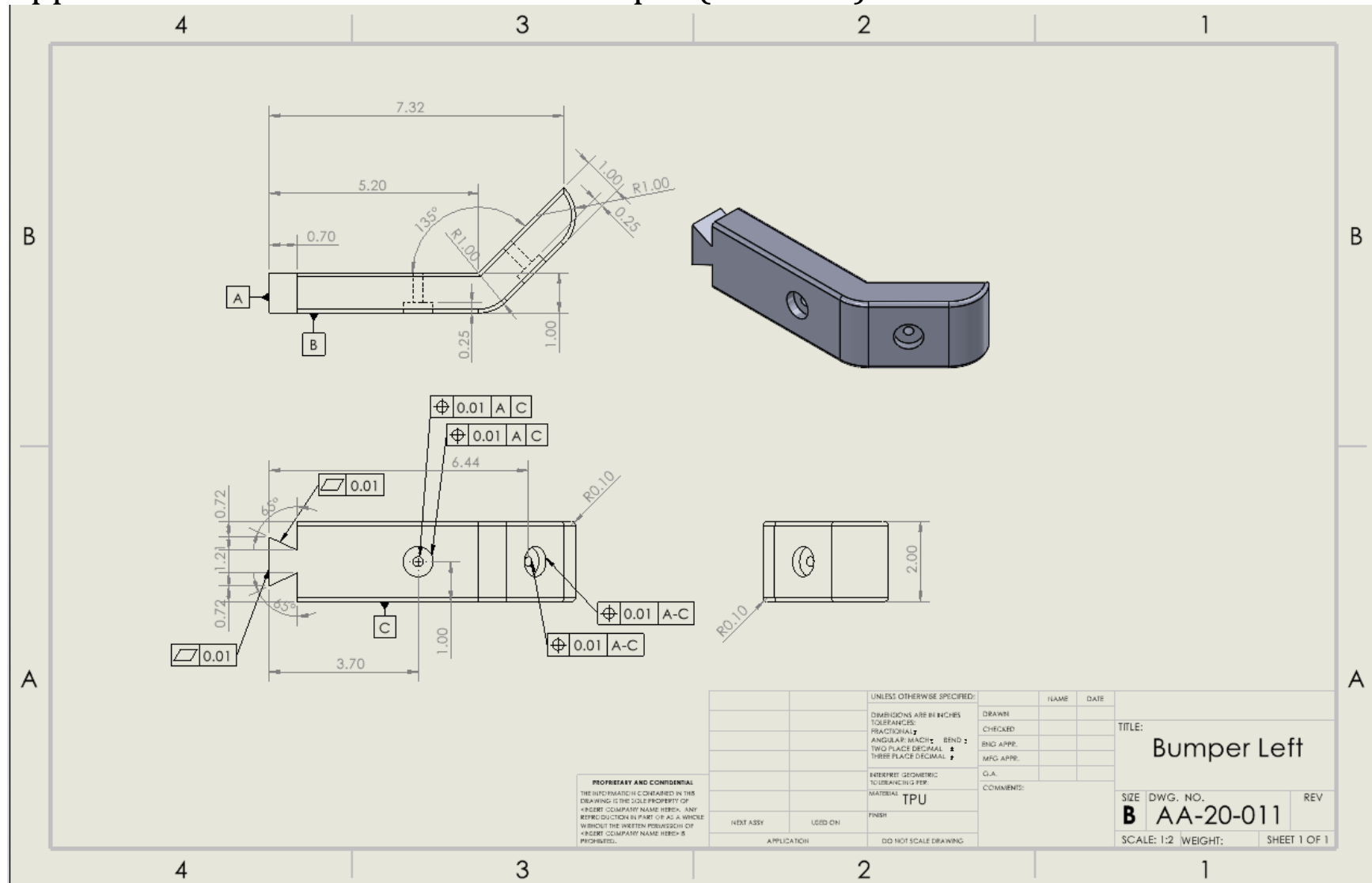
Appendix B29 – AA-20-010 – Chassis Aerodynamic Cover



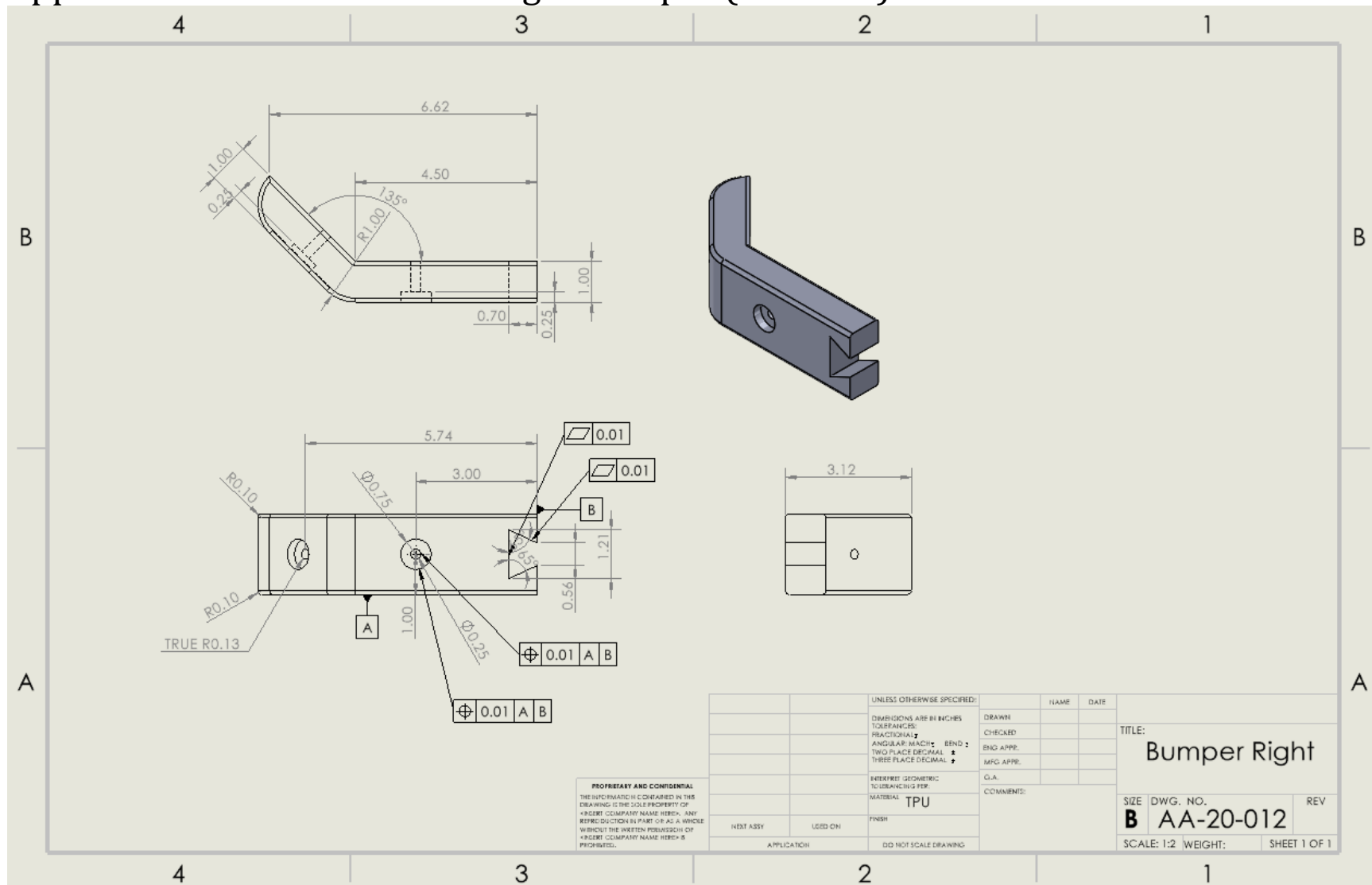
Appendix B30 – AA-20-010 – Chassis Aerodynamic Cover (Revision 1)



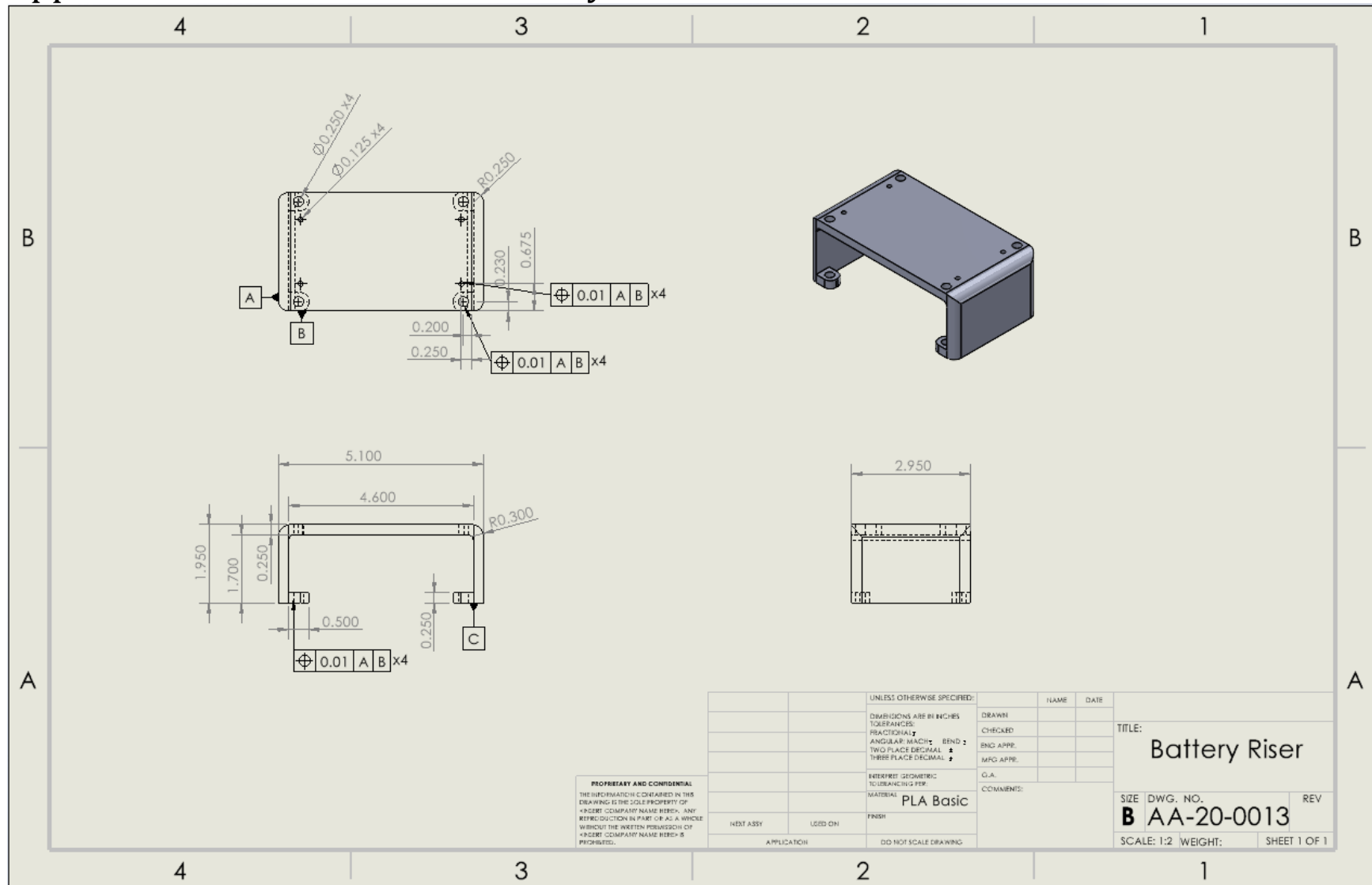
Appendix B31 – AA-20-011 – Left Bumper (Obsolete)



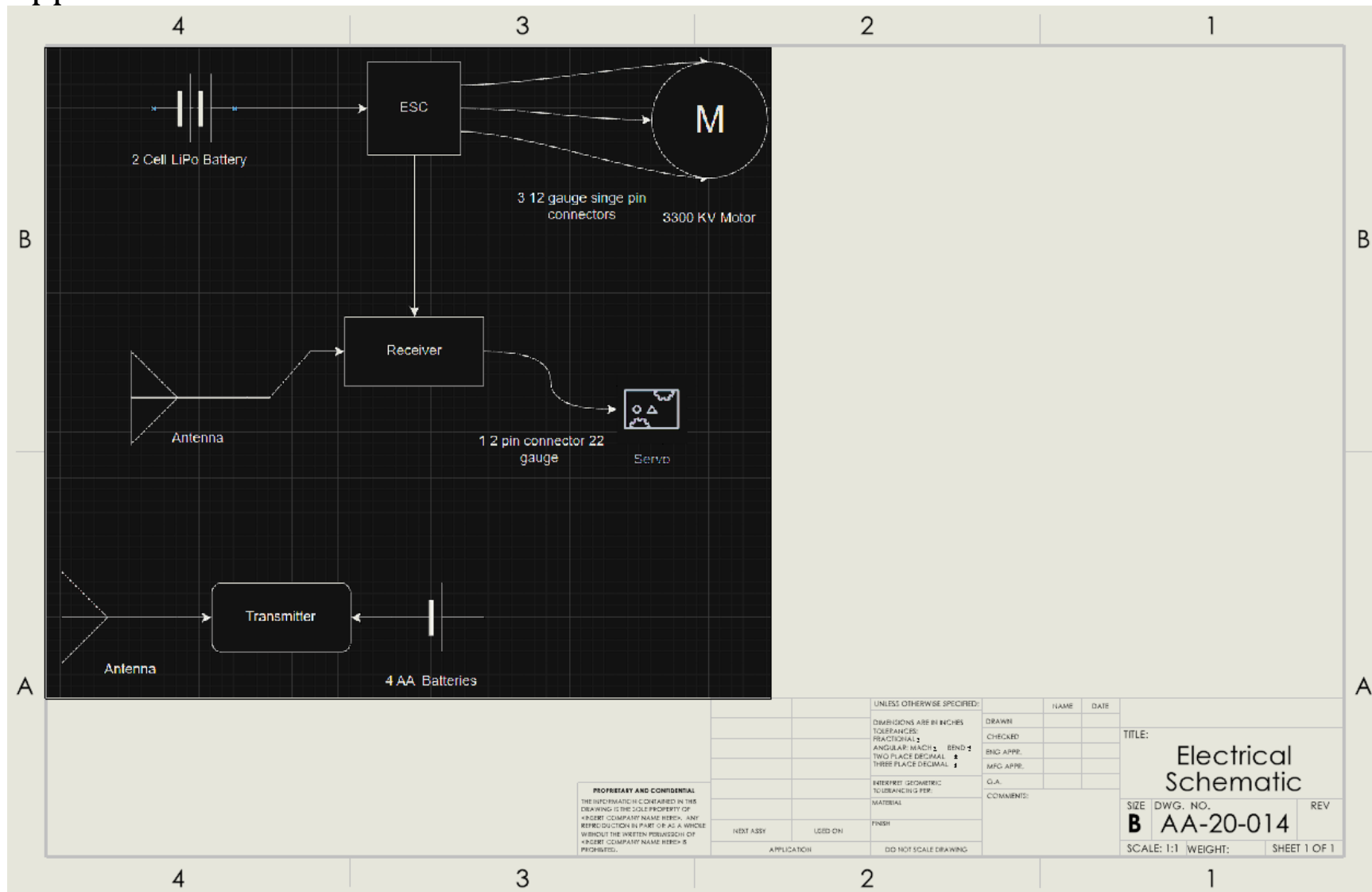
Appendix B32 – AA-20-012 – Right Bumper (obsolete)



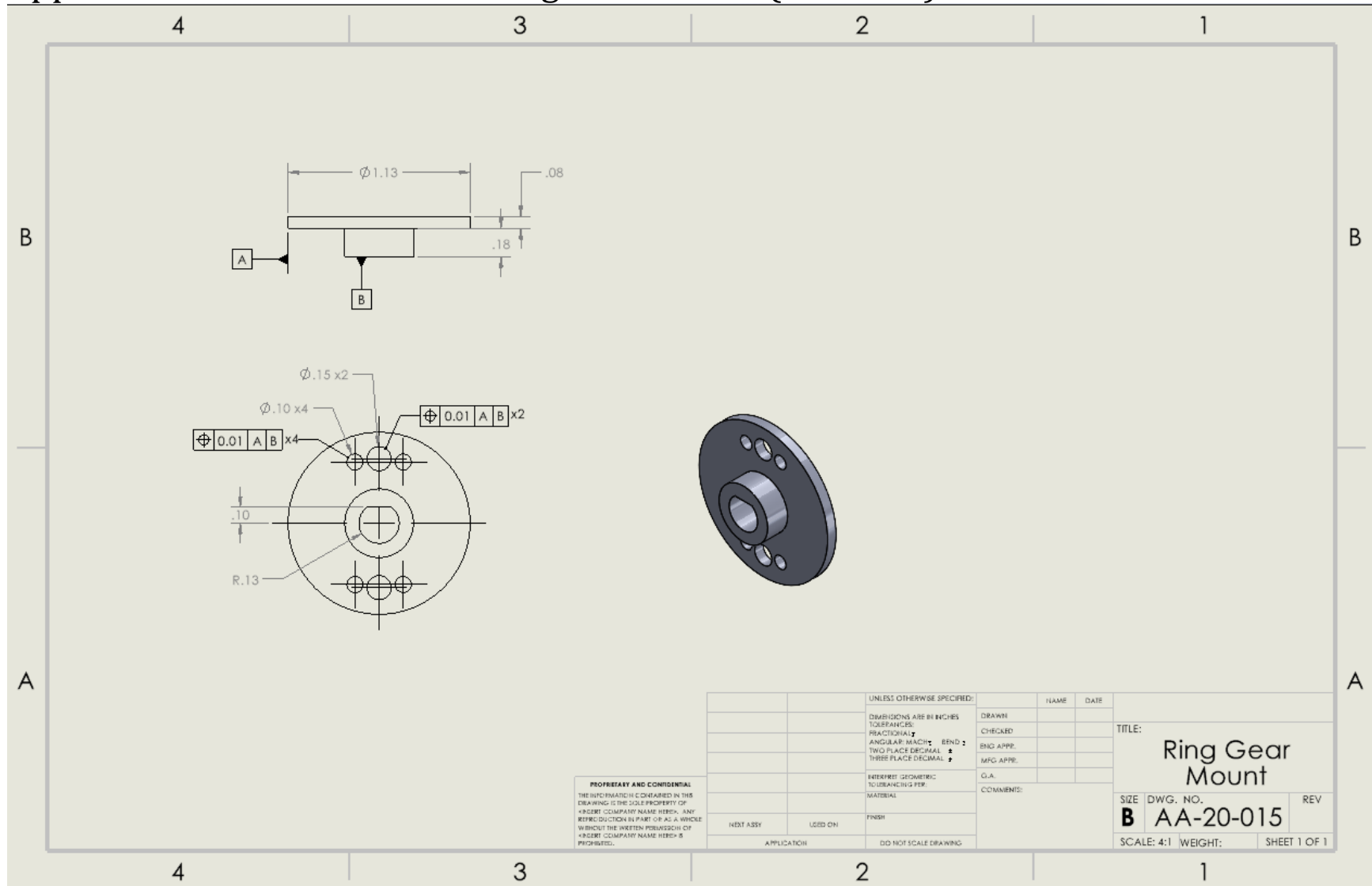
Appendix B33 – AA-20-013 – Battery Riser



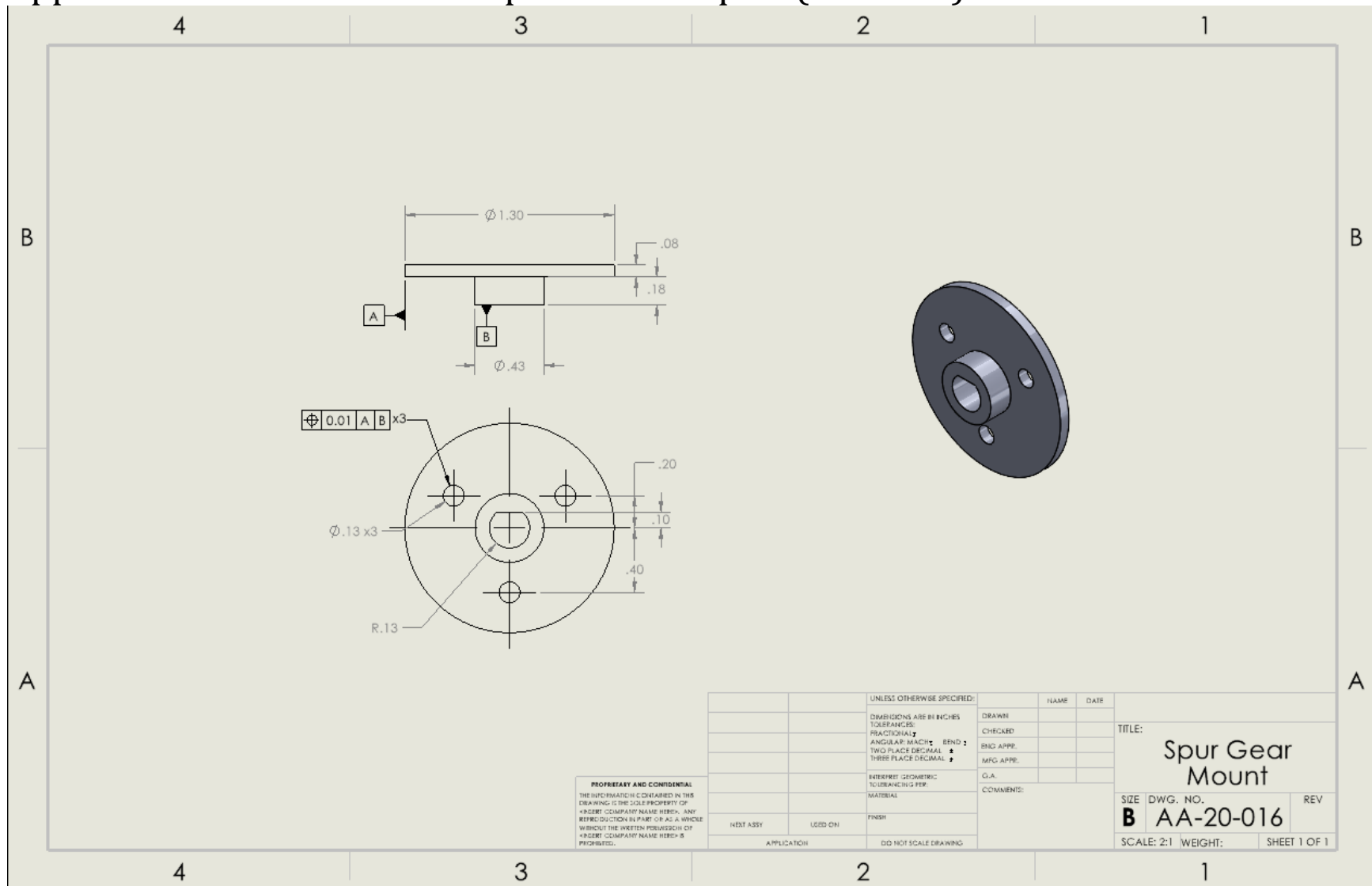
Appendix B34 – AA-20-014 – Electrical Schematic



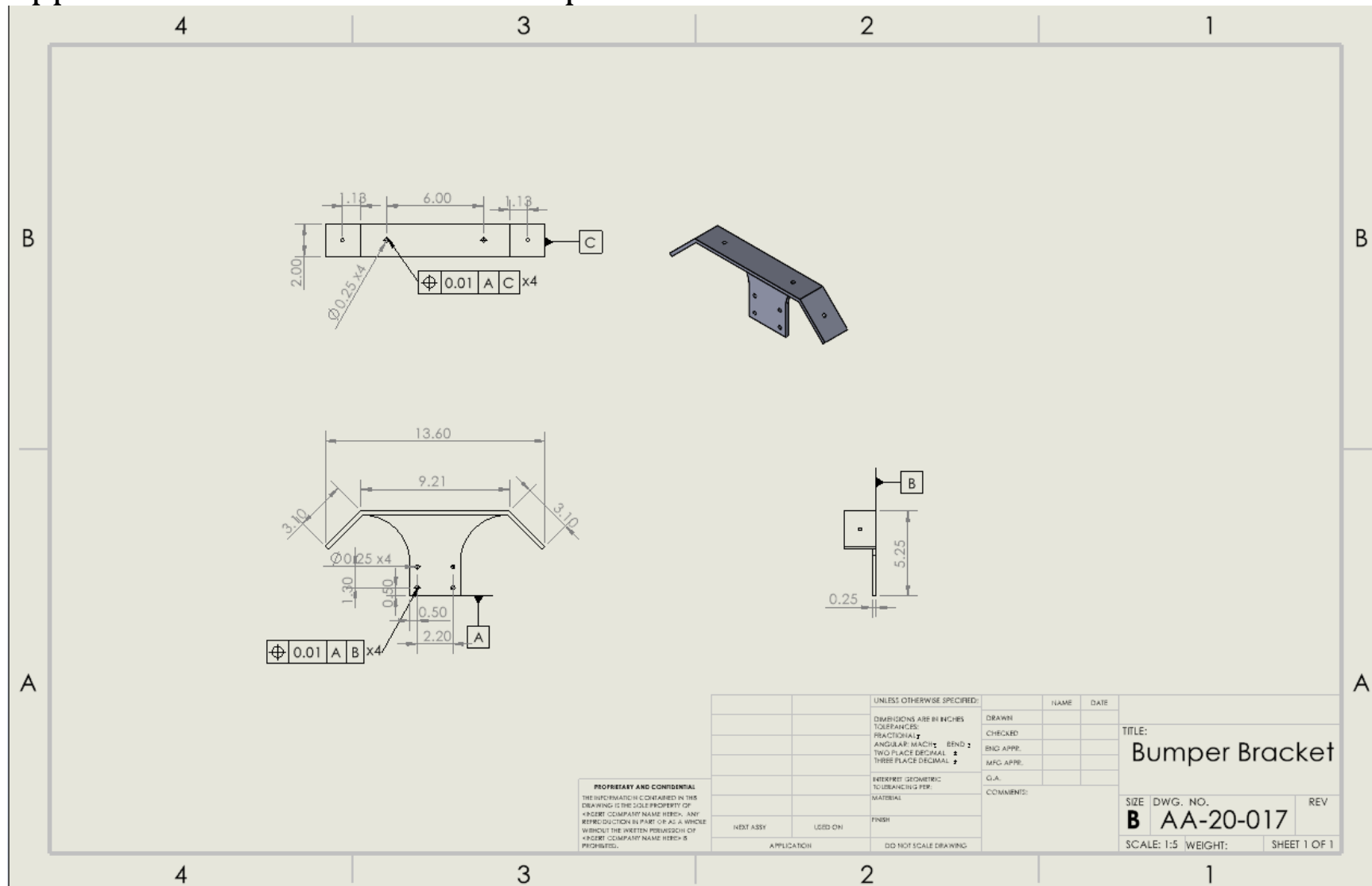
Appendix B35 – AA-20-015 – Ring Gear Mount (Obsolete)



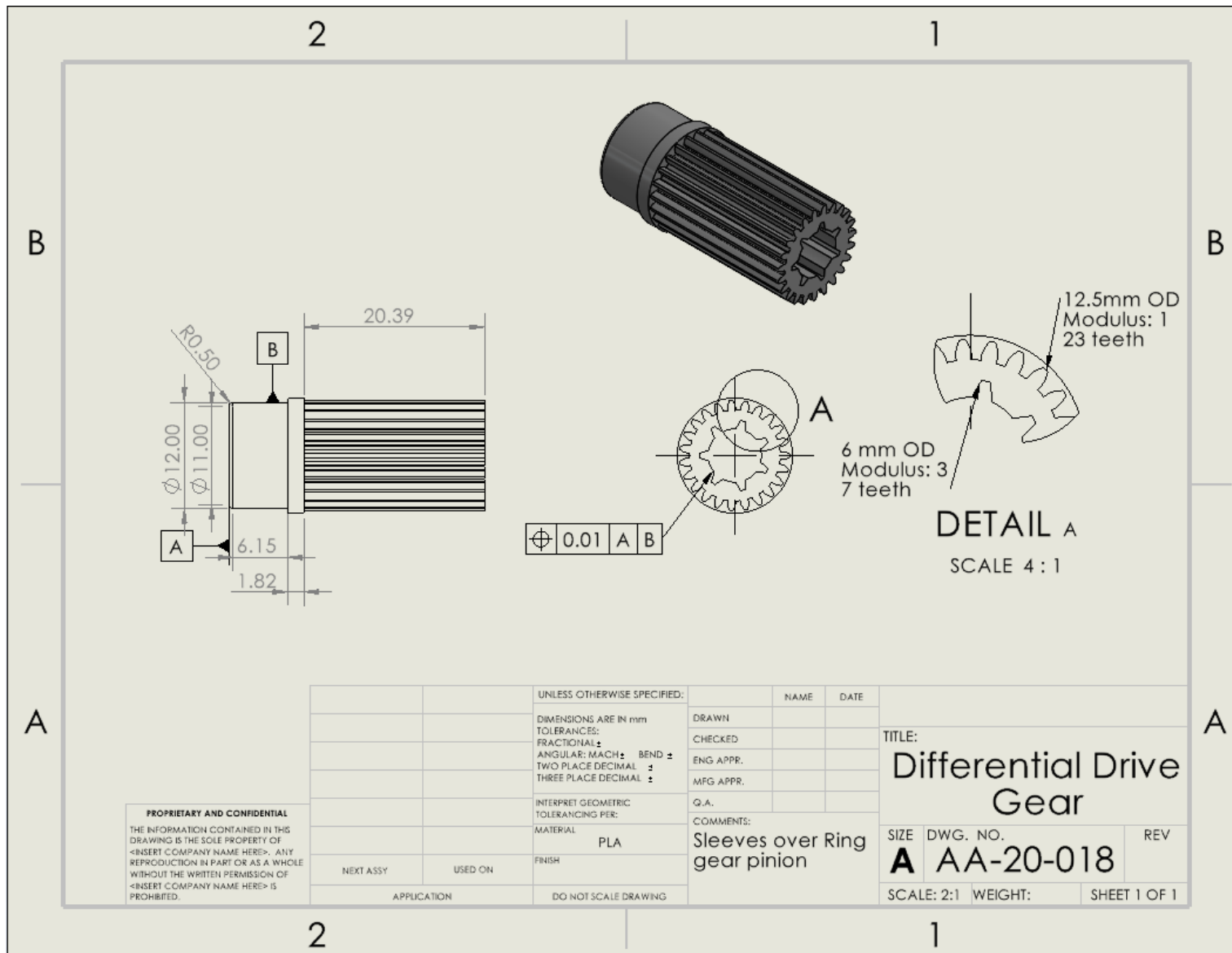
Appendix B36 – AA-20-016 – Spur Gear Adapter (Obsolete)



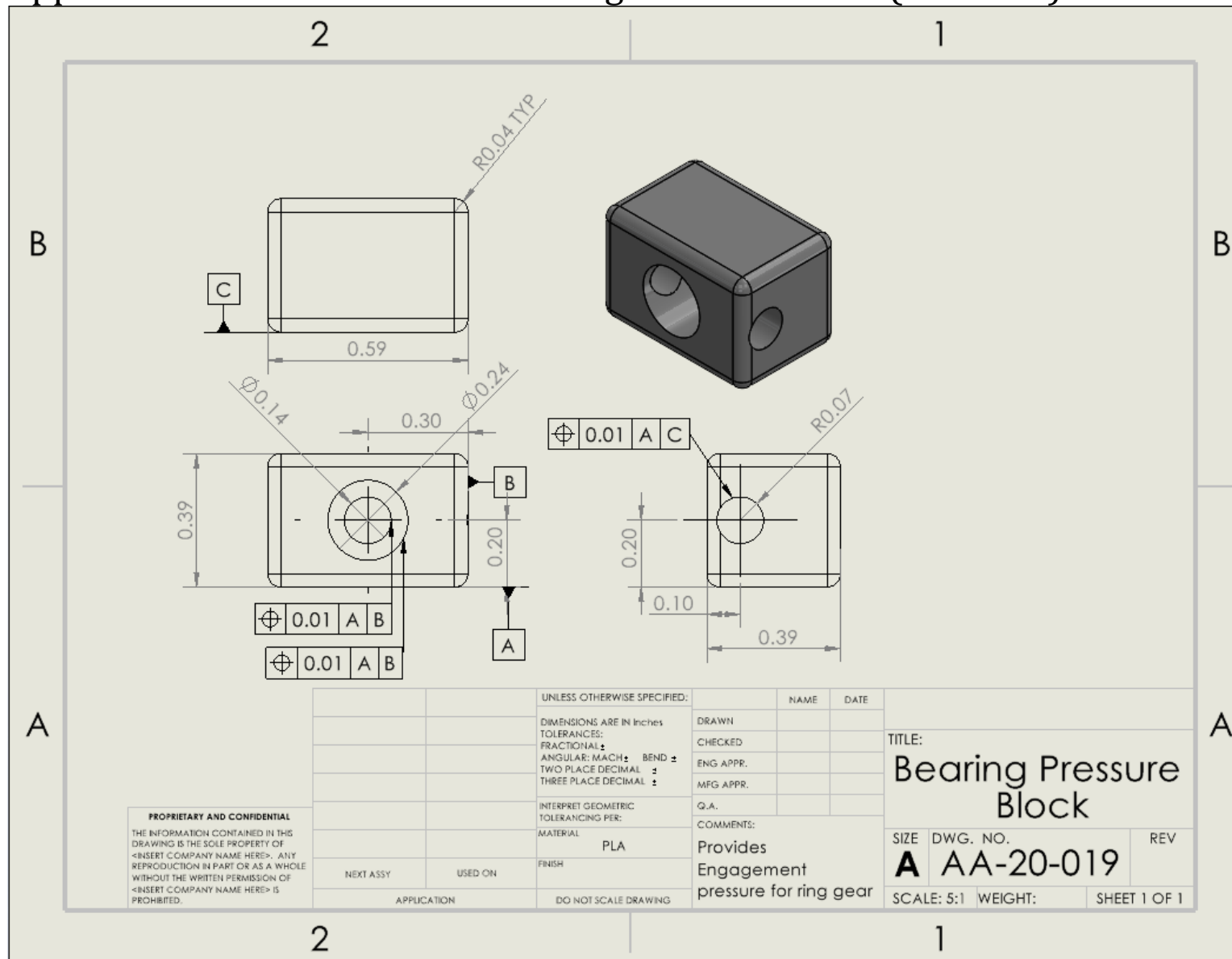
Appendix B37 – AA-20-017 – Bumper Bracket



Appendix B38 – AA-20-018 – Differential Drive Gear



Appendix B39 – AA-20-019 – Bearing Pressure Block (Obsolete)



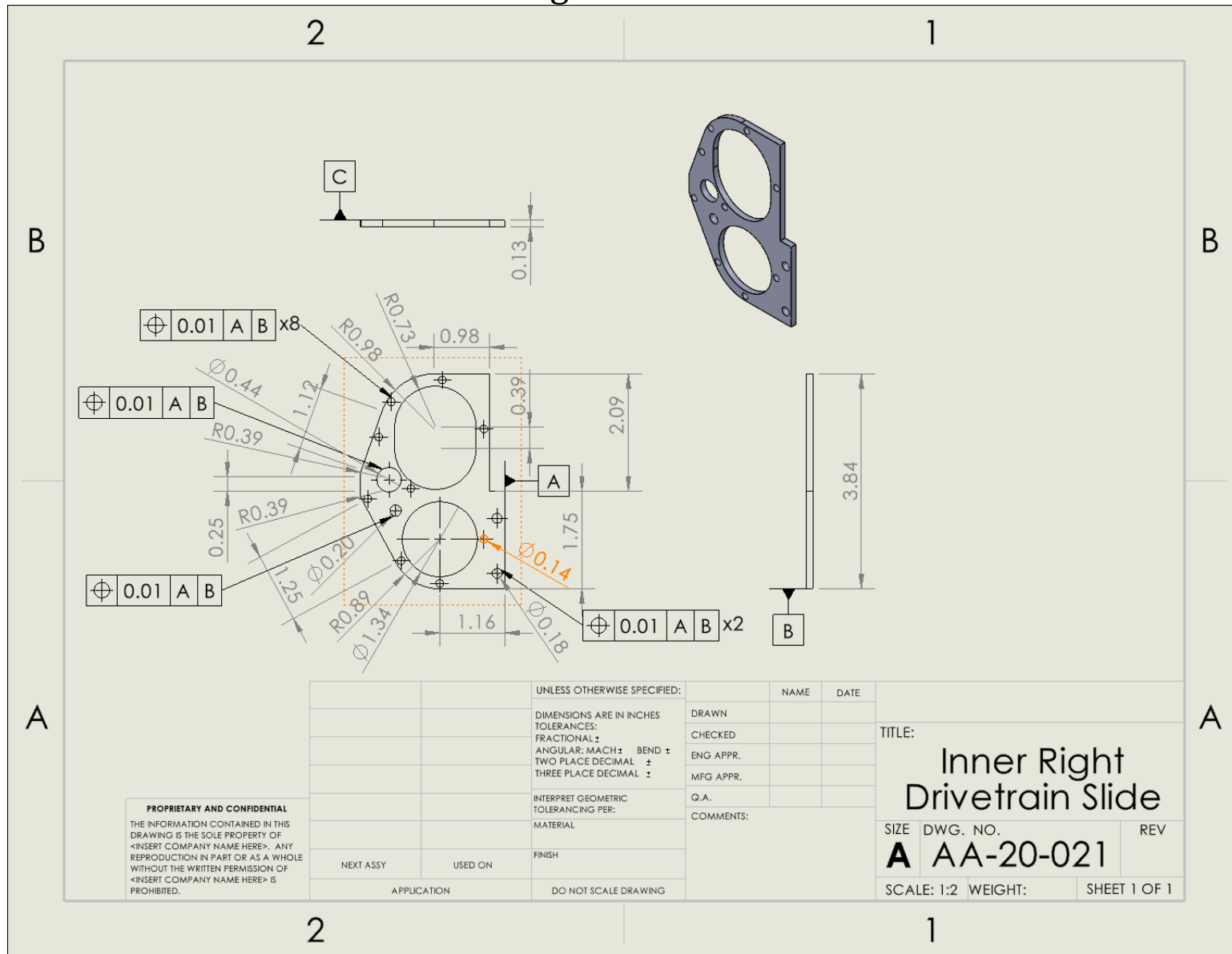
2 1

B B

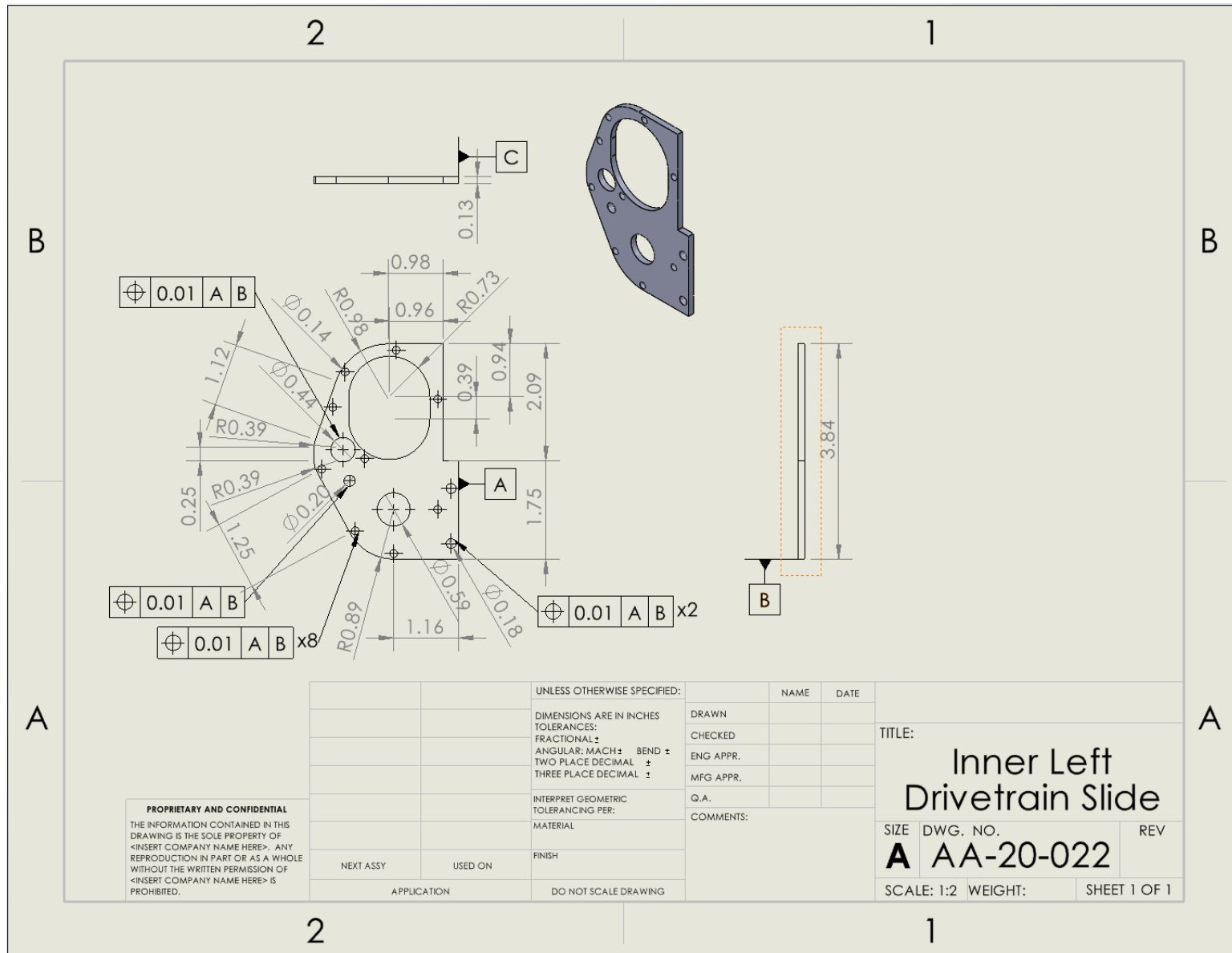
A A

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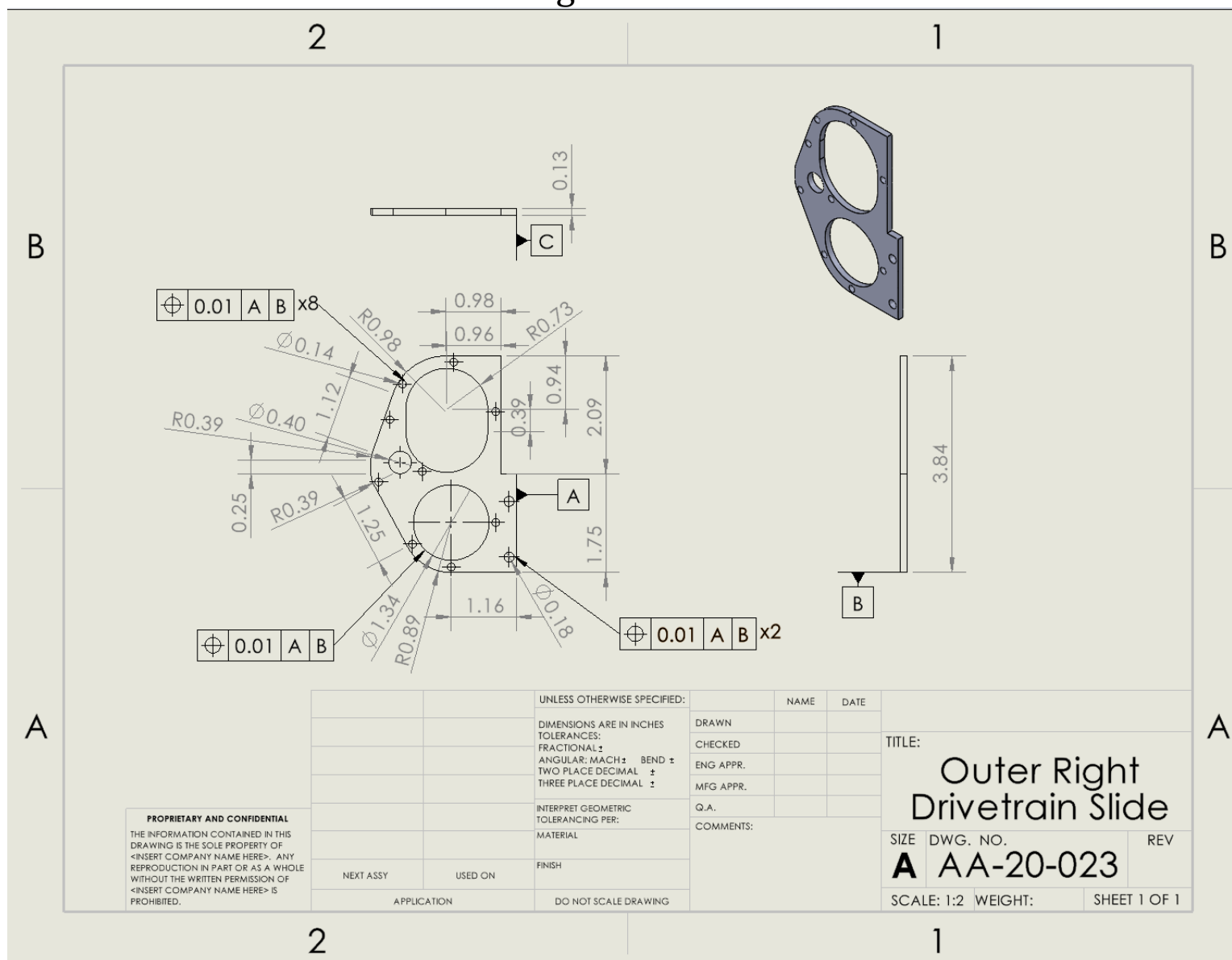
Appendix B41 – AA-20-021 – Inner Right Differential Slide



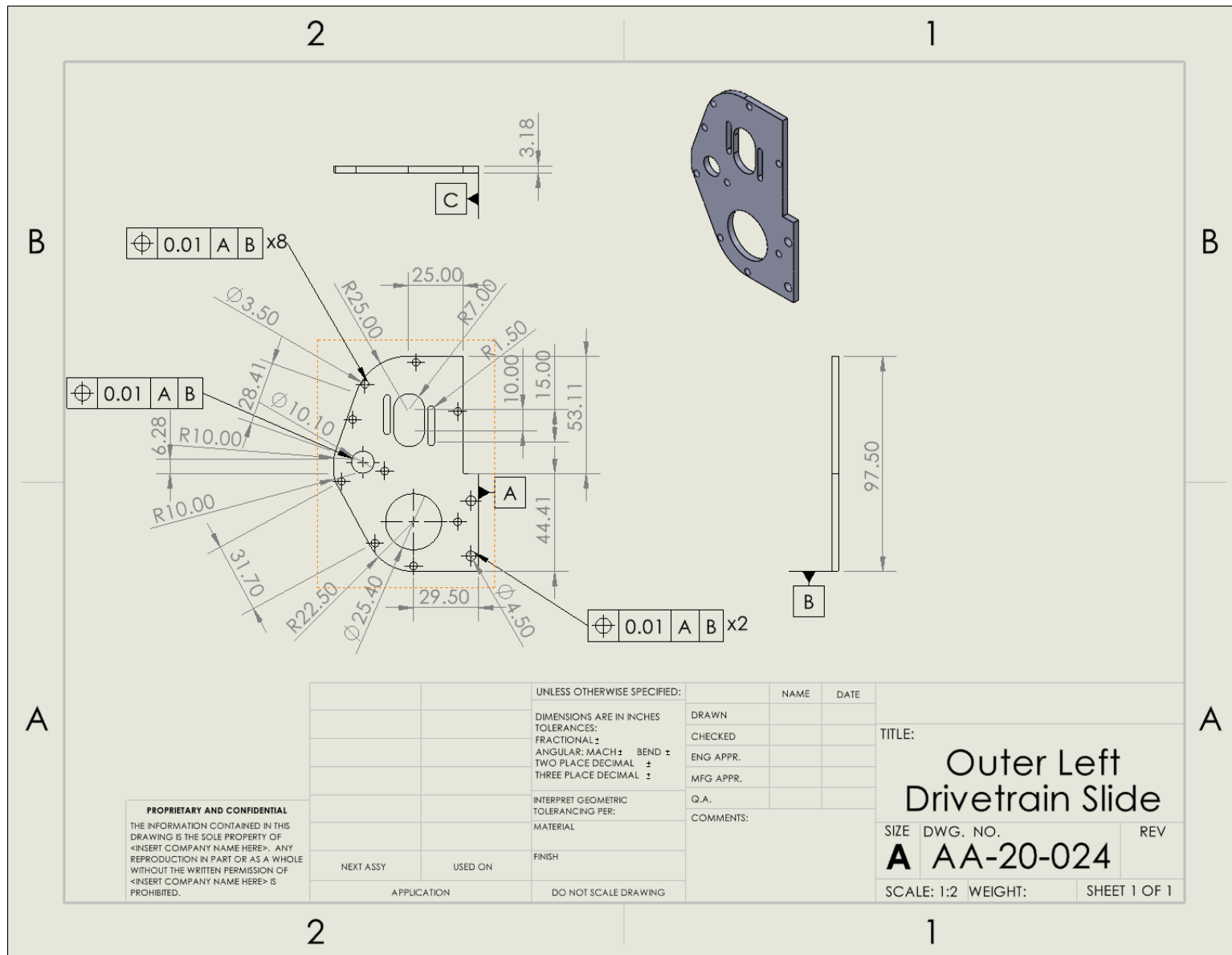
Appendix B42 – AA-20-022 – Inner Left Differential Slide



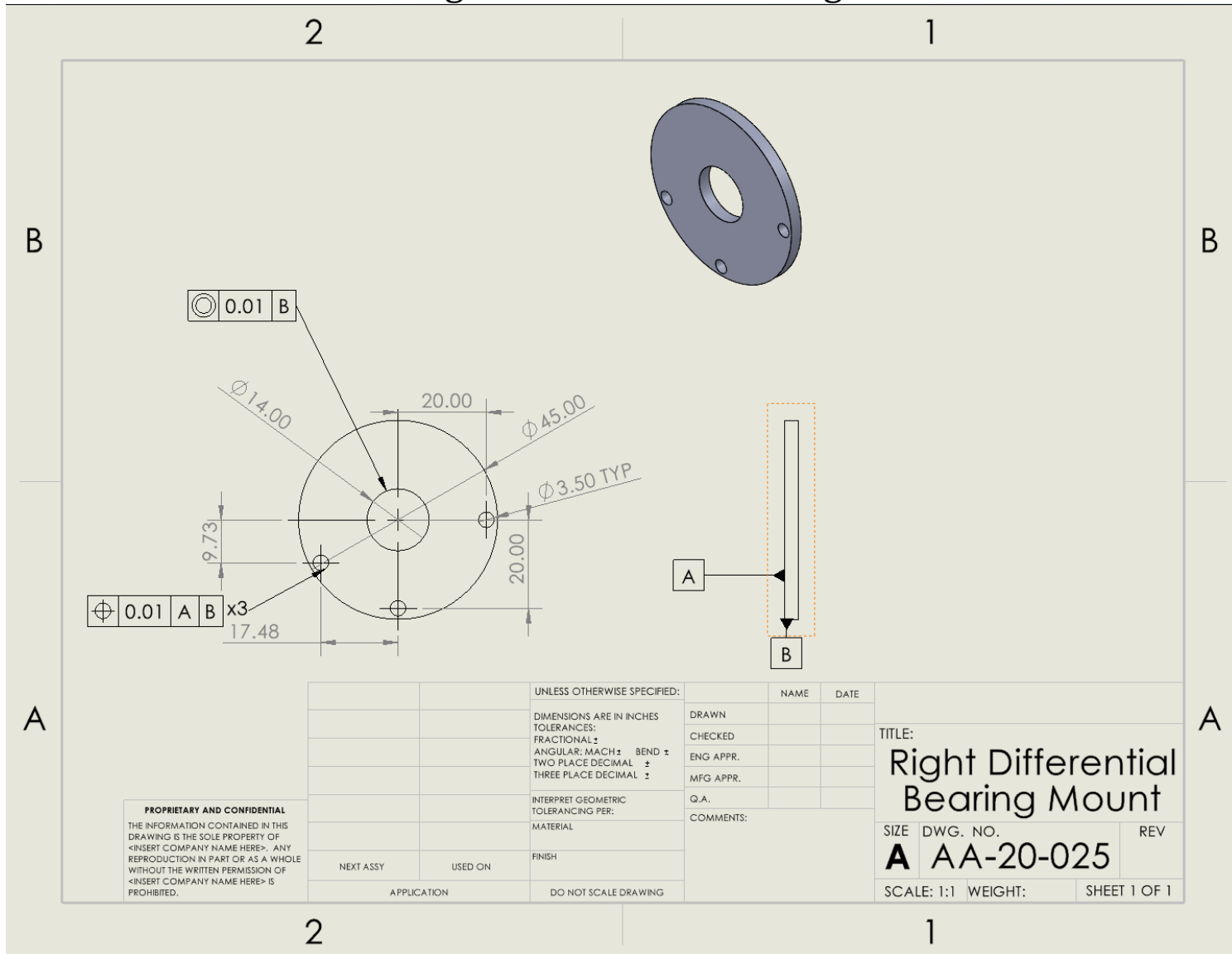
Appendix B43 – AA-20-023 – Outer Right Differential Slide



Appendix B44 – AA-20-024 – Outer Left Differential Slide



Appendix B45 – AA-20-025 – Right Differential Bearing Mount



APPENDIX C – Parts List and Costs

Table C1. Parts List

Part Number	Qty	Part Description	Source	Cost	Disposition
20-001	1	Chassis Frame	Made in house	~\$10*	To be made 1/12/2026
20-002	1	Driveshaft	Made in house	~\$6*	To be made 1/19/2026
20-003	2	Rear Axle	Traxxas	\$10	To be made 1/19/2026
20-004	1	Motor Housing	Made in house	\$8.59**	To be made 1/6/2026
20-005	1	Battery Housing	Made in house	\$3.29**	Made 1/31/2026
20-007	1	Differential lower housing	Made in house	\$2.19**	Made 1/31/2026
20-008	1	Differential top cover	Made in house	\$2.19**	Made 2/7/2026
20-010	1	Aerodynamic cover	Made in house	\$15.27**	Made 2/5/2026
20-011	1	Left Bumper	Made in house	\$14.31**	Made 2/1/2026
20-012	1	Right Bumper	Made in house	\$14.31**	Made 2/1/2026
20-013	1	Battery Riser	Made in house	\$5.27**	Made 1/31/2026
55-001	1	Brushless BL-2s 3300 KV motor + ESC	Traxxas	\$99.99	To be ordered during winter quarter
55-003	5	Pinion and Spur Gears for Drive Train	Traxxas	\$30	To be order during winter quarter
			Cost Total:	\$ 221.41	
			Total Parts:	18	

Notes:

*: about the price of metal used

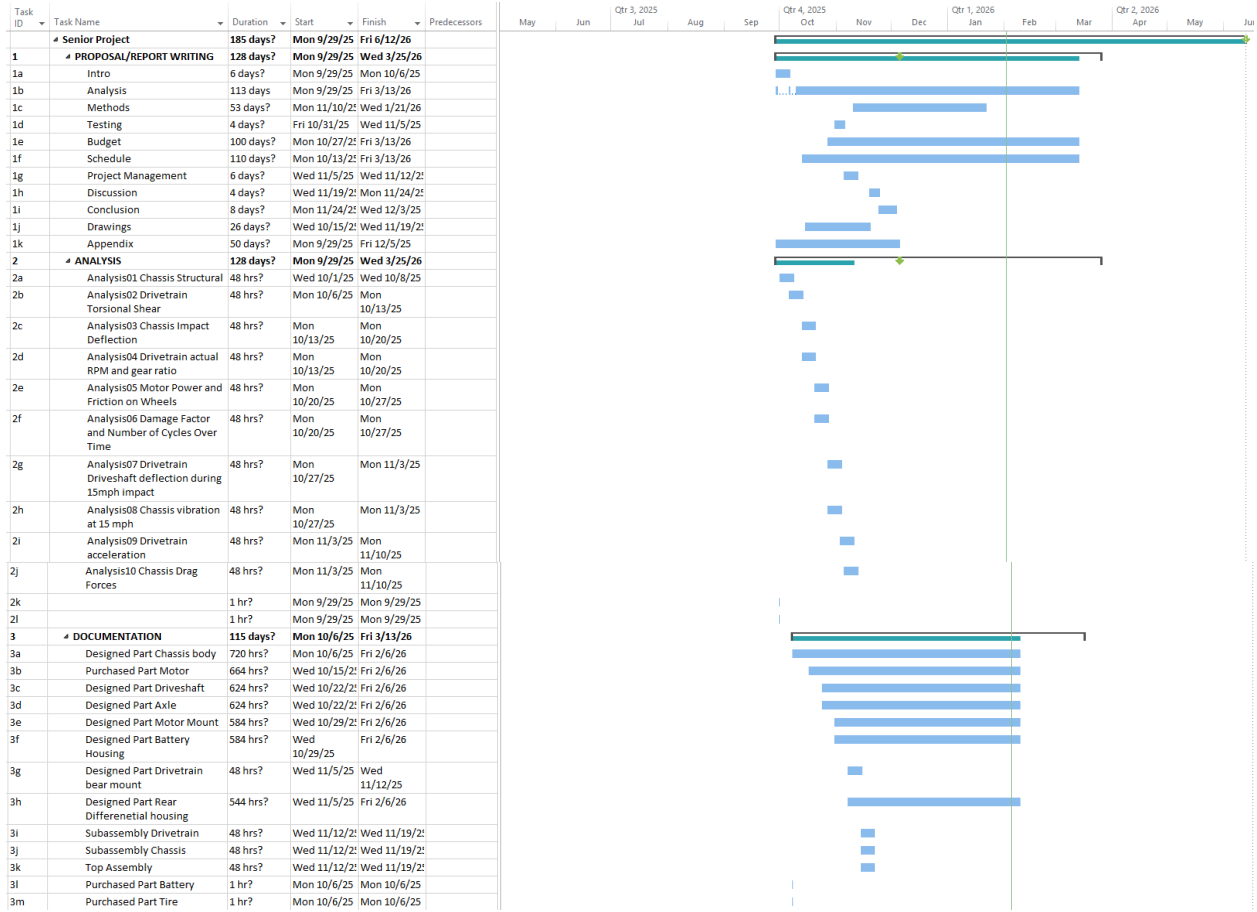
**: The cost of Filament used

APPENDIX D – Budget

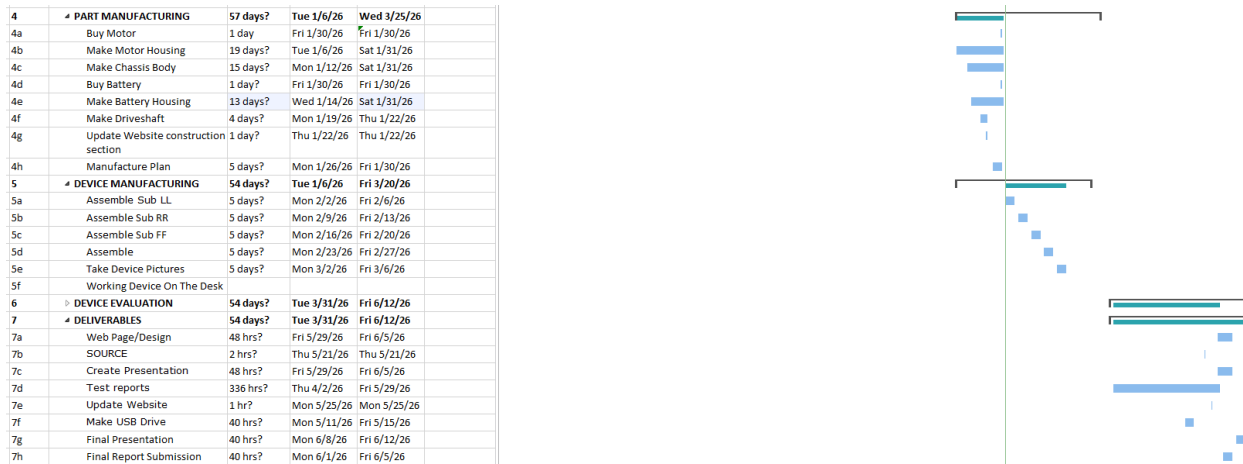
Table D1. Project Budget.

Item	Qty	Description	Cost
.125inx2ftx2ft 6061-T6 Aluminum	1	From Metalsdepot.com Used for part 20-001 and 20-004	\$110.84
¾inx36ft Aluminum Round Bar 2024-T351	1	From Oninemetals.com Used for part 20-002 and 20-003	\$37.23
Brushless BL-2s 3300 KV motor + ESC	1	Traxxas	\$99.99
Partner Budget	5	List of purchased budget items from RC Baja project partner	\$192.49
			Total: \$440.55
Manufacturing Labor	4 hours	Spent on manufacturing	\$48
Partner Labor	7 hours	Spent on manufacturing	\$200
			Total: \$248

APPENDIX E - Schedule

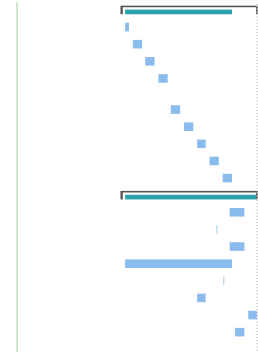


Fall Quarter Gantt Chart



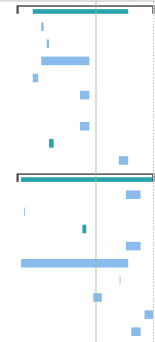
Winter Quarter

6	➤ DEVICE EVALUATION	54 days?	Tue 3/31/26	Fri 6/12/26	
6a	Deflection Test	2 days?	Thu 4/2/26	Fri 4/3/26	
6b	Acceleration Testing	5 days?	Mon 4/6/26	Fri 4/10/26	
6c	Friction Testing	5 days?	Mon 4/13/26	Fri 4/17/26	
6d	Sudden stop and Start Testing	5 days?	Mon 4/20/26	Fri 4/24/26	
6e		5 days?	Mon 4/27/26	Fri 5/1/26	
6f		5 days?	Mon 5/4/26	Fri 5/8/26	
6g		5 days?	Mon 5/11/26	Fri 5/15/26	
6h		5 days?	Mon 5/18/26	Fri 5/22/26	
6i	Take Device Photos	5 days?	Mon 5/25/26	Fri 5/29/26	
7	➤ DELIVERABLES	54 days?	Tue 3/31/26	Fri 6/12/26	
7a	Web Page/Design	48 hrs?	Fri 5/29/26	Fri 6/5/26	
7b	SOURCE	2 hrs?	Thu 5/21/26	Thu 5/21/26	
7c	Create Presentation	48 hrs?	Fri 5/29/26	Fri 6/5/26	
7d	Test reports	336 hrs?	Thu 4/2/26	Fri 5/29/26	
7e	Update Website	1 hr?	Mon 5/25/26	Mon 5/25/26	
7f	Make USB Drive	40 hrs?	Mon 5/11/26	Fri 5/15/26	
7g	Final Presentation	40 hrs?	Mon 6/8/26	Fri 6/12/26	
7h	Final Report Submission	40 hrs?	Mon 6/1/26	Fri 6/5/26	



Spring Quarter

Task ID	Task Name	Duration	Start	Finish	Predecessors	May	Jun	Qtr 3, 2025	Aug	Sep	Qtr 4, 2025	Nov	Dec	Qtr 1, 2026	Jan	Feb	Mar	Qtr 2, 2026	Apr	May	Ju
6	➤ DEVICE EVALUATION	54 days?	Tue 3/31/26	Fri 6/12/26																	
6a	Deflection Test	1 day?	Mon 4/13/26	Mon 4/13/26																	
6b	Acceleration Testing	1 day?	Thu 4/16/26	Thu 4/16/26																	
6c	Friction Testing	20 days?	Mon 4/13/26	Fri 5/8/26																	
6d	Deflection Testing Procedure	3 days?	Wed 4/8/26	Fri 4/10/26																	
6e	Acceleration Testing Procedure	5 days?	Mon 5/4/26	Fri 5/8/26																	
6f	Friction Testing Procedure	5 days?	Mon 5/4/26	Fri 5/8/26																	
6g	RC Baja Competition	1 day	Sat 4/18/26	Sat 4/18/26																	
6h	Take Device Photos	5 days?	Mon 5/25/26	Fri 5/29/26																	
7	➤ DELIVERABLES	54 days?	Tue 3/31/26	Fri 6/12/26																	
7a	Web Page/Design	48 hrs?	Fri 5/29/26	Fri 6/5/26																	
7b	SOURCE Sign Up	2 hrs?	Fri 4/3/26	Fri 4/3/26																	
7c	SOURCE POSTER	1 day	Wed 5/6/26	Wed 5/6/26																	
7d	Create Presentation	48 hrs?	Fri 5/29/26	Fri 6/5/26																	
7e	Test reports	336 hrs?	Thu 4/2/26	Fri 5/29/26																	
7f	Update Website	1 hr?	Mon 5/25/26	Mon 5/25/26																	
7g	Make USB Drive	40 hrs?	Mon 5/11/26	Fri 5/15/26																	
7h	Final Presentation	40 hrs?	Mon 6/8/26	Fri 6/12/26																	
7i	Final Report Submission	40 hrs?	Mon 6/1/26	Fri 6/5/26																	



APPENDIX F – Expertise and Resources

Appendix F01 – Drivetrain Decision Matrix

Criterion	Weight 1 to 3	Best Possible 3	Front Wheel Score x Wt	Rear Wheel Score x Wt	All Wheel Score x Wt
Cost	2	6	1	2	3
Weight	3	9	2	6	2
Manufacturability	3	9	2	6	3
Drivetrain Impact Resistance	1	3	2	3	2
Chassis Impact Resistance	2	6	2	4	1
Drivetrain Speed	3	9	1	3	2
Controlability	3	9	2	6	3

Appendix F02 – Metal Construction Method Decision Matrix

Criterion	Weight 1 to 3	Best Possible 3	Laser Cut	Score x Wt CNC	Score x Wt Mix of Both	Score x Wt
Cost	1	3	1	1	1	2
Weight	2	6	2	4	2	4
Able to get to spec	3	9	3	9	6	9
Postprocessing	3	9	1	3	6	6
Time Spent on manufacturing	2	6	2	4	2	4

Appendix F03 – Plastic Construction Method Decision Matrix

Criterion	Weight 1 to 3	Best Possible 3	3D printing	Score x Wt	Vaccum	Score x Wt	Heat mold	Score x Wt
Cost	1	3	2	2	2	2	2	2
Weight	2	6	3	6	2	4	2	4
Percision of Tolerance	3	9	3	9	1	3	1	3
Machine Availability	2	6	3	6	1	2	1	2
Material Availability	3	9	2	6	1	3	1	3

Appendix F04 – Chassis Material Selection Decision Matrix

Criterion	Weight 1 to 3	Best Possible 3	Aluminum	Score x Wt	PLA	Score x Wt	ABS	Score x Wt
cost	2	6	1	2	2	4	1	2
Weight	3	9	2	6	2	6	2	6
Availability	2	6	3	6	1	2	1	2
Strength	3	9	3	9	2	6	3	9
Torsional Resistance	3	9	2	6	1	3	1	3
					</			

APPENDIX G – Testing Report

Appendix G1 Head-on impact Testing

Introduction

This test was done to confirm test requirement 1d.2, the chassis must deflect less than 0.6 inches due to a head on impact at 15 mph. The parameter of interest for this test is the deflection of the chassis. This test uses a camera and to see exactly where the deflection starts and ends then using a caliper to find the total deflection. This test is predicted to have a total deflection of 0.6 inches from normal. This is based on calculations done in Appendix A03. The schedule for this test can be seen in Appendix E as well as at the end of this document.

Method/Approach

This test used mostly hardware and a camera to be able to see the test data for this test. Tape was used to line up the car to be able to get as close as possible to a head-on impact. The test involves drive the car to reach 15mph then slamming it into a curb using a long line of tape to line up the car at the end. The data was written down on a sheet of paper during the test, then was later reported in a table that can be seen in Appendix G1.3 later in this document. The speedometer used for this test was inaccurate, as such a range of speeds was accepted for the speed of impact. The range used was plus or minus 2 mph.

Test Procedure

Summary:

This Procedure documents the set-up, process, and recording of the destructive crash testing. The idea of doing this test before other tests was to get the most destructive test done first to allow for time to reprint or change the car's design, should the car not survive the entire testing process. This test was done to test the bumper and the chassis deflection to a head on impact. This test met the testable requirement for requirement 2 in section 1d.



Figure 1. Testing set-up

Time: This test was conducted on Friday April 10, 2026, from 2:30 pm to 4:30 pm outside of Wendell Hill Hall. There a period of a half hour was dedicated to collecting equipment and tracking out a path of where the test will take place, as shown in figure 1 above. There were

two plans for after the test is conducted. First if the car didn't survive, there was to be a half hour of assessment of damage and assessing if parts needed to be reconstructed and reprinted, and second If the car survives this test, there will be a half hour of assessment of why it survived.

Place: Outside of Wendell Hill Hall, Central Washington University campus in Ellensburg, WA

Required Equipment includes:

- Tron-Car
- Camera
- Speedometer (app on cellphone)
- A Curb
- RC Transmitter (Controller)
- A Charged Battery
- Blue Tape
- Measuring Square

Risk: This test was a destructive test, so there were a few risks that came with it. The first risk is that there was a possibility of plastic or metal being expelled after impact. To mitigate this risk, safety glasses must be worn throughout the entire test. The second risk is that the speedometer might not have been accurate, so the test had a possibility of not being done at the correct speed, either over or under the testing speed. To mitigate this, the testing speed had a range (+ or – 2 mph) of acceptable speed values. The third risk in this test is that the test is done in a public environment, therefore pedestrians and other vehicles could have entered the testing zone. The final risk involved in this test is that the camera could be blurry, preventing data from being evaluated accurately, or the camera isn't recording at the time of impact. Additional personals were not necessary for the successful completion of this test.

The test procedure is as follows:

1. To set up for the testing, apply about 5 feet of tape to the ground starting from the intended impact curb and moving outward, ensure perpendicularity with a measuring square.
2. Charge the battery using a compatible charging device, as well as place AA batteries into the transmitter if necessary.
3. Turn on the Electronic Speed Controller (ESC), located to the front right of the car as shown in the figures 2 by the red box, by holding down the blue button, shown in figure 3, for a minimum of 4 seconds. The Receiver will flash between red and green; this indicates the receiver is looking for a signal from the transmitter.

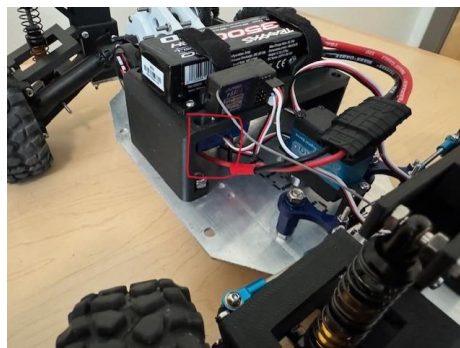


Figure 2. ESC location



Figure 3. ESC On Button

4. On the matching transmitter or controller, flip the switch labeled on/off as shown in figure 4 , on the side with the steering dial, to the on position. The red and green light on the receiver should turn orange, this indicates that the receiver and the transmitter are paired.
 - a. If the red and green light doesn't switch over, then turn the ESC by pressing the button for 4 seconds, then turn it back on while simultaneously turning the steering dial forward and pressing the throttle trigger forward. This will pair the transmitter and the receiver.



Figure 4. Controller On Switch

5. Prior to the test starting, ensure the testing area is safe for testing to begin. This includes putting eye protection on, as well as pedestrians and vehicles are not going to interfere with the test or get harmed by flying debris if the event occurs.
6. On the transmitter, ensure the transmitter is set to supply 50% of the motor power, using the knob shown in figure 5 by the red box, to ensure control and safety. The motor power dial is located on the bottom right of the output controller panel, under the liftable flap.



Figure 5. Motor Power Dial

7. Take a top-down picture of the car before starting the test, the picture should be close enough that the car fills in the frame with a 1-inch margin. This is to ensure as close of an observation as possible.
8. Begin the test by depressing the throttle trigger to achieve 15 mph, depress trigger slowly as to not over torque the gears.
9. Using the speedometer, ensure the car reaches a minimum of 15 mph, this value can be between 13 – 17 mph. This range is determined by the space of the test.
10. Once the car has reached a speed of 15 mph, steer the car towards the curb using the previously applied blue tape as a guideline up to impact, this ensures that the car hits the curb mostly head-on, with a 2–5-degree margin of error.
11. Take a second picture of the RC car after the impact using the same 1-inch margin.
12. Record Observations and measure the deflection of the car after the impact.
13. Repeat steps 7-11 until either the bumper exceeds its limit and breaks or 5 trials are conducted.

Deliverables

This testing was initially a failure, due to the higher deflection value in the bumper, however after redesigning the bumper to better absorb the impact, the deflection value was reduced to under the requirement. The parameter value was found to be a deflection of 0.59 inches from the starting value. The calculated value for this parameter was 0.6 inches, and this is the requirement for this car. To consider this test to be a successful test, the value had to have been under the calculated value. The first run of this test resulted in failure; however, the rerun resulted in a successful test. This test was conducted to determine if the car would be able to survive a head-on impact with a wall should the controller lose control of the car during operation. This test is determined to be a successful test after the redesign.

Appendix G1.1 – Procedure Checklist

The speedometer was downloaded. The measuring square and tape were collected from the operator's primary residence. The test was ready to begin after ensuring the area was clear of potential obstacles.

Appendix G1.2 – Data Forms

Test #1

Impact Number	Amount Deflected from Normal (in)
0	
1	
2	

Test #2, rerun with modified bumper

Impact Number	Amount Deflected from Normal (in)
0	
1	
2	
3	

Appendix G1.3 – Raw Data

Test #1

Impact Number	Amount Deflected from Normal (in)
0	No Deflection
1	0.632
2	Failure

Test #2, rerun with modified bumper

Impact Number	Amount Deflected from Normal (in)
0	No Deflection
1	0.582
2	0.59
3	Failure

Appendix G1.4 – Evaluation Sheet

No evaluation sheet was needed for this test

Appendix G1.5 – Schedule (Testing)

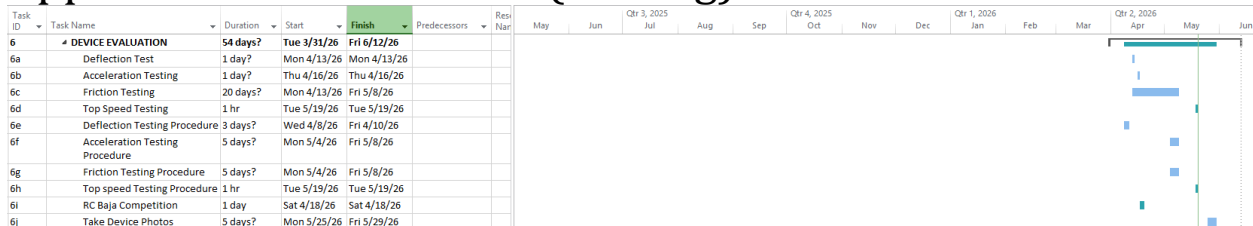


Figure 6. Testing Schedule

Appendix G2 Acceleration Testing

Introduction

This test was conducted to test requirement 8 in section 1d, the drivetrain must reach 10 mph in under 10 seconds. The parameter of interest for this test is the time that it takes for the car to reach 10 mph. The calculation to confirm that the car can reach the required speed can be seen in Appendix A09, however the predicted time is 7 seconds. This prediction is based on the Baja sprint completion time, this turned out to be incorrect as the distance for the test and the distance for the sprint were not comparable. The data was collected using the stopwatch found on most cellular devices. The schedule for this test can be seen in the Gantt chart in Appendix E as well as at the end of this document.

Method/Approach

This test used blue tape to determine the start and end of the testing area. Data capture was done when the car crossed the starting line and ended when the car crossed the finish. This test uses a calculated distance to determine when exactly the car would reach 10 mph. This ensures that the car will reach 10 mph every time without needing to look at an inaccurate speedometer. This, however, resulted in an inaccurate predicted value for the time. Each point of time was written down during the test, then later reported in a table format seen in Appendix G2.3.

Test Procedure

Summary:

This Procedure documents the set-up, process, and recording for the acceleration testing. This test would have been a better first test rather than the destructive test, as this test makes certain that the car can reach the required speed for the slalom-and-sprint. This test was done to test the drivetrains' ability to reach 10 mph within a certain time frame. This test met the testable requirement for requirement 8 in section 1d, the drivetrain must reach a speed of 10 mph within 10 seconds.



Figure 1. Testing set-up

Time: This test was performed on April 26, 2026, at 6:30 pm. The test took a total duration of 2 hours. This time included the 20 minutes to set up the test and the 10 minutes to remove testing material from the parking lot.

Place: Outside of Wendell Hill Hall, Central Washington University campus in Ellensburg, WA

Required Equipment includes:

- Tron-Car
- Timer
- RC Transmitter (Controller)
- A Charged Battery
- Blue Tape
- Measuring Square
- Tape Measure

Risk:

There are only 2 risks that were associated for this test. The first risk is that this is done in the parking lot, so ensuring that pedestrians and cars were not present during testing was critical. This was solved by pausing testing and moving while cars and people moved out of the way. The second risk for this test was reaction speed. The ability to start and stop the timer when the car started and stopped. This was unfortunately something that was not able to easily fix so there is a plus or minus .2 seconds margin of error in the data to account for this reaction speed.

The test procedure is as follows:

1. To set up for the testing, apply about 3 feet of tape to the ground. This acts as the starting point for the test. Then place another piece of tap about 1.5 feet perpendicular to the starting piece, use a square to ensure perpendicularity.
2. Place the car so that the front bumper is behind the starting piece, and that both front and rear tires are on the perpendicular tape.
3. Run a tape measure 14.30 feet from the starting position, if the tape measure does not go full distance, than place a piece of tape on the ground at the furthest point and continue until 14.3 feet. This values calculation can be seen in appendix G2.4. The parallelism of the second piece is not as important as the starting point.
4. Charge the battery using a compatible charging device, as well as place AA batteries into the transmitter if necessary.
5. Turn on the Electronic Speed Controller (ESC), located to the front right of the car as shown in figure 2 by the red box, by holding down the blue button, shown in figure 3, for a minimum of 4 seconds. The Receiver will flash between red and green; this indicates the receiver is looking for a signal from the transmitter.

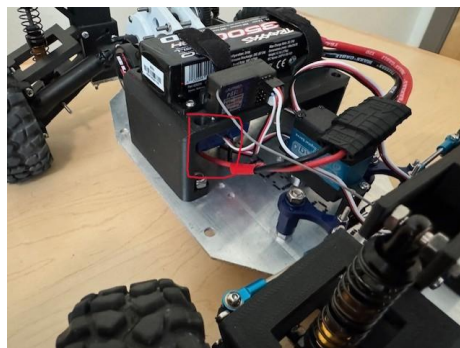


Figure 2. ESC location



Figure 3. ESC On Button

6. On the matching transmitter or controller, flip the switch labeled on/off as shown in figure 4, on the side with the steering dial, to the on position. The red and green light on the receiver should turn orange, this indicates that the receiver and the transmitter are paired.
 - a. If the red and green light doesn't switch over, then turn the ESC by pressing the button for 4 seconds, then turn it back on while simultaneously turning the steering dial forward and pressing the throttle trigger forward. This will pair the transmitter and the receiver.



Figure 4. Controller On Switch

7. Prior to the test starting, ensure the testing area is safe for testing to begin. This includes putting eye protection on, as well as pedestrians and vehicles are not going to interfere with the test or get harmed by flying debris if the event occurs.
8. On the transmitter, ensure the transmitter is set to supply 25% of the motor power, using the knob shown in figure 5 by the red box, to ensure control and safety. The motor power dial is located on the bottom right of the output controller panel, under the liftable flap.



Figure 5. Motor Power Dial

9. Move to the ending point of the test, while having a timer on standby.
10. Begin the test by depressing the throttle trigger. Start the timer at the same time.
11. Stop the timer the moment the RC crosses the finishing piece of tape. Release the throttle trigger at the same time.
12. Record the time for the trial.
13. Repeat Steps 10-12 a total of 5 times.

Deliverables

This testing was a Successful test, with some margin of error for mistakes in calculations. The parameter value was the time that the car took to reach the end of the testing track, this value was an average of 2.48 seconds from start to finish. There was no calculated value for this parameter, however the requirement for this test was under 10 seconds. To consider this test to be a successful test, the value had to have been below the required value. There was a massive jump in the requirement verses the required value, this difference is likely due to changes in the drivetrain design that allowed for less torque loss than the original design. This test was determined to be a successful test.

Appendix G2.1 – Procedure Checklist

A measuring square and tape were reused from test #1 in Appendix G1. The test was ready to begin after ensuring the site was clear of obstacles.

Appendix G2.2 – Data Forms

Trial Number	Seconds
1	
2	
3	
4	
5	
Average	

Appendix G2.3 – Raw Data

Trial Number	Seconds
1	2.4
2	2.3
3	2.5
4	2.8
5	2.4
Average	2.48

Appendix G2.4 – Evaluation Sheet

Alex Amadio | MET 489-C | 4/25/2026

Acceleration test calculations:

Given: 10 mph = 4.4704 m/s

find: distance for RC car to reach 10 mph

assume: moderate exponential rise ($K=1.0$)

$T = 7 \text{ seconds}$

method: moderate exponential rise

$$A(t) = C \times (e^{(Kt)} - 1)$$

$$V(t) = C(e^{(Kt)} - 1)$$

Solution

$$4.4704 = C(e^7 - 1)$$

$$C = \frac{4.4704}{(e^7 - 1)} = \frac{4.4704}{1064.13316} = 0.0041642$$

$$a(t) = 0.0041642(e^7 - 1) = 4.499 \text{ m/s}^2$$

$$Distance = (1/K^2) e^{(Kt)} - 1/2 T^2 - (1/K) \cdot T - 1/K^2$$

$$e^7 - 0.5 \times 4.4 - 7 - 1 = 106.63316 - 20.5 - 7 - 1 = 1064.13316$$

$$S = C \times 1064.13316 \approx 0.0041642 \times 1064.13316 = 4.370 \text{ meters}$$

$$4.370 \times 3.28084 = \boxed{14.34 \text{ ft}}$$

Figure 1. Testing distance

Appendix G2.5 – Schedule (Testing)

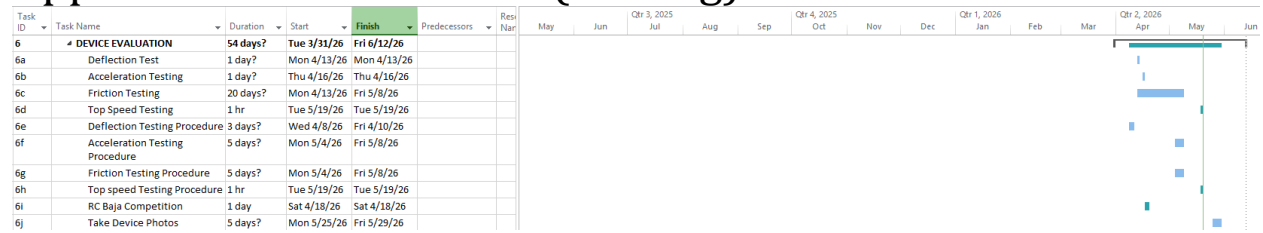


Figure 6. Testing Schedule

Appendix G3 Ramp Test

Introduction

This test was conducted to test requirement 10 in section 1d, the drivetrain must provide enough torque to climb a 45-degree ramp. The parameter of interest for this test is whether or not the car can climb the 45-degree incline. The calculation to confirm that the car can reach the required speed can be seen in Appendix A05, based off of this calculation the car is predicted to be able to climb the ramp. The data was collected using observational analysis. The schedule for this test can be seen in the Gantt chart in Appendix E as well as at the end of this document.

Method/Approach

This test used a ramp that was constructed by the student to ensure the accuracy of the angle. Data capture was done when the car reached the top of the ramp, for this test if the car got stuck climbing over a lip at the top of the ramp, it would have been considered a passing test. The car was placed about 6 feet away from the ramp then driven towards the ramp with the objective of climbing over it. Each observation was written down during the test, then later reported in a table format seen in Appendix G3.3.

Test Procedure

This Procedure documents the set-up, process, and recording for the ramp testing. This test was done to test the drivetrains' ability to climb a 45-degree ramp. This test met the testable requirement for requirement 10 in section 1d, the drivetrain must provide enough torque to climb a 45-degree ramp.



Figure 1. Testing set-up

Time: This test was performed on May 5, 2026, at 11:50 pm. The test took a total duration of 1 hour. This time included the 10 minutes to set up the test and the 10 minutes to remove testing material from the parking lot.

Place: Outside of Wendell Hill Hall, Central Washington University campus in Ellensburg, WA

Required Equipment includes:

- Tron-Car
- RC Transmitter (Controller)
- A Charged Battery
- Built Wooden Ramp
- Tape Measure

Risk:

The only risk for this test is if the car hits the ramp too fast there is a possibility of the car flying off the ramp. This would put the car into an uncontrolled fall and could damage components if it hit the ground too hard.

The test procedure is as follows:

1. To set up for the testing, construct a wooden ramp that has a base and height of 20 inches to get a 45-degree angle, as seen in Figure 1.
2. Place the car so that the front bumper is 6 feet from the ramp.
3. Charge the battery using a compatible charging device, as well as place AA batteries into the transmitter if necessary.
4. Turn on the Electronic Speed Controller (ESC), located to the front right of the car as shown in figure 2 by the red box, by holding down the blue button, shown in figure 3, for a minimum of 4 seconds. The Receiver will flash between red and green; this indicates the receiver is looking for a signal from the transmitter.

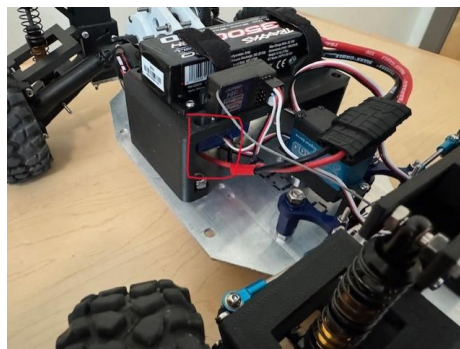


Figure 2. ESC location



Figure 3. ESC On Button

5. On the matching transmitter or controller, flip the switch labeled on/off as shown in figure 4, on the side with the steering dial, to the on position. The red and green light on the receiver should turn orange, this indicates that the receiver and the transmitter are paired.
 - a. If the red and green light doesn't switch over, then turn the ESC by pressing the button for 4 seconds, then turn it back on while simultaneously turning the steering dial forward and pressing the throttle trigger forward. This will pair the transmitter and the receiver.



Figure 4. Controller On Switch

6. Prior to the test starting, ensure the testing area is safe for testing to begin. This includes putting eye protection on, as well as pedestrians and vehicles are not going to interfere with the test or get harmed by flying debris if the event occurs.
7. On the transmitter, ensure the transmitter is set to supply 60% of the motor power, using the knob shown in figure 5 by the red box, to ensure control and safety. The motor power dial is located on the bottom right of the output controller panel, under the liftable flap.



Figure 5. Motor Power Dial

8. Begin the test by slowly depressing the trigger while driving towards to the ramp.
9. Drive the car up the ramp and onto the landing platform.
10. Repeat steps 8 and 9, five times to ensure consistency.
11. If needed remove bumper, then repeat steps 8 and 9, five times.

Deliverables

This testing was a Successful test. The parameter value was whether this car was able to climb over a 45-degree ramp. The calculated friction and torque value found in Appendix A05, states that the car should have been able to climb the ramp without must issue. To consider this test to be a successful test, the car needed to climb the ramp. The car was able to climb the ramp; however, two tests were done for this requirement, the first test was with the bumper attached and the second test was with the bumper detached from the car. While the car was able to climb the ramp in both tests, the test without the bumper was more successful than the test with the bumper. This difference shows that the bumper was either too large for the front of the car or the suspension doesn't have enough room to articulate. This test was determined to be a successful test.

Appendix G3.1 – Procedure Checklist

The ramp was constructed using scrape 2x4s and panels at the operator's primary residence. This ramp can be seen in Figure 1 for the testing set-up. The Test was ready to begin after ensuring that the testing area was clear of pedestrians.

Appendix G3.2 – Data Forms

Test 1 with bumper	
Trial Number	Pass/Fail
1	
2	
3	
4	
5	

Test 2 without bumper	
Trial Number	Pass/Fail
1	
2	
3	
4	
5	

Appendix G3.3 – Raw Data

Test 1 with bumper	
Trial Number	Pass/Fail
1	Pass
2	Pass
3	Pass
4	Pass
5	pass

Test 2 without bumper	
Trial Number	Pass/Fail
1	Pass
2	Pass
3	Pass
4	Pass
5	Pass

Appendix G3.4 – Evaluation Sheet

No evaluation sheet was used for this test.

Appendix G3.5 – Schedule (Testing)

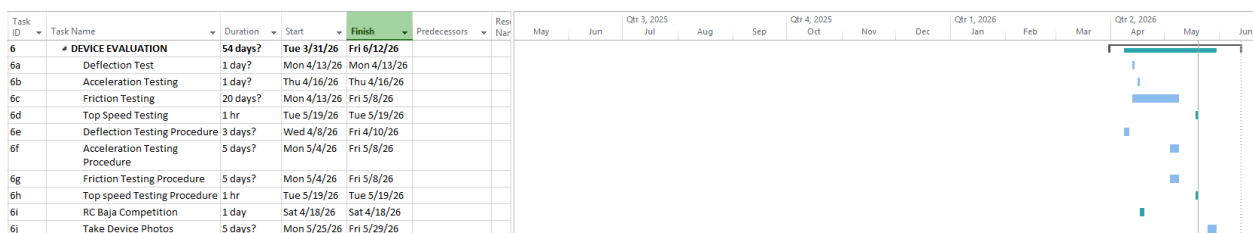


Figure 6. Testing Schedule

Appendix G4 Top Speed Test

Introduction

This test was conducted to test requirement 4 in section 1d, the drivetrain be able to achieve a maximum speed of 30 mph. The parameter of interest for this test is the RPM achieved by the wheels when the controller is completely depressed. The calculation to confirm that the car can achieve the required speed can be seen in Appendix A04 and A05. With the gear reduction the car was predicted to be able to achieve the required 30 mph. The schedule for this test can be seen in the Gantt chart in Appendix E as well as at the end of this document.

Method/Approach

This test used a light-based RPM measuring device to gather accurate RPM readings. Data capture was done after 10 seconds of a continuous testing period. The car was placed upside down for this test and then run by pressing the trigger on the controller to, the measuring device was placed to read the wheel rotation. The final RPM was taken after 10 seconds then written down during the test, then later reported in a table format seen in Appendix G4.3.

Test Procedure

This Procedure documents the set-up, process, and recording for the top speed testing. This test was done to test the drivetrains' ability to reach a maximum speed of 30 mph. This test met the testable requirement for requirement 4 in section 1d, the drivetrain be able to achieve a maximum speed of 30 mph.



Figure 1. Testing set-up

Time: This test was performed on May 19, 2026, at 8:00 pm. The test took a total duration of 30 minutes. This time included the 10 minutes to set up the test and the 10 minutes to remove testing material from the parking lot.

Place: Inside of Wendell Hill Hall, Central Washington University campus in Ellensburg, WA

Required Equipment includes:

- Tron-Car
- RC Transmitter (Controller)
- A Charged Battery

- A Small Piece of Blue tape
- An RPM measuring device

Risk:

The only risk for this test is if the car's wheels are not secured to the axle well enough, there is a chance that the wheel gets ejected from the testing area and causing injury. To prevent injury, wearing safety glasses will prevent irreversible damage.

The test procedure is as follows:

1. To set up for the testing, place the car against a wall upside down to or right side up with the wheel elevated slightly, as seen in Figure 1.
2. Place a Piece of blue tape on the wheel to allow the RPM measuring device to be able to give a reading.
3. Charge the battery using a compatible charging device, as well as place AA batteries into the transmitter if necessary.
4. Turn on the Electronic Speed Controller (ESC), located to the front right of the car as shown in figure 2 by the red box, by holding down the blue button, shown in figure 3, for a minimum of 4 seconds. The Receiver will flash between red and green; this indicates the receiver is looking for a signal from the transmitter.

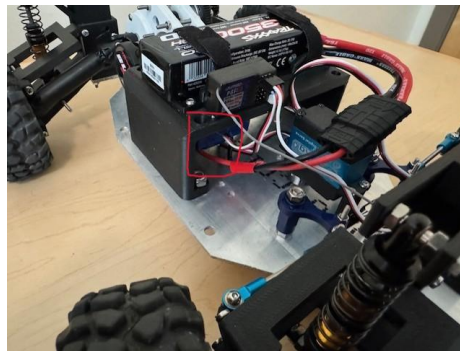


Figure 2. ESC location



Figure 3. ESC On Button

5. On the matching transmitter or controller, flip the switch labeled on/off as shown in figure 4, on the side with the steering dial, to the on position. The red and green light on the receiver should turn orange, this indicates that the receiver and the transmitter are paired.

- a. If the red and green light doesn't switch over, then turn the ESC by pressing the button for 4 seconds, then turn it back on while simultaneously turning the steering dial forward and pressing the throttle trigger forward. This will pair the transmitter and the receiver.



Figure 4. Controller On Switch

6. Prior to the test starting, ensure the testing area is safe for testing to begin. This includes putting eye protection on. Then turn the lights off to allow for better RPM reading
7. Then hold the RPM measuring device up to the piece of blue tape. Turn on the RPM measuring device using the provided instructions for the available device.
8. On the transmitter, ensure the transmitter is set to supply 50% of the motor power, using the knob shown in figure 5 by the red box, to ensure control and safety. The motor power dial is located on the bottom right of the output controller panel, under the liftable flap.



Figure 5. Motor Power Dial

9. Begin the test by depressing the trigger while the RPM measuring device is pointed at the piece of blue tape.

10. Keep this test running for a minimum of 10 seconds to allow for the drivetrain to give stabilized reading.
11. Mark down the result
12. Change the motor power to 75%, then repeat steps 9 through 11.
13. Change the motor power to 100%, then repeat steps 9 through 11.

Deliverables

This testing was not a successful test; The parameter value was the RPM of the wheels during operation; at 100% motor power the RPM was 2235 RPM for a 4-inch wheel. The calculated value for this test was to be able to achieve 30 mph according to Appendix A05. To consider this test to be a successful test, the value had to have been within 5% of 30 mph without going over the requirement. The reason that these values were not within the parameter is likely due to friction losses somewhere along the drivetrain, this would explain the close value for 100% motor power. This test was determined to be an unsuccessful test for this reason, while the car will not experience the maximum speed during normal usage this inability to be within range of the maximum speed value was why this test was not successful.

Appendix G3.1 – Procedure Checklist

An optical RPM measuring device was borrowed from Hogue Hall to be returned the next day. This test was done indoors, therefore there was no need to wait until the testing area was clear to begin testing.

Appendix G3.2 – Data Forms

Motor Power Output	RPM Value	Converted MPH Value
50%		
75%		
100%		

Appendix G3.3 – Raw Data

Motor Power Output	RPM Value (RPM)	Converted mph Value (mph)
50%	1000	11.9
75%	1329	15.82
100%	2235	26.6

Appendix G3.4 – Evaluation Sheet

An RPM converter application was used to convert the RPM values to mph, so there is no evaluation sheet for this test.

Appendix G3.5 – Schedule (Testing)

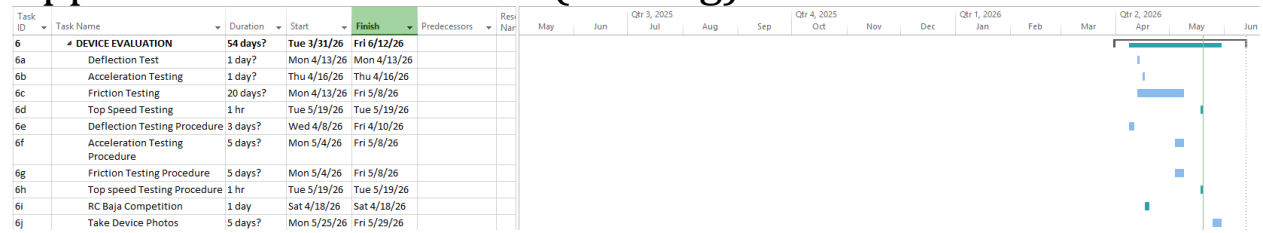


Figure 6. Testing Schedule

APPENDIX H – Resume

ALEX AMADIO

| alex.amadio@outlook.com

Objective

Actively looking to gain experience for a career with a Mechanical engineering background.

Skills & Abilities

Abilities

- Flexibility for complex situations
- On-the-spot learning
- Problem solving in creative and cost-effective ways.

Skills

- 5+ years' experience in 3D modeling in AutoCAD Fusion 360
- SolidWorks Associate Certified
- 5+ years' experience 3D printing
- Project cost analysis
- Public Speaking

Experience

June 2021- september 2021	Courtesy Clerk, <i>Safeway</i> • Worked to help customer satisfaction, and bagging for cashiers.
September 2021- december 2021	Deli Clerk, <i>Safeway</i> • Helped with housekeeping in House Deli, with primary job in cooking in the morning.
June 2022 – September 2022	Deli Clerk, <i>Safeway</i> • Helped with housekeeping in House Deli, with primary job producing products and customer satisfaction.
June 2025 – August 2025	Mechanical Engineering Intern, <i>H.F. Hauff</i>

June 2021-
september 2021

Courtesy Clerk, *Safeway*

- Worked to help customer satisfaction, and bagging for cashiers.
- Helped with designing the frame of a project to compete with competitors' products. This work used SolidWorks to confirm calculations

Education

Dates From-
Current

Mechanical Engineering Technology, Ellensburg, *Central Washington University*

Classes	Proficiency
Computer Aided Design	Solidworks Associate Certification
Project Cost Analysis	Proficient
Statics	Proficient
Machining	Proficient
Lean Manufacturing	Proficient